

Exercises for Mathematics of Linguistic Reality

A companion workbook

Cem Bozşahin

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Cem Bozşahin

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Preface

This workbook accompanies *Mathematics of Linguistic Reality: An Introduction*. Its organization follows the text exactly. Chapter, section, and subsection numbers therefore identify the corresponding portion of the main book rather than page numbers, which may change between versions.

The exercises ask students not only to calculate, but also to construct, diagnose, compare, and apply mathematical descriptions to linguistic objects.

The book has been generated by ChatGPT 5.6 Sol under my guidance to strictly follow the book that I wrote myself, which included some brainstorming with it for its structuring—pardon the anthropomorphism. I have guided it toward the exercises in the directions that I wanted, one section at a time, after providing it with the main text in whole. I have checked the soundness of the exercises personally. I am solely responsible for the adequacy of the questions and their relevance to the book's material and purpose, and for all remaining errors and inaccuracies.

How to use this workbook

Exercise numbers encode their source section. For example, Exercise 1.4.2.3 is the third exercise associated with §1.4.2 of *Mathematics of Linguistic Reality*. Exercises within a section may be attempted selectively; their order normally moves from direct use of definitions to construction, diagnosis, and linguistic application.

In this world of AI-generated mathematics and science, over and above human-generated mathematics and science on which it is based, it seems pointless to keep a complete solutions list private for instructors to compare with student solutions or to grade them. Anybody can ask these questions to an AI and get a reasonable answer. But delusion is a teacherly thing. I have asked an AI to do just that, and I make it available to instructors for a bit of help. I have done the wetware check myself for the solutions. I wish my fellow teachers the best in hoping to face another wetware when they are marking.

The point of the exercises is self study and self examination to better understand the concepts and ideas behind methods so that we can ask better questions at the end of the class, to ourselves, colleagues and AIs, and, to think of better concepts and methods for ourselves. Where an exercise has more than one defensible answer, students should state the assumptions under which their answer is admissible.

1 Basics revisited

1.1 Sets

Exercise 1.1.1 (Recognition) For each set below, state its cardinality. Explain any cases in which the number of written expressions differs from the number of elements.

1. $\{a, b, c, d\}$
2. $\{a, b, a, c, b\}$
3. $\{\emptyset\}$
4. $\{\emptyset, \{\emptyset\}\}$
5. $\{0, \{0\}, \{\{0\}\}\}$
6. $\{\{a, b\}, \{b, a\}\}$

Exercise 1.1.2 (Membership and inclusion) Let

$$A = \{a, b\}, \quad B = \{a, \{a, b\}\}, \quad C = \{\emptyset, A\}.$$

Determine whether each statement is true or false. Justify each answer by distinguishing membership from inclusion.

1. $a \in A$
2. $a \subseteq A$
3. $A \in B$
4. $A \subseteq B$
5. $\emptyset \in C$

1 Basics revisited

6. $\emptyset \subseteq C$

7. $A \in C$

8. $\{A\} \subseteq C$

Exercise 1.1.3 (Diagnosis) A student writes

$$\{a, b, c\} \neq \{c, b, a\}$$

because the elements occur in different orders. Another student writes

$$\{a, a, b\} \neq \{a, b\}$$

because the first expression contains three written items.

Evaluate both claims. What two properties of set abstraction are at issue? Give a kind of mathematical object for which the first student's distinction would matter.

Exercise 1.1.4 (Set operations) Let

$$A = \{a, b, c, d\}, \quad B = \{c, d, e\}, \quad C = \{a, c, e\}.$$

Calculate the following:

1. $A \cup B$

2. $A \cap B$

3. $A - B$

4. $B - A$

5. $(A \cup B) - C$

6. $(A - C) \cap (B - C)$

7. $(A \cap C) \cup (B \cap C)$

For the final two expressions, say whether changing the order of the sets changes the result.

Exercise 1.1.5 (Construction) Write each of the following as a set by listing its elements and, where appropriate, in set-builder notation.

1. The nonnegative even numbers smaller than 12.
2. The natural numbers greater than 3 and smaller than 9.
3. The members of $\{1, 2, 3, 4, 5, 6\}$ that are divisible by 2 or by 3.
4. The members of $\{1, 2, 3, 4, 5, 6\}$ that are divisible by both 2 and 3.

Explain how the contrast between the last two sets is represented in the conditions on membership.

Exercise 1.1.6 (*Arbitrary largeness*) Consider the set

$$E = \{n \mid n \in \mathbb{N} \text{ and } n \text{ is even}\}.$$

Show that no proposed member of E can be its largest member. Your argument should specify, for an arbitrary $n \in E$, a larger number that must also belong to E .

Would the same argument establish that the following set is arbitrarily large? Why or why not?

$$F = \{n \mid n \in \mathbb{N} \text{ and } n < 10^{100}\}.$$

Exercise 1.1.7 (*Cartesian product*) Let $A = \{a, b\}$ and $B = \{0, 1, 2\}$.

1. List the members of $A \times B$.
2. List the members of $B \times A$.
3. Is $A \times B = B \times A$?
4. Compare $|A \times B|$ and $|B \times A|$.
5. State a general relation between $|A \times B|$, $|A|$, and $|B|$ for finite sets.

Explain why the order within an ordered pair matters although the order in which the members of a set are written does not.

Exercise 1.1.8 (*Power sets*) List the members of each power set.

1. $\mathcal{P}(\emptyset)$
2. $\mathcal{P}(\{a\})$
3. $\mathcal{P}(\{a, b\})$

1 Basics revisited

4. $\mathcal{P}(\{\emptyset, \{\emptyset\}\})$

On the basis of your results, formulate a general rule for $|\mathcal{P}(A)|$ when $|A| = n$.

Exercise 1.1.9 (Diagnosis) Consider the following proposed definition:

$$\mathcal{P}(A) = \{\text{all subsets } x \text{ of } A \mid A \subseteq A, \emptyset \subseteq A\}.$$

The two conditions following the vertical bar are true for every set A , but they do not determine which values of x belong to the power set.

Identify the defective condition and rewrite the definition so that membership in $\mathcal{P}(A)$ depends on the possible value of x .

Exercise 1.1.10 (Representation) Compare the following representations:

$$\{n \mid n \in \mathbb{N}\} \quad \text{and} \quad \{n \mid n \in \mathbb{R}\}.$$

1. What does each representation characterize?
2. Does either expression, by itself, state how to obtain a next member?
3. Give a rule involving 0 and the successor operation that can be used to generate the natural numbers.
4. Explain why replacing $\mathbb{N}$ with $\mathbb{R}$ in that rule would not generate all the real numbers.

Distinguish clearly between characterizing the members of a set and providing a procedure that generates them.

Exercise 1.1.11 (Conceptual distinction) Evaluate the following claims.

1. Every very large finite set is uncountable.
2. Every arbitrarily large set is uncountable.
3. If the elements of a set are real numbers, the set must be uncountable.
4. If a set is countable, its members can be put into correspondence with some or all of the natural numbers.
5. If a set is countable, it must have a largest member.

For every false claim, give a counterexample.

Exercise 1.1.12 (Linguistic application) Suppose that a small lexicon is classified as follows:

$$N = \{\text{book}, \text{fish}, \text{walk}\},$$

$$V = \{\text{fish}, \text{walk}, \text{read}\},$$

$$T = \{\text{book}, \text{read}\},$$

where N is the set of items with a nominal use, V the set of items with a verbal use, and T the set of obligatorily transitive items under the analysis being considered.

Calculate and interpret:

1. $N \cap V$
2. $N - V$
3. $V - N$
4. $V \cap T$
5. $(N \cup V) - T$

Does membership in both N and V require two copies of the same written word in the lexicon considered as a set? Explain.

Exercise 1.1.13 (Modeling) A corpus contains 10,000 word tokens but only 1,200 distinct word forms. Consider two abstractions:

$$A = \{\text{word tokens in the corpus}\} \quad \text{and} \quad B = \{\text{word forms occurring in the corpus}\}.$$

1. Under the ordinary notion of set membership, can A preserve repeated occurrences of the same token value?
2. Can either set preserve the order in which words occurred?
3. What information is lost when the corpus is represented only by B ?
4. What additional mathematical structure would be needed to preserve frequency?
5. What additional mathematical structure would be needed to preserve linear order?

Explain why choosing a set as a representation is already a theoretical decision about which properties are treated as relevant.

1.2 Strings

Exercise 1.2.1 (*Sets and strings*) Compare the following objects:

$$\{a, b\}, \quad \{b, a\}, \quad ab, \quad ba, \quad aa.$$

1. Which expressions denote the same object?
2. State the cardinality of each set.
3. State the length of each string.
4. In which objects is repetition significant?
5. In which objects is linear order significant?

Explain what information is added when we move from sets to strings.

Exercise 1.2.2 (*Set concatenation*) Let

$$A = \{a, ab\}, \quad B = \{b, \epsilon\}.$$

Calculate:

1. AB
2. BA
3. AA
4. BB

Remember that duplicate results are counted only once because the result of set concatenation is itself a set. Is set concatenation commutative in this example?

Exercise 1.2.3 (*Empty set and empty string*) Let $A = \{a, b\}$. Calculate or simplify:

1. $A\{\epsilon\}$
2. $\{\epsilon\}A$
3. $\{\epsilon\}\{\epsilon\}$
4. $A\emptyset$

5. $\emptyset A$

6. A^0

7. $\emptyset^0$

8. $\emptyset^1$

Use the results to explain why $\{\epsilon\} \neq \emptyset$.

Exercise 1.2.4 (Identity) For an arbitrary string u , evaluate:

$$\epsilon u, \quad u \epsilon, \quad \epsilon \epsilon, \quad \epsilon^n \quad (n \in \mathbb{N}).$$

What property makes ϵ the identity element of string concatenation? Why is it the only string whose repeated concatenation does not increase length?

Exercise 1.2.5 (Powers of an alphabet) Let $\Sigma = \{a, b\}$. List all members of:

$$\Sigma^0, \quad \Sigma^1, \quad \Sigma^2, \quad \Sigma^3.$$

Then answer the following:

1. What is the length of every string in Σ^n ?
2. What is $|\Sigma^n|$ for $n = 0, 1, 2, 3$?
3. Formulate a general rule for $|\Sigma^n|$.
4. How would that rule change if $|\Sigma| = 3$?

Distinguish the length of a string from the cardinality of a set of strings.

Exercise 1.2.6 (Atomicity) Consider the following two alphabets:

$$A = \{a, b\}, \quad B = \{ab\}.$$

1. Is ab a member of A ?
2. Is ab a member of A^2 ?
3. Is ab a member of B ?
4. What is the length of ab when it is treated as a string over A ?

1 Basics revisited

5. What is its length when it is treated as a single symbol from B ?

Explain why string length depends on the choice of atomic elements in the alphabet.

Exercise 1.2.7 (Kleene star) Let $\Sigma = \{a, b\}$. Determine whether each string belongs to Σ^* . If it does, state the value of n for which it belongs to Σ^n .

ϵ , a , bb , aba , $baab$, c , abc .

Can a string belong to both Σ^m and Σ^n when $m \neq n$? Explain.

Exercise 1.2.8 (Special closures) Determine:

$\emptyset^*$ and $\{\epsilon\}^*$.

Although $\emptyset$ and $\{\epsilon\}$ are distinct sets, their Kleene closures are identical. Explain why.

Exercise 1.2.9 (Diagnosis) A student proposes the following definition:

$$A^* = \{x^i \mid x \in A, i \in \mathbb{N}\}.$$

Let $A = \{a, b\}$.

1. List some strings admitted by the proposed definition.
2. Give four strings that must belong to A^* but are excluded by it.
3. Explain what restriction the proposed definition accidentally imposes.
4. Repair the definition using the sets A^i .

Your corrected definition must admit every finite string constructed from members of A , including the empty string.

Exercise 1.2.10 (Concatenation and Cartesian product) Let $A = \{a, b\}$ and $B = \{0, 1\}$. Compare:

AB and $A \times B$.

1. List the members of each set.
2. What kind of object is a member of AB ?

3. What kind of object is a member of $A \times B$?
4. Does $a0$ have the same internal structure as $(a, 0)$?
5. What would be lost if strings were identified with ordered pairs?

Explain why the fact that both constructions preserve order does not make them the same abstraction.

Exercise 1.2.11 (*Associativity and order*) Let

$$A = \{a, b\}, \quad B = \{0, 1\}, \quad C = \{x, y\}.$$

Calculate enough of the following sets to determine whether the stated equalities hold:

$$(AB)C = A(BC), \quad AB = BA.$$

What does the result show about the difference between associativity and commutativity of concatenation?

Exercise 1.2.12 (*Arbitrarily long strings*) Let Σ be a nonempty alphabet containing at least one nonempty symbol. Show that for every string $u \in \Sigma^*$, there is a string in Σ^* that is longer than u .

Does this establish that every individual string is arbitrarily long? Distinguish the size of Σ^* from the length of each of its members.

Exercise 1.2.13 (*Linguistic application*) Consider the words *dogs*, *chase*, and *cats*.

1. Write the set containing these three word forms.
2. Write all six strings in which each word occurs exactly once.
3. Which distinctions among these strings disappear when they are represented only by the set of their words?
4. Which of the strings would ordinarily be admissible as an English declarative sentence?
5. Does the set of words alone determine the admissibility judgment?

Explain why grammatical admissibility concerns ordered structures rather than collections alone.

Exercise 1.2.14 (*Empty strings and empty categories*) Evaluate the following inference:

1 Basics revisited

An expression is realized by a string of length zero. Therefore its grammatical category contains no information and may be treated as an empty category.

Distinguish:

1. the length of a string;
2. the membership of ϵ in a stringset;
3. the grammatical category assigned to a position or expression.

What additional theory would be required before the third property could be inferred from either of the first two?

Exercise 1.2.15 (Modeling) Suppose two procedures generate exactly the same set of strings. The first constructs each string from left to right; the second constructs it from right to left.

1. Must the resulting strings differ?
2. Does the direction of construction alter the order internal to a completed string?
3. Is the generation procedure a property of the string or of the method used to obtain it?
4. Which part of the description is invariant, and which part is a variant?

Relate your answer to the distinction between the admissibility of strings and a method for generating them.

1.3 Relations

Exercise 1.3.1 (Relations as subsets) Let

$$A = \{a, b\}, \quad B = \{0, 1, 2\}.$$

1. List the members of $A \times B$.
2. Construct a relation $R \subseteq A \times B$ containing exactly three ordered pairs.

3. Construct a different relation $S \subseteq A \times B$ containing exactly three ordered pairs.
4. Is either relation determined merely by knowing A and B ?
5. How many distinct binary relations from A to B are possible?

Explain why defining a relation requires selecting a subset of the Cartesian product.

Exercise 1.3.2 (Tuple and infix notation) Let

$$R = \{(a, b), (a, c), (c, a)\}.$$

Rewrite R entirely in infix notation. Then determine whether each statement is true:

$$aRb, \quad bRa, \quad aRc, \quad cRa, \quad bRc.$$

What information is expressed by the order of the two values in each tuple?

Exercise 1.3.3 (Ariety) Let

$$S = \{\text{student names}\}, \quad C = \{\text{course names}\}, \quad T = \{\text{academic terms}\}.$$

Consider the constraint expressed by

$$(s, c, t) \in E$$

if student s is enrolled in course c during term t .

1. Of which Cartesian product is E a subset?
2. Is E unary, binary, or ternary?
3. What information would be lost if E were replaced by a binary relation between students and courses?
4. Give one way to turn the ternary relation into a binary relation without discarding the term information.

1 Basics revisited

Exercise 1.3.4 (Recognizing properties) Let $A = \{a, b, c\}$, and consider the following relations on A :

$$R_1 = \{(a, a), (b, b), (c, c)\},$$

$$R_2 = \{(a, b), (b, a), (b, c), (c, b)\},$$

$$R_3 = \{(a, b), (b, c), (a, c)\},$$

$$R_4 = \{(a, a), (b, b), (c, c), (a, b), (b, a)\},$$

$$R_5 = \emptyset.$$

For each relation, determine whether it is:

1. reflexive;
2. irreflexive;
3. symmetric;
4. asymmetric;
5. antisymmetric;
6. transitive.

A relation may have more than one of these properties.

Exercise 1.3.5 (Counterexamples) For each false implication below, construct the smallest counterexample you can.

1. Every symmetric relation is reflexive.
2. Every reflexive relation is symmetric.
3. Every antisymmetric relation is asymmetric.
4. Every asymmetric relation is antisymmetric.
5. Every transitive relation is reflexive.
6. Every irreflexive relation is asymmetric.

For any implication that is true, give a proof from the definitions.

Exercise 1.3.6 (Asymmetry and antisymmetry) Let R be a relation on a set A .

1. State in words what asymmetry prohibits.
2. State in words what antisymmetry permits when $a = b$.
3. Can an asymmetric relation contain a pair (a, a) ?
4. Can an antisymmetric relation contain a pair (a, a) ?
5. Can a relation be both asymmetric and antisymmetric?
6. Can a relation be symmetric and antisymmetric?

Give an explicit relation illustrating each possible combination.

Exercise 1.3.7 (Everyday relations) Consider the following relations on an appropriate set of people:

1. has the same birthday as;
2. is a sibling of;
3. is a parent of;
4. is an ancestor of;
5. is older than;
6. is no older than;
7. is the same person as.

Classify each relation as reflexive or irreflexive, symmetric, asymmetric or antisymmetric, and transitive or nontransitive.

State any assumptions required by your classification. In particular, consider whether a relation may have properties in addition to the one used to introduce it.

Exercise 1.3.8 (Constraint versus mapping) Let

$$W = \{\text{book}, \text{fish}, \text{walk}\}, \quad C = \{N, V\},$$

and let

$$R = \{(\text{book}, N), (\text{fish}, N), (\text{fish}, V), (\text{walk}, N), (\text{walk}, V)\}.$$

1. What constraint does R express?

1 Basics revisited

1. Does every member of W participate in the relation?
2. May one member of W be related to more than one member of C ?
3. Does either fact prevent R from being a relation?
4. Does either fact prevent R from being a relation?
5. What additional discipline would be required if R were to be treated as a function from W to C ?

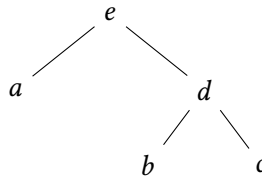
Explain what is obscured by calling R merely a defective function.

Exercise 1.3.9 (*A relation may be neither*) Construct a relation on $\{a, b, c\}$ that is:

1. neither reflexive nor irreflexive;
2. neither symmetric nor asymmetric;
3. neither symmetric nor antisymmetric;
4. neither transitive nor intransitive under the ordinary negation of transitivity.

Explain why the named relation types do not constitute a list of mutually exclusive possibilities.

Exercise 1.3.10 (*Dominance*) Consider the following tree:



Let D be reflexive dominance: every node dominates itself, and a node dominates every node below it.

1. List D as a set of ordered pairs.
2. Show that D is reflexive.
3. Show that D is antisymmetric.
4. Show that D is transitive.

5. Is D symmetric?

6. Is D asymmetric?

For every negative answer, identify a pair or collection of pairs that provides a counterexample.

Exercise 1.3.11 (Strict and reflexive dominance) Define strict dominance D^+ by removing every pair (x, x) from the reflexive dominance relation D in Exercise 1.3.10.

1. List the members of D^+ .
2. Is D^+ reflexive or irreflexive?
3. Is it asymmetric?
4. Is it antisymmetric?
5. Is it transitive?

Which properties change when the self-pairs are removed, and which remain?

Exercise 1.3.12 (C-command) For the tree in Exercise 1.3.10, use the following definition:

A node x c-commands a node y if x does not dominate y and the first branching node dominating x also dominates y .

1. List the c-command relation C as a set of ordered pairs.
2. Is C reflexive?
3. Is C symmetric?
4. Is C antisymmetric?
5. Is C transitive?
6. For each property that fails, give a counterexample from the tree.

Compare the relational properties of c-command with those of dominance.

Exercise 1.3.13 (First branching node) Suppose that “first branching node dominating x ” could associate x with two distinct nodes in the same tree.

1 Basics revisited

1. Would the definition of c-command give a unique structural criterion?
2. Could the two candidate branching nodes dominate different sets of nodes?
3. How could this change the extension of the c-command relation?
4. What uniqueness requirement is therefore imposed on “first branching node”?

Explain why c-command is a relation although identifying the first branching node requires a function-like association.

Exercise 1.3.14 (*Tree admissibility*) For each structure described below, determine whether it is admitted as a tree under the conditions in §1.3 of *Mathematics of Linguistic Reality*.

1. A single node with no links.
2. Two disconnected rooted structures.
3. A nonroot node with two parents.
4. A cycle in which a is the parent of b , b of c , and c of a .
5. A root with three leaf children.
6. Two roots sharing one child.

For every inadmissible structure, identify the condition that it violates. Do not appeal to a particular drawing convention.

Exercise 1.3.15 (*Parenthood and dominance*) Using the tree in Exercise 1.3.10, define:

$$P = \{(x, y) \mid x \text{ is the parent of } y\}$$

and

$$D^+ = \{(x, y) \mid x \text{ strictly dominates } y\}.$$

1. List P and D^+ .
2. Which pairs belong to D^+ but not to P ?
3. Is parenthood transitive?
4. Is strict dominance transitive?

5. Explain how strict dominance extends the local parenthood constraint.

Exercise 1.3.16 (*Linguistic relations*) Consider the following relations on linguistic objects:

1. precedes, on positions in a string;
2. immediately precedes, on positions in a string;
3. has the same phonological form as, on word tokens;
4. agrees in number with, on expressions;
5. is a constituent of, on syntactic objects;
6. is the immediate parent of, on nodes in a tree.

For each relation, determine which of the properties in §1.3 of *Mathematics of Linguistic Reality* it has. State the domain and any analytical assumptions on which the answer depends.

Exercise 1.3.17 (*Invariant and variant*) Classify each proposed property as:

1. a candidate condition on the admissibility of trees;
 2. a further relation definable only after a tree has been admitted; or
 3. a feature of a particular notation for displaying trees.
1. There is one overall root.
 2. Every nonroot node has one parent.
 3. Sisters are drawn at the same vertical height.
 4. Nodes are connected by straight lines.
 5. Leaves have no children.
 6. One node c-commands another.
 7. Branches do not cross on the page.
 8. Every node is connected to the root.

1 Basics revisited

Explain why a relation may be mathematically well-defined without being an invariant of tree admissibility.

Exercise 1.3.18 (*Conceptual synthesis*) Evaluate the following argument:

Functions are special kinds of relations. Therefore relations should be introduced as functions for which uniqueness or totality has been relaxed.

Distinguish the following two perspectives:

1. the extensional fact that a function can be represented as a set of ordered pairs;
2. the conceptual distinction between a constraint and a mapping.

Why might the first perspective be mathematically convenient while the second is more revealing for the study of linguistic admissibility?

1.4 Functions: named and nameless

Exercise 1.4.1 (*Relation or function*) Let $A = \{a, b, c\}$ and $B = \{0, 1\}$. Determine which of the following relations are partial functions from A to B , which are total functions, and which are not functions.

$$R_1 = \{(a, 0), (b, 1), (c, 0)\},$$

$$R_2 = \{(a, 0), (a, 1), (b, 0)\},$$

$$R_3 = \{(a, 1), (c, 0)\},$$

$$R_4 = \emptyset,$$

$$R_5 = \{(a, 0), (b, 0), (c, 0)\}.$$

For every relation that is not a function, identify the violated discipline.

Exercise 1.4.2 (*Domain, codomain, and range*) Let

$$f : \mathbb{N} \longrightarrow \mathbb{N}, \quad f(n) = 2n.$$

State:

1. the domain of f ;
2. the codomain of f ;

3. the range of f ;
4. whether f is total;
5. whether f is one-to-one;
6. whether f is onto.

Explain why the codomain and range are not identical in this example.

Exercise 1.4.3 (*Changing the declared sets*) Consider the rule $f(n) = n + 1$ under each declaration:

$$\begin{aligned} f_1 &: \mathbb{N} \longrightarrow \mathbb{N}, \\ f_2 &: \mathbb{N} \longrightarrow \{1, 2, 3, \dots\}, \\ f_3 &: \mathbb{Z} \longrightarrow \mathbb{Z}, \\ f_4 &: \mathbb{N} \longrightarrow \{0, 1, 2\}. \end{aligned}$$

For each declaration, determine whether the rule defines a total function and, if so, whether it is one-to-one and onto.

Explain why the algebraic expression alone does not completely specify a function.

Exercise 1.4.4 (*Functions with product domains*) Give the domain and codomain of each function:

$$\begin{aligned} f(x, y) &= x + y, \\ g(x, y) &= (x + y, x - y), \quad x, y, z \in \mathbb{Z}. \\ h(x, y, z) &= xyz, \end{aligned}$$

Then answer:

1. How many arguments does each function take?
2. How many values does each function return?
3. Why does g return one value rather than two?
4. What is the mathematical type of the value returned by g ?

Exercise 1.4.5 (*Characterization rather than computation*) Consider

$$f(2n) = n, \quad n \in \mathbb{N}.$$

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1. Describe the domain characterized by the argument expression $2n$.
2. State an appropriate codomain.
3. Is f total on the even natural numbers?
4. Would the same rule be total if its domain were declared to be $\mathbb{N}$?
5. What does this example show about treating a function merely as a computational recipe?

Exercise 1.4.6 (One-to-one and onto) For each function, determine whether it is one-to-one and whether it is onto:

$$\begin{aligned}f &: \mathbb{Z} \longrightarrow \mathbb{Z}, & f(n) &= n + 1, \\g &: \mathbb{Z} \longrightarrow \mathbb{Z}, & g(n) &= n^2, \\h &: \mathbb{N} \longrightarrow \mathbb{N}, & h(n) &= 2n, \\k &: \mathbb{Z} \longrightarrow \mathbb{N}, & k(n) &= |n|.\end{aligned}$$

For every failure of one-to-one-ness or onto-ness, provide a witness.

Exercise 1.4.7 (Function spaces) Let $A = \{a, b, c\}$ and $B = \{0, 1\}$.

1. List every total function in B^A .
2. How many such functions are there?
3. How many are one-to-one?
4. How many are onto?
5. Formulate a rule for $|B^A|$ when A and B are finite.

Explain why B^A denotes a set whose members are themselves functions.

Exercise 1.4.8 (Partial versus total function spaces) Let $A = \{a, b\}$ and $B = \{0, 1\}$.

1. How many total functions are there from A to B ?
2. How many partial functions are there from A to B ?
3. Why does allowing an argument to be unmapped add an effective third possibility for each member of A ?

4. Does the conventional notation B^A include these partial functions?

Exercise 1.4.9 (*Odd and even numbers*) Let

$$\mathbb{O} = \{2n + 1 \mid n \in \mathbb{N}\}, \quad \mathbb{E} = \{2n \mid n \in \mathbb{N}\},$$

and define

$$f_o(2n + 1) = 2n, \quad f_e(2n) = 2n + 1.$$

1. State the domain and codomain of each function.
2. Show that both functions are one-to-one.
3. Show that both are onto their stated codomains.
4. Show that $f_e \circ f_o$ is the identity on $\mathbb{O}$.
5. Show that $f_o \circ f_e$ is the identity on $\mathbb{E}$.

What establishes the isomorphism between the two sets?

Exercise 1.4.10 (*Homomorphic preservation*) Let

$$g : \mathbb{N} \longrightarrow \{0, 1, 2\}, \quad g(n) = n \bmod 3.$$

1. Calculate $g(4)$, $g(7)$, $g(9)$, and $g(14)$.
2. Give three distinct inputs mapped to the same output.
3. Is g one-to-one?
4. Is it onto?
5. Verify for three pairs of natural numbers that

$$g(a + b) = (g(a) + g(b)) \bmod 3.$$

6. What operation is preserved by the mapping?

Explain why preservation of an operation does not require isomorphism.

Exercise 1.4.11 (*Degenerate homomorphism*) Let

$$h : \mathbb{N} \longrightarrow \mathbb{N}, \quad h(n) = 0.$$

1 Basics revisited

1. Show that $h(a + b) = h(a) + h(b)$.
2. Is h one-to-one?
3. How much information about the input survives the mapping?
4. In what formal sense is h structure-preserving?
5. Why might it nevertheless be useless as an empirical model?

Explain why being a homomorphism is not by itself a guarantee of an informative abstraction.

Exercise 1.4.12 (*Isomorphism and knowledge*) Evaluate the following proposal:

A syntactic structure and a semantic structure should be assumed to be isomorphic because both exhibit composition.

What would have to be established about the two structures and the mapping between them before an isomorphism could be claimed? What weaker claim would a homomorphism support?

1.4.1 Trees as functions

Exercise 1.4.1.1 (*Reading a parent function*) Consider the parent function:

$$p(a) = d, \quad p(b) = d, \quad p(c) = d, \quad p(d) = e, \quad p(f) = e.$$

1. Identify the root.
2. Identify the leaves.
3. Identify the internal nonroot node.
4. List the children of d .
5. List the children of e .
6. Draw the tree determined by p .

Explain why the equations determine more than a collection of node names.

Exercise 1.4.1.2 (*From function to relation*) Using the parent function in Exercise 1.4.1.1, define the child-of relation C by:

$$xCy \text{ iff } p(x) = y.$$

1. List C as a set of ordered pairs.
2. Is C reflexive?
3. Is C symmetric?
4. Is C antisymmetric?
5. Is C transitive?
6. Is every member of the tree in the first coordinate of some pair in C ?

Relate the final answer to the special status of the root.

Exercise 1.4.1.3 (*Deriving sisterhood*) Define sisterhood by:

$$xSy \text{ iff } p(x) = p(y).$$

Using the parent function in Exercise 1.4.1.1:

1. List every ordered pair in S .
2. Is every node a sister of itself under this definition?
3. Is S symmetric?
4. Is S transitive?
5. Is S an equivalence relation on the domain of p ?
6. What are its equivalence classes?

Why is the root absent from these classes when p is undefined for it?

Exercise 1.4.1.4 (*Strict sisterhood*) A linguist may prefer to define strict sisterhood:

$$xS^+y \text{ iff } p(x) = p(y) \text{ and } x \neq y.$$

1. List S^+ for the running tree.
2. Is S^+ reflexive or irreflexive?
3. Is it symmetric?
4. Is it transitive?

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5. Does it remain an equivalence relation?

Explain what changes when self-sisterhood is excluded.

Exercise 1.4.1.5 (Iteration) Calculate all defined values of:

$$p(p(a)), \quad p(p(b)), \quad p(p(c)), \quad p(p(d)), \quad p(p(f)).$$

1. Which nodes have a grandparent?
2. Which do not?
3. Is $p \circ p$ total on the domain of p ?
4. What is the domain of $p \circ p$ in this tree?
5. What is its range?

Explain why composition may yield a partial function even when the individual applications used at an earlier level are defined.

Exercise 1.4.1.6 (Parenthood and dominance) Let P be immediate parenthood:

$$xPy \text{ iff } p(y) = x,$$

and let D^+ be strict dominance.

1. List P for the running tree.
2. List D^+ .
3. Which pairs belong to D^+ but not to P ?
4. Is P transitive?
5. Is D^+ transitive?
6. How does iteration of p derive the nonlocal pairs in D^+ ?

Exercise 1.4.1.7 (Reconstructing a tree) Let

$$q(a) = c, \quad q(b) = c, \quad q(c) = f, \quad q(d) = e, \quad q(e) = f.$$

1. Draw the tree determined by q .

2. Identify the root.
3. Identify the leaves.
4. Identify all sister groups.
5. Calculate every defined value of $q \circ q$.
6. Calculate every defined value of $q \circ q \circ q$.

Does the parent function determine the tree uniquely, apart from how it is drawn on the page?

Exercise 1.4.1.8 (Function discipline) Consider the proposed parent assignments:

$$p(a) = d, \quad p(a) = e, \quad p(b) = d, \quad p(d) = e.$$

1. Why do these assignments fail to define a function?
2. What structural configuration would they require for a ?
3. Which condition on tree admissibility would be violated?
4. Could the same assignments still define a relation?
5. Why is being a relation insufficient for constituting a tree?

Exercise 1.4.1.9 (Incomplete parent information) Let the node set be

$$N = \{a, b, c, d, e\}$$

and suppose only the following values are given:

$$p(a) = d, \quad p(b) = d, \quad p(d) = e.$$

1. Can the position of c be recovered?
2. Can we determine whether c is a leaf?
3. Can we determine whether the complete structure is connected?
4. Give two different trees compatible with the stated equations.
5. What additional value of p would distinguish your two trees?

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Explain the difference between a partial description of a tree and a partial parent function within a completely specified tree.

Exercise 1.4.1.10 (*Domain and range of the parent function*) For the parent function in Exercise 1.4.1.1, state:

1. the complete set of nodes;
2. the domain of p ;
3. the range of p ;
4. the set of leaves;
5. the set of internal nodes;
6. the root.

Then evaluate:

1. Must every leaf belong to the domain of p ?
2. May a leaf belong to the range of p ?
3. Must every nonroot internal node belong to both the domain and range?
4. Why is the root in the range but not in the domain under the present definition?

Exercise 1.4.1.11 (*Direction of abstraction*) The same link between a and d can be described as:

a is a child of d or d is the parent of a .

1. Which direction associates each nonroot node with at most one value?
2. Which direction may associate one node with several values?
3. Which direction therefore gives the parent function p ?
4. Does reversing the verbal description merely change notation?

Explain why looking at the hierarchy bottom-up reveals a function that is not available in the same form top-down.

Exercise 1.4.1.12 (*Relations do not automatically compose*) A student reasons:

Since a is a child of d and d is a child of e , the expression $C(C(a))$ should return e .

Explain what is wrong with this notation when C is a relation. Then show how the intended result is expressed using:

$$p(p(a)).$$

Could a separate operation of relational composition recover a grandchild relation? Why would that still differ from treating C itself as an ordinary one-place function?

Exercise 1.4.1.13 (*Constitutive and derived structure*) Suppose we are given only the sister groups:

$$\{a, b, c\} \quad \text{and} \quad \{d, f\}.$$

1. Can we infer that a , b , and c share a parent?
2. Can we identify that parent?
3. Can we determine whether the parent of a, b, c is below, above, or unrelated to the parent of d, f ?
4. Can we identify the root?
5. What additional information does the parent function supply?

Explain why sisterhood can be derived from p , whereas p cannot in general be recovered from sisterhood alone.

Exercise 1.4.1.14 (*Alternative drawings*) Draw two visibly different diagrams representing the same parent function. You may change:

1. the left-to-right order of sisters;
2. branch lengths;
3. angles between branches;

4. the vertical spacing between levels.

Which properties remain invariant across the drawings? Which properties belong only to the graphical variants?

Exercise 1.4.1.15 (*Linguistic application*) Consider the syllable structure:

$$p(\text{Onset}) = \sigma, \quad p(\text{Rhyme}) = \sigma, \quad p(\text{Nucleus}) = \text{Rhyme}, \quad p(\text{Coda}) = \text{Rhyme}.$$

1. Draw the hierarchy.
2. Identify the sister relations.
3. Calculate $p(p(\text{Nucleus}))$.
4. Calculate $p(p(\text{Coda}))$.
5. Which nodes are leaves under this analysis?
6. Which aspects of the analysis follow from p , and which require the linguistic labels?

Exercise 1.4.1.16 (*Conceptual synthesis*) Evaluate the following statement:

A tree is fundamentally a picture. The parent function merely records information already present in the picture.

In your answer, distinguish:

1. the admissible hierarchy;
2. a graphical notation for displaying it;
3. the constitutive parent mapping;
4. relations derived from that mapping;
5. functions obtained by iterating that mapping.

Why does the functional characterization reveal connectedness more directly than the picture alone?

1.4.2 Function composition

Exercise 1.4.2.1 (*Admissibility of composition*) Let

$$f : B \longrightarrow C, \quad g : A \longrightarrow B, \quad h : C \longrightarrow D, \quad k : D \longrightarrow A.$$

For each expression, determine whether the composition is licensed by the declared domains and codomains:

$$f \circ g, \quad g \circ f, \quad h \circ f, \quad h \circ f \circ g, \quad g \circ k, \quad k \circ h, \quad f \circ k.$$

For every licensed composition, state its domain and codomain. For every unlicensed composition, identify the mismatched intermediate sets.

Exercise 1.4.2.2 (*Reading the order*) Suppose

$$g : A \longrightarrow B \quad \text{and} \quad f : B \longrightarrow C.$$

Complete the following description of $f \circ g$:

Function _____ first maps a value from A to a value in B . Function _____ then maps that value to a value in C . Function _____ determines the final kind of result.

Why is the function performed first written on the right in the conventional notation?

Exercise 1.4.2.3 (*Calculating composition*) Let

$$f(x) = x^2, \quad g(x) = x + 1, \quad h(x) = 2x.$$

Calculate:

$$\begin{array}{ll} (f \circ g)(x), & (g \circ f)(x), \\ (f \circ h)(x), & (h \circ f)(x), \\ (g \circ h)(x), & (h \circ g)(x). \end{array}$$

Which pairs of functions commute? For every pair that does not commute, give a numerical input witnessing the difference.

Exercise 1.4.2.4 (*Composition as a new function*) Let

$$f : \mathbb{N} \longrightarrow \mathbb{N}, \quad f(n) = n^2,$$

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and

$$g : \mathbb{N} \longrightarrow \mathbb{N}, \quad g(n) = n + 1.$$

1. Give an explicit rule for $f \circ g$.
2. State its domain and codomain.
3. Give an explicit rule for $g \circ f$.
4. State its domain and codomain.
5. Are the two resulting functions identical merely because they have the same domain and codomain?

Exercise 1.4.2.5 (*Intermediate types*) Let

$$g : \mathbb{N} \longrightarrow \mathbb{Z}, \quad g(n) = -n,$$

and

$$f : \mathbb{Z} \longrightarrow \mathbb{N}, \quad f(z) = |z|.$$

1. Is $f \circ g$ defined?
2. Give its type and explicit rule.
3. Is $g \circ f$ defined?
4. Give its type and explicit rule.
5. Are either of the compositions identity functions?

Explain why the intermediate type must be checked even when the algebraic expressions appear easy to combine.

Exercise 1.4.2.6 (*Partial composition*) Let

$$g : \mathbb{Z} \rightarrow \mathbb{N}, \quad g(z) = \sqrt{z},$$

where g is defined only when z is a nonnegative perfect square, and let

$$f : \mathbb{N} \longrightarrow \mathbb{N}, \quad f(n) = n + 1.$$

1. What is the domain of g ?

2. Is $f \circ g$ a function?
3. Is it total on $\mathbb{Z}$?
4. Give an explicit rule for $f \circ g$.
5. What is the domain of the composition?

Explain how partiality is inherited through composition.

Exercise 1.4.2.7 (Associativity) Let

$$f(x) = x + 1, \quad g(x) = 2x, \quad h(x) = x^2.$$

Calculate:

$$((h \circ g) \circ f)(x) \quad \text{and} \quad (h \circ (g \circ f))(x).$$

Then:

1. evaluate both expressions at $x = 3$;
2. identify the order in which the three functions are performed;
3. explain what associativity permits us to change;
4. explain what associativity does not permit us to change.

Exercise 1.4.2.8 (Associativity is not commutativity) A student argues:

Since function composition is associative, the functions in a composition may be performed in any order.

Use two of the functions in Exercise 1.4.2.7 to construct a counterexample. Distinguish changing the grouping of operations from changing their order.

Exercise 1.4.2.9 (Composition with product values) Let

$$g : \mathbb{Z} \longrightarrow \mathbb{Z} \times \mathbb{Z}, \quad g(x) = (x, x + 1),$$

and

$$f : \mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad f(y, z) = yz.$$

1. Show that $f \circ g$ is defined.

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2. State the type of $f \circ g$.
3. Calculate $(f \circ g)(x)$.
4. Evaluate the composition at $x = 4$.
5. Why does g supply one value rather than two?

Exercise 1.4.2.10 (Endofunctions) For each function, determine whether it is an endofunction on the declared set:

$$\begin{aligned} f : \mathbb{N} &\longrightarrow \mathbb{N}, & f(n) &= n + 1, \\ g : \mathbb{N} &\longrightarrow \mathbb{Z}, & g(n) &= -n, \\ h : \mathbb{Z} &\longrightarrow \mathbb{Z}, & h(n) &= -n, \\ k : \{0, 1, 2\} &\longrightarrow \{0, 1, 2\}, & k(n) &= (n + 1) \bmod 3. \end{aligned}$$

1. Which functions may be composed with themselves?
2. Give explicit rules for every defined self-composition.
3. Which functions may be iterated arbitrarily many times without changing their declared type?

Exercise 1.4.2.11 (Fixpoints) Find every fixpoint of each function on its stated domain:

$$\begin{aligned} f : \mathbb{Z} &\longrightarrow \mathbb{Z}, & f(x) &= x, \\ g : \mathbb{Z} &\longrightarrow \mathbb{Z}, & g(x) &= -x, \\ h : \mathbb{R} &\longrightarrow \mathbb{R}, & h(x) &= x^2, \\ k : \mathbb{N} &\longrightarrow \mathbb{N}, & k(x) &= x + 1, \\ m : \{0, 1, 2\} &\longrightarrow \{0, 1, 2\}, & m(x) &= 0. \end{aligned}$$

Does every endofunction have a fixpoint? Can a function have more than one?

Exercise 1.4.2.12 (Higher-order fixpoints) Let $\mathcal{F}$ be a set of functions from $\mathbb{N}$ to $\mathbb{N}$. Consider the functionals:

$$I(F) = F$$

and

$$K(F) = Z,$$

where $Z(n) = 0$ for every $n \in \mathbb{N}$.

1. Which functions are fixpoints of I ?
2. Show that Z is a fixpoint of K .
3. Can any function other than Z be a fixpoint of K ?
4. What kind of object is the input to a functional?
5. What kind of object is its fixpoint?

Distinguish a fixpoint value from a fixpoint function.

Exercise 1.4.2.13 (*Parent function as an endofunction*) Let

$$N = \{a, b, c, d, e, f\}$$

and define:

$$p(a) = d, \quad p(b) = d, \quad p(c) = d, \quad p(d) = e, \quad p(f) = e, \quad p(e) = e.$$

1. Show that $p : N \rightarrow N$ is an endofunction.
2. Identify its fixpoint.
3. Calculate $p^2(x) = (p \circ p)(x)$ for every $x \in N$.
4. Calculate $p^3(x)$ for every $x \in N$.
5. For which values does $p^2(x) = p(x)$?
6. For which values does $p^3(x) = p^2(x)$?

What does stabilization under repeated composition reveal about the root?

Exercise 1.4.2.14 (*Cycles*) Consider each proposed endofunction on $N = \{a, b, c, r\}$:

$$p_1 = \{a \mapsto b, b \mapsto c, c \mapsto r, r \mapsto r\},$$

$$p_2 = \{a \mapsto b, b \mapsto a, c \mapsto r, r \mapsto r\},$$

$$p_3 = \{a \mapsto b, b \mapsto c, c \mapsto a, r \mapsto r\}.$$

1. Identify every fixpoint.
2. Identify every nontrivial cycle.

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3. Calculate several successive iterates beginning with a .
4. Which function can serve as the parent function of a connected rooted tree?
5. Which tree-admissibility property is violated by each rejected function?

Exercise 1.4.2.15 (*Finite convergence to the root*) Let $p : N \rightarrow N$ be a parent endofunction on a finite set of nodes. Assume:

1. p has exactly one fixpoint r ;
2. p has no cycles other than $p(r) = r$;
3. every node is connected to r by repeated application of p .

Explain why, for every $x \in N$, there must be some $n \in \mathbb{N}$ such that

$$p^n(x) = r.$$

Why is the finiteness of N relevant to the argument?

Exercise 1.4.2.16 (*Application and composition*) Let p be a parent endofunction and a a node. Distinguish:

$$p(a), \quad (p \circ p)(a), \quad (p \circ p \circ p)(a).$$

1. Which expressions contain function application?
2. Which contain function composition?
3. What hierarchical relation does each expression reveal?
4. Why does the innermost occurrence $p(a)$ make application appear indispensable?

State the question that remains to be answered in §1.4.5 of *Mathematics of Linguistic Reality*. Do not attempt its answer yet.

Exercise 1.4.2.17 (*Empirical associativity*) Evaluate the following inference:

Function composition is associative. Therefore every linguistic operation represented by a function must itself be associative.

Distinguish:

1. associativity of the composition operation;
2. the internal properties of the particular functions being composed;
3. admissibility of a chain of compositions;
4. empirical claims about linguistic objects represented by those functions.

Why does associativity of composition not settle the empirical question?

Exercise 1.4.2.18 (*Connectedness*) Explain the mathematical content of each statement:

1. g supplies the kind of value required by f .
2. The codomain of g matches the domain of f .
3. $f \circ g$ is admissible.
4. The two mappings are connected.

Are these four independent claims, or four perspectives on the same condition? Why is the shared intermediate set structurally significant?

Exercise 1.4.2.19 (*Conceptual synthesis*) Starting with a parent function p , reconstruct the following progression:

$$p(x) \longrightarrow p(p(x)) \longrightarrow (p \circ p)(x) \longrightarrow p^n(x) \longrightarrow \text{root}.$$

For each step, state:

1. the function being used;
2. its domain and codomain;
3. whether application, composition, or both are visible;
4. the hierarchical information obtained;
5. the role of the root fixpoint.

Conclude by explaining why composition is a candidate mathematical invariant of connectedness.

1.4.3 Diagrams of functions

Exercise 1.4.3.1 (*Three distinct notions*) Match each statement with *associativity*, *commutativity*, or *commuting*:

1. Changing the grouping of three operations does not change the result.
2. Exchanging the order of two arguments does not change the result.
3. Two paths with the same starting and ending points give the same result.
4. $(f \circ g) \circ h = f \circ (g \circ h)$.
5. $x \star y = y \star x$.
6. $f \circ g = h \circ k$.

Explain why the three notions address different kinds of comparison.

Exercise 1.4.3.2 (*Associative but not commutative*) Let x , y , and z be strings.

1. Show that

$$x(yz) = (xy)z.$$

2. Give strings x and y for which

$$xy \neq yx.$$

3. Does the failure of commutativity affect associativity?
4. What information is changed by reordering?
5. What information is unchanged by regrouping?

Exercise 1.4.3.3 (*Prefixing functions*) Let

$$f(w) = aw, \quad h(w) = bw, \quad k(w) = cw.$$

Calculate:

$$\begin{array}{ll} (h \circ f)(w), & (f \circ h)(w), \\ ((k \circ h) \circ f)(w), & (k \circ (h \circ f))(w). \end{array}$$

1. Do f and h commute?

2. Are f , h , and k associative under composition?
3. What changes when f and h are exchanged?
4. What changes when the three functions are regrouped?

Exercise 1.4.3.4 (*Functions that commute*) Let

$$f(w) = aw \quad \text{and} \quad g(w) = wb.$$

1. Calculate $(f \circ g)(w)$.
2. Calculate $(g \circ f)(w)$.
3. Do f and g commute?
4. Compare this result with two prefixing functions.
5. What structural property permits prefixing and suffixing to commute in this example?

Exercise 1.4.3.5 (*Same types do not guarantee commuting*) Let

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(x) = x + 1,$$

and

$$g : \mathbb{Z} \rightarrow \mathbb{Z}, \quad g(x) = 2x.$$

1. Show that both $f \circ g$ and $g \circ f$ are well-defined.
2. Calculate both compositions.
3. Do the functions commute?
4. Give a numerical witness.
5. Why is type compatibility necessary but insufficient for commuting?

Exercise 1.4.3.6 (*A commuting square*) Let

$$g : \mathbb{Z} \rightarrow \mathbb{Z}, \quad g(x) = x + 1,$$

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(x) = 2x,$$

$$k : \mathbb{Z} \rightarrow \mathbb{Z}, \quad k(x) = 2x,$$

$$h : \mathbb{Z} \rightarrow \mathbb{Z}, \quad h(x) = x + 2.$$

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1. Calculate $(f \circ g)(x)$.
2. Calculate $(h \circ k)(x)$.
3. Show that $f \circ g = h \circ k$.
4. Draw a square whose upper path is g followed by f and whose lower path is k followed by h .
5. Identify the common starting and ending sets.

Explain why the diagram commutes although the two paths contain different functions.

Exercise 1.4.3.7 (*Breaking a commuting square*) Modify exactly one function in Exercise 1.4.3.6 so that the diagram no longer commutes.

1. Give the modified function.
2. Calculate both paths.
3. Give an input witnessing the difference.
4. Do the two paths still have the same starting and ending sets?
5. Why is that fact insufficient for commutation?

Exercise 1.4.3.8 (*Repairing a diagram*) Let

$$g(x) = x + 1, \quad f(x) = 3x, \quad k(x) = 3x.$$

Find a function h such that:

$$f \circ g = h \circ k.$$

1. Calculate $f \circ g$.
2. Use the result to determine h .
3. Verify the equality for an arbitrary input.
4. Is your solution unique if all functions have domain and codomain $\mathbb{Z}$?

Exercise 1.4.3.9 (*A commuting triangle*) Let

$$f : A \rightarrow B, \quad g : B \rightarrow C, \quad h : A \rightarrow C.$$

1. What two paths lead from A to C ?
2. State the equation required for the triangle to commute.
3. Which path contains a composition?
4. Which path is direct?
5. Does commutation require f , g , and h to be the same function?

Exercise 1.4.3.10 (*Comparing paths*) Suppose a diagram contains:

$$f : A \rightarrow B, \quad g : B \rightarrow D, \quad h : A \rightarrow C, \quad k : C \rightarrow D, \quad m : D \rightarrow E.$$

1. Write the two paths from A to D .
2. State the equation required for those paths to agree.
3. Write the corresponding paths from A to E .
4. If the paths from A to D agree, show that the extended paths to E agree.
5. Which function is shared by the extended paths?

Exercise 1.4.3.11 (*Path typing*) Let

$$f : A \rightarrow B, \quad g : B \rightarrow C, \quad h : A \rightarrow D, \quad k : D \rightarrow E.$$

A student proposes comparing $g \circ f$ with $k \circ h$.

1. State the type of $g \circ f$.
2. State the type of $k \circ h$.
3. Under what condition can the two paths be compared for equality?
4. Why must commuting paths have both the same starting point and the same ending point?
5. Can two paths be individually well-formed but ineligible for comparison?

Exercise 1.4.3.12 (*Local and global commutation*) Consider a diagram with paths:

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

and

$$A \xrightarrow{k} C \xrightarrow{h} D.$$

Suppose $g \circ f = k$.

1. Show that the triangle from A to C commutes.
2. Show that the two paths from A to D also agree.
3. Does the converse necessarily hold if h is not one-to-one?
4. Construct an example in which

$$h \circ g \circ f = h \circ k$$

but

$$g \circ f \neq k.$$

What does this show about information lost along a path?

Exercise 1.4.3.13 (*Diagram versus drawing*) Two diagrams contain the same sets, functions, and compositional paths, but one is drawn as a square and the other as a diamond.

1. Are they necessarily different mathematical diagrams?
2. Which properties belong to the functions and paths?
3. Which properties belong only to the graphical arrangement?
4. Can changing the drawing make a noncommuting diagram commute?

Relate the distinction to invariant structure and notational variants.

Exercise 1.4.3.14 (*Linguistic paths*) Suppose a linguistic expression can be associated with an interpretation by two analytical paths:

$$\text{expression} \longrightarrow \text{syntactic structure} \longrightarrow \text{interpretation}$$

and

$$\text{expression} \longrightarrow \text{surface cues} \longrightarrow \text{interpretation}.$$

1. Represent the two paths as compositions of named functions.
2. What would it mean for the resulting diagram to commute?
3. Would commutation show that syntactic structure and surface cues are identical?
4. Would it show that the two models make the same prediction for every expression in the domain?
5. What empirical evidence might distinguish the models despite commutation on the observed data?

Exercise 1.4.3.15 (Model selection) Two models map the same observations to the same predictions through different intermediate representations.

1. Draw the situation as a commuting diagram.
2. What equality of composed functions expresses observational agreement?
3. Does observational agreement establish that the intermediate representations are isomorphic?
4. Could one path preserve distinctions that the other eliminates?
5. What additional criteria might be used in model selection?

Explain why commuting diagrams represent comparability rather than mere commutativity.

Exercise 1.4.3.16 (Conceptual synthesis) For each equation, state exactly what is being asserted:

$$\begin{aligned}
 x(yz) &= (xy)z, \\
 x \star y &= y \star x, \\
 (f \circ g) \circ h &= f \circ (g \circ h), \\
 f \circ g &= g \circ f, \\
 f \circ g &= h \circ k.
 \end{aligned}$$

Classify each as an instance of associativity, commutativity, functions commuting with one another, or paths in a diagram commuting. Explain why the last two may share an equation-like appearance without expressing the same conceptual comparison.

1.4.4 Lambda calculus

Exercise 1.4.4.1 (*Anatomy of a lambda term*) Consider:

$$\lambda x.x + 1.$$

Identify:

1. the lambda operator;
2. the placeholder bound by it;
3. the dot separating the placeholder from the body;
4. the body of the function;
5. the free occurrences of x ;
6. the bound occurrences of x .

Express in words what the term does when supplied with an argument.

Exercise 1.4.4.2 (*From named to nameless*) Rewrite each named function as a lambda term:

$$f(x) = x + 1,$$

$$g(y) = y^2,$$

$$h(z) = 2z - 3,$$

$$k(x, y) = x + y,$$

$$m(x, y) = x - y,$$

$$n(x, y, z) = x + y + z.$$

What information disappears when the function name is removed? What information remains explicit?

Exercise 1.4.4.3 (*From nameless to named*) Give an arbitrary name and an ordinary function equation for each term:

$$\lambda x.x^2 + 1,$$

$$\lambda y.3y,$$

$$\lambda x.\lambda y.xy,$$

$$\lambda x.\lambda y.x,$$

$$\lambda f.\lambda x.fx.$$

Does the choice of function name affect what the function does?

Exercise 1.4.4.4 (*Scope of the dot*) Lambda abstraction extends as far to the right as possible. Insert parentheses to make the structure explicit:

$\lambda x.x + 1,$
 $\lambda x.\lambda y.x + y,$
 $\lambda x.fx,$
 $\lambda f.\lambda x.f(fx),$
 $(\lambda x.x + 1)3,$
 $(\lambda x.\lambda y.x + y)3.$

Which terms have abstraction as their outermost structure, and which have application as their outermost structure?

Exercise 1.4.4.5 (*Application associates to the left*) By convention, a sequence of applications associates to the left:

$$MNP = (MN)P.$$

Insert parentheses into:

$fx y,$
 $MNPQ,$
 $(\lambda x.\lambda y.x + y)3\ 4,$
 $(\lambda f.\lambda x.fx)gh.$

Explain why

$$(\lambda x.\lambda y.x + y)3\ 4$$

does not mean

$$(\lambda x.\lambda y.x + y)(3\ 4).$$

Exercise 1.4.4.6 (*One argument at a time*) Consider:

$$\lambda x.\lambda y.x + y.$$

1. Supply the first argument 3.
2. What remains after the first beta-reduction?

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3. Is the result a number or a function?
4. Supply the second argument 4.
5. What expression remains?

Explain how a two-argument function is represented by a succession of one-argument functions.

Exercise 1.4.4.7 (*Order of abstraction*) Compare:

$$M = \lambda x. \lambda y. x - y$$

and

$$N = \lambda y. \lambda x. x - y.$$

1. Reduce $M3$.
2. Reduce $N3$.
3. Describe the partial function obtained in each case.
4. Reduce $M3\ 2$.
5. Reduce $N3\ 2$.
6. Do the two terms receive the supplied values in the same roles?

Exercise 1.4.4.8 (*Currying and types*) Let

$$f : A \times B \longrightarrow C.$$

1. What is the Curried type when an A -value is supplied first?
2. What is the Curried type when a B -value is supplied first?
3. In each case, what kind of object is returned after the first argument?
4. What kind of object is returned after the second?

Express the two Curried types using the exponential notation for function spaces.

Exercise 1.4.4.9 (*Permuting arguments*) Let

$$f(x, y, z) = x - y + z.$$

Write lambda terms that receive the arguments in the following orders:

$$x, y, z; \quad y, x, z; \quad z, y, x.$$

Supply the values 10, 3, and 2 to the appropriate placeholders in each term so that all three ultimately represent:

$$10 - 3 + 2.$$

Why must the order of supplied values track the order of abstraction?

Exercise 1.4.4.10 (*Recognizing lambda terms*) Determine which expressions are well-formed lambda terms:

$x,$
 $1,$
 $xy,$
 $\lambda x.x,$
 $\lambda x.xy,$
 $\lambda.x,$
 $\lambda x.,$
 $(\lambda x.x)y,$
 $\lambda x.\lambda y.xyz.$

For each well-formed expression, identify the clause in the definition of lambda terms that licenses its outermost structure.

Exercise 1.4.4.11 (*Finding redexes*) Identify every beta-redex in:

$(\lambda x.x + 1)3,$
 $\lambda x.x + 1,$
 $(\lambda x.\lambda y.x + y)3,$
 $(\lambda x.x + 1)((\lambda y.y^2)3),$
 $f((\lambda x.x)a),$
 $(\lambda f.fx)(\lambda y.y^2).$

Why is a lambda abstraction without a supplied argument not itself a redex?

Exercise 1.4.4.12 (*First beta-reductions*) Beta-reduce until no redex remains:

$(\lambda x.x)a,$
 $(\lambda x.x + 1)3,$
 $(\lambda x.x^2)4,$
 $(\lambda x.5)9,$
 $(\lambda x.xx)a,$
 $(\lambda x.\lambda y.x + y)3\ 4.$

Do not perform arithmetic. Distinguish beta-reduction from calculation.

Exercise 1.4.4.13 (*Unused placeholders*) Reduce:

$(\lambda x.5)a,$
 $(\lambda x.\lambda y.y)a,$
 $(\lambda x.\lambda y.x)a,$
 $(\lambda x.\lambda y.\lambda z.x) a\ b\ c.$

What happens when the bound placeholder has no occurrence in the body?

Exercise 1.4.4.14 (*Repeated occurrences*) Reduce:

$(\lambda x.x + x)3,$
 $(\lambda x.xx)M,$
 $(\lambda x.fxx)a,$
 $(\lambda x.\lambda y.xy x)a.$

What happens when a bound placeholder occurs more than once in the body?

Does beta-reduction replace only the first occurrence?

Exercise 1.4.4.15 (*Free and bound occurrences*) Identify every free and bound occurrence of a placeholder:

$\lambda x.x,$
 $\lambda x.xy,$
 $(\lambda x.x)y,$
 $\lambda x.\lambda y.xy,$
 $\lambda x.(\lambda y.xy)z,$
 $\lambda x.(\lambda x.x)x.$

Do not classify a letter globally: classify each occurrence according to its position and binder.

Exercise 1.4.4.16 (Shadowing) Consider:

$$\lambda x.(\lambda x.x^2)x.$$

1. Which x is bound by the outer lambda?
2. Which x occurrences are bound by the inner lambda?
3. Rewrite the term by renaming the inner placeholder.
4. Reduce:

$$(\lambda x.(\lambda x.x^2)x)3.$$

5. Why is the result not $\lambda x.(3^2)3$?

Exercise 1.4.4.17 (Alpha-conversion) Determine which pairs differ only by consistent renaming of bound placeholders:

$\lambda x.x$	and	$\lambda y.y,$
$\lambda x.xy$	and	$\lambda z.zy,$
$\lambda x.xy$	and	$\lambda y.yy,$
$\lambda x.\lambda y.xy$	and	$\lambda a.\lambda b.ab.$

Why may a free placeholder not be renamed as part of alpha-conversion?

Exercise 1.4.4.18 (Avoiding variable capture) Consider:

$$(\lambda x.\lambda y.xy)y.$$

Naive textual replacement would produce:

$$\lambda y.yy,$$

causing the free y in the argument to become bound.

1. Alpha-convert the bound y before beta-reduction.
2. Perform the reduction.

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3. Identify the occurrence that remains free.
4. Explain why the naive result changes the structure of the term.

Exercise 1.4.4.19 (*Structured substitution*) Reduce:

$$(\lambda x.(\lambda y.yx)z)a.$$

Proceed in the following order:

1. identify the outer redex;
2. identify the occurrences bound by its lambda;
3. substitute the argument only for those occurrences;
4. identify the remaining redex;
5. complete the reduction.

Why is beta-reduction more than replacing matching letters?

Exercise 1.4.4.20 (*Outermost and innermost reduction*) Consider:

$$(\lambda x.x + 1)((\lambda y.y^2)3).$$

1. Reduce it using outermost-leftmost reduction.
2. Reduce it using innermost-leftmost reduction.
3. Continue both derivations until no redex remains.
4. Compare the final terms.
5. At which stages do the two derivations differ?

Exercise 1.4.4.21 (*A discarded argument*) Consider:

$$(\lambda x.a)((\lambda y.yy)(\lambda y.yy)).$$

1. Reduce the outermost redex first.
2. What result is obtained?

3. Examine what happens if reduction begins with the argument.
4. Must the argument be reduced in order to obtain the result?

What does this example show about the difference between normal-order and applicative-order reduction?

Exercise 1.4.4.22 (*Functions as arguments*) Let:

$$F = \lambda y. y + 1.$$

Reduce:

$$\begin{aligned} &(\lambda f. f 3)F, \\ &(\lambda f. \lambda x. f x)F, \\ &(\lambda f. \lambda x. f(fx))F, \\ &(\lambda f. \lambda x. f(fx))F 3. \end{aligned}$$

At which points is a function used as an argument? At which points is it returned as a value?

Exercise 1.4.4.23 (*No arithmetic interpreter*) Beta-reduce:

$$\begin{aligned} &(\lambda x. x^2 + 1)3, \\ &(\lambda x. \# @ x)3, \\ &(\lambda x. \text{glorp}(x))a. \end{aligned}$$

For each result, distinguish:

1. what lambda calculus determines by substitution;
2. what requires an interpretation of constants or operations;
3. what requires an additional computational method.

Exercise 1.4.4.24 (*Beta-abstraction*) For each expression, introduce a lambda abstraction and an argument whose beta-reduction returns the original expression:

$$3 + 1, \quad f(a), \quad a + b, \quad g(h(x)).$$

More than one answer may be possible. Explain why beta-abstraction enlarges a term whereas beta-reduction simplifies it.

Exercise 1.4.4.25 (*Linguistic currying*) Suppose the interpretation of a transitive verb is represented schematically as:

$$\text{chase}(x, y),$$

where x is the chaser and y is the chased participant.

1. Curry the function with x supplied first.
2. Curry it with y supplied first.
3. Supply *dogs* first in the first representation.
4. Supply *cats* first in the second representation.
5. What partial function results in each case?
6. Does lambda calculus itself determine which argument order a language realizes?

Distinguish the admissible functional possibilities from their linguistic variants.

Exercise 1.4.4.26 (*Why namelessness matters*) Evaluate the claim:

Removing function names is merely an economical notational choice.

In your answer, consider:

1. reference to arbitrarily many functions;
2. explicit representation of argument order;
3. functions returned as values;
4. functions supplied as arguments;
5. the representation of every multiargument function as successive one-argument functions.

Exercise 1.4.4.27 (*Conceptual synthesis*) Explain the progression:

$$f(x, y) = b \longrightarrow \lambda x. \lambda y. b \longrightarrow (\lambda x. \lambda y. b)a \longrightarrow \lambda y. b[x/a].$$

For every stage, identify:

1. the function body;
2. the bound placeholders;
3. any free placeholders;
4. the outermost structure;
5. whether a function or a nonfunctional value is produced.

Conclude by explaining why lambda calculus is a theory of structured substitution rather than a theory of what particular functions mean.

1.4.5 Function composition and application: one connectedness

Exercise 1.4.5.1 Let

$$B = \lambda f. \lambda g. \lambda x. f(gx).$$

Beta-reduce BFG and show that the resulting lambda term represents $F \circ G$.

Exercise 1.4.5.2 Let

$$f = \lambda x. \lambda y. x + y \quad \text{and} \quad g = \lambda z. z^2.$$

1. Beta-reduce Bfg .
2. Apply the resulting function to 3.
3. Explain why one argument of f remains unsupplied.

Exercise 1.4.5.3 Let

$$f = \lambda y. y^2 \quad \text{and} \quad g = \lambda x. \lambda y. x - y.$$

Attempt to form the ordinary composition $f \circ g$.

1. Beta-reduce $\lambda x. f(gx)$ as far as possible.
2. What kind of object does gx denote?
3. Why does squaring that object not amount to squaring a numerical argument?

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4. What additional structure would be needed to select one of the arguments of g ?

Exercise 1.4.5.4 Let

$$g = \lambda x. \lambda y. x - y.$$

Raise the argument 2 to $\lambda p. p2$ and beta-reduce

$$\lambda z. (\lambda p. p2)(gz).$$

Explain why the result fixes the second argument of g , rather than its first argument.

Exercise 1.4.5.5 Use the method of Exercise 1.4.5.4 to fix the second argument of each function.

1. $\lambda x. \lambda y. x + y$, fixing y as 5
2. $\lambda x. \lambda y. xy$, fixing y as 3
3. $\lambda x. \lambda y. \lambda z. x + y + z$, fixing z as 4

Show the beta-reductions.

Exercise 1.4.5.6 Let

$$T = \lambda x_1. \lambda x_2. x_2 x_1.$$

Beta-reduce the following.

1. Ta
2. Taf
3. Tf
4. $(Tf)h$

Which reduction exhibits the function–argument orientation supplied by T ?

Exercise 1.4.5.7 Show that raising an object with T gives

$$Ta =_{\beta} \lambda q. qa.$$

Then show that

$$Tf =_{\beta} \lambda p. pf.$$

Explain why the same combinator can raise both an argument and a function, although the raised objects subsequently play different roles.

Exercise 1.4.5.8 Recall that

$$(F \circ G)x = F(Gx).$$

Expand the composition in

$$((Tf) \circ (Ta))T$$

and beta-reduce it step by step until fa is obtained. Identify:

1. the raised function,
2. the raised argument,
3. the point at which they are composed, and
4. the step at which T restores the function–argument orientation.

Exercise 1.4.5.9 Let

$$f = \lambda y. y^2 \quad \text{and} \quad a = 4.$$

Evaluate

$$((Tf) \circ (Ta))T$$

by beta-conversion. Why is the result the lambda-calculus expression 4^2 rather than the computed number 16?

Exercise 1.4.5.10 Parentheses determine whether a lambda term contains a redex. For each term, identify the scope of the lambda abstraction, state whether the term is a redex, and beta-reduce it if possible.

1. $(\lambda x. xa)f$
2. $\lambda x. xaf$
3. $(\lambda x. xaf)g$
4. $(\lambda x. xa)(\lambda y. y)$

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Explain why $\lambda x.xaf$ by itself cannot beta-reduce to fa .

Exercise 1.4.5.11 Consider the two representations

$$fa \quad \text{and} \quad ((Tf) \circ (Ta))T.$$

1. Show that they are beta-equivalent.
2. Does this mean that their written structures are identical?
3. Which representation takes application as immediately given?
4. Which representation makes composition and function–argument orientation explicit?

Exercise 1.4.5.12 A student proposes the following derivation:

$$(\lambda x.xaf) =_{\beta} fa.$$

Diagnose the error. Supply two corrected terms:

1. one in which ordinary beta-reduction produces fa , and
2. one in which composition of raised terms, mediated by T , produces fa .

Exercise 1.4.5.13 Explain why the identity

$$((Tf) \circ (Ta))T =_{\beta} fa$$

does not show that function–argument orientation is unnecessary. What would be lost if the occurrences of T were removed?

Exercise 1.4.5.14 Assess the following two claims separately.

1. Function composition can be represented by applications in lambda calculus.
2. Application is therefore constitutive of function composition.

Why does the first claim not by itself establish the second?

Exercise 1.4.5.15 Section 1.4.5 treats application as a derivative manifestation of connectedness. In a short paragraph, explain the respective contributions of:

1. raising,
2. function composition, and
3. the combinator T .

Your explanation should make clear why composition alone is insufficient when the function–argument orientation has not already been supplied.

1.5 Recursion: named and nameless

Exercise 1.5.1 Compute the following from the recursive definitions in (45). Show every expansion.

1. $4!$
2. $\text{fib}(3)$
3. $\text{fib}(4)$

Exercise 1.5.2 For each expression, state whether it is a recursive definition. If it is, identify the occurrence or occurrences responsible for the recursion.

1. $f(n) = n + 1$
2. $f(n) = f(n - 1) + 1$
3. $g(n) = g(n - 1) + g(n - 2)$
4. $h(n) = 1 \times 2 \times \cdots \times n$

Exercise 1.5.3 Compare the two definitions of factorial in (45a) and (46a).

1. Which definition is recursive?
2. Do the two definitions determine different values?
3. What does their equivalence show about the difference between a mathematical object and a description of that object?

Exercise 1.5.4 The Fibonacci function in (45b) refers to itself twice. Does this mean that it exhibits “more recursion” than the factorial function? Explain the difference between the number of recursive calls in a description and the kind of values determined by that description.

Exercise 1.5.5 Consider the following definition:

$$s(0) = 0, \quad s(n) = s(n-1) + 2.$$

1. Compute $s(1)$, $s(2)$, $s(3)$ and $s(4)$.
2. Give a nonrecursive definition of $s(n)$.
3. Does the nonrecursive definition change the function?

Exercise 1.5.6 Consider the sentence in (44):

I think you think she thinks I like candies.

Identify the clauses and the verb that selects each subordinate clause. Explain why the successive clauses instantiate dependencies of the same general kind without being identical copies of one another.

Exercise 1.5.7 Explain the structural difference between (47a) and (47b).

1. What would have to remain identical during the expansion in (47a)?
2. What new object is introduced in (47b)?
3. Why can the dependency in (47b) be represented by ordinary subcategorization?

Exercise 1.5.8 Construct another sentence containing at least three levels of clausal embedding. Draw only enough structure to show the dependency of each embedding predicate on its clausal argument. Is the resulting structure more like (47a) or (47b)? Justify your answer.

Exercise 1.5.9 Suppose that a language expresses a sequence of propositions as separate sentences rather than by clausal embedding.

1. Would this fact alone show that the language cannot express arbitrarily long sequences?

2. Would it show that its grammar contains no recursive descriptions?
3. Why must the two questions be distinguished?

Exercise 1.5.10 In (48), let

$$h = \lambda f. \lambda n. \text{if } n \text{ eq } 0 \text{ then } 1 \text{ else } n \times f(n - 1).$$

Beta-reduce $h(\text{fact})$. Why does the result support the equation

$$h(\text{fact}) = \text{fact}?$$

Exercise 1.5.11 In the definition of h above, distinguish the roles of f and n .

1. Which placeholder ranges over functions?
2. Which placeholder ranges over natural numbers?
3. Why is h called a higher-order function?

Exercise 1.5.12 A fixpoint of a function g is a value a such that $g(a) = a$. Find the fixpoints, if any, of the following functions over the indicated domains.

1. $g(x) = x + 0$ over $\mathbb{N}$
2. $g(x) = x + 1$ over $\mathbb{N}$
3. $g(x) = x^2$ over $\mathbb{N}$
4. $g(x) = 1 - x$ over $\mathbb{R}$

Exercise 1.5.13 Let Y satisfy the fixpoint equation

$$YH = H(YH).$$

Show the first three expansions of YH . Why does this equation make YH a fixpoint of H without giving the recursive object a name?

Exercise 1.5.14 Compare

$$YH = H(H(H(\dots)))$$

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with the sequence of clauses in (44). Why does the repeated appearance of *H* in the first expression not correspond to the successive occurrences of *think* in (44)? Consider both the identity of the functions and their arguments.

Exercise 1.5.15 Assess the following claim:

A language permits arbitrarily deep clausal embedding; therefore recursion must be a constitutive property of linguistic objects.

In your answer, distinguish among:

1. recursive values or dependencies,
2. recursive descriptions,
3. clausal embedding, and
4. fixpoint computation.

Exercise 1.5.16 Section 1.5 claims that named recursion is a form of descriptive compression. Give one argument for this claim using factorial, Fibonacci numbers or linguistic embedding. Then explain why descriptive convenience does not by itself establish mathematical necessity.

1.6 Probability

Exercise 1.6.1 For each of the following investigations, identify:

1. the experiment or process of observation,
 2. one possible outcome,
 3. the sample space, and
 4. one event.
-
1. Tossing a coin once
 2. Rolling a six-sided die once
 3. Recording whether a transitive clause places the object before or after the verb

- Recording the first word produced by each child in a study

Exercise 1.6.2 Let

$$T = \{t, ht, hht, hhht, \dots\}$$

be the sample space in (49), where every string ends with the first occurrence of t .

- Is hth an element of T ? Why?
- Is hhh an element of T ? Why?
- Describe the general form of every member of T .
- Explain why T is an abstraction rather than a collection of uninterpreted observations.

Exercise 1.6.3 Using the sample space T in (49), define the following events by listing their members or by giving a set description.

- The first tail occurs in at most four tosses.
- The first tail occurs on an even-numbered toss.
- The first tail occurs after at least three heads.
- The first tail occurs on the first toss.

Exercise 1.6.4 Let

$$A = \{t, ht, hht, hhht\}$$

and let

$$B = \{ht, hhht, hhhhht, \dots\}$$

be events in T . Find:

- $A \cap B$,
- $A \cup B$,
- $A \setminus B$, and
- $B \setminus A$.

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Is either event a subset of the other?

Exercise 1.6.5 Assume that the coin in (49) is fair and that successive tosses are independent.

1. Find $P(t)$, $P(ht)$, $P(hht)$ and $P(hhht)$.
2. State a general formula for $P(h^n t)$, where $n \geq 0$.
3. Find the probability that the first tail occurs in at most four tosses.
4. Find the probability that the first tail occurs on an even-numbered toss.

Exercise 1.6.6 Let A and B be the events in Exercise 1.6.4. Using the probabilities obtained in Exercise 1.6.5, calculate

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Verify the result by calculating the probability of the complement of $A \cup B$.

Exercise 1.6.7 Distinguish *mutual exclusivity* from *probabilistic independence*.

1. What equation characterizes mutually exclusive events?
2. What equation characterizes independent events?
3. Can two events with positive probability be both mutually exclusive and independent? Prove your answer.
4. Give an example of two events that are independent but not mutually exclusive.

Exercise 1.6.8 A corpus contains 200 clauses, classified as follows:

	active	passive
transitive	120	30
intransitive	45	5

Assuming that a clause is selected uniformly at random, calculate:

1. the joint probability that it is transitive and active,

2. the joint probability that it is transitive and passive,
3. the marginal probability that it is transitive,
4. the marginal probability that it is passive,
5. the conditional probability that it is passive given that it is transitive, and
6. the conditional probability that it is transitive given that it is passive.

Exercise 1.6.9 Use the table in Exercise 1.6.8 to determine whether transitivity and voice are probabilistically independent. Compare

$$P(\text{transitive} \cap \text{passive})$$

with

$$P(\text{transitive})P(\text{passive}).$$

What would equality between these quantities mean?

Exercise 1.6.10 The following is a joint probability distribution:

	$Y = y_1$	$Y = y_2$	$P(X)$
$X = x_1$	0.30	0.20	
$X = x_2$	0.10	0.40	
$P(Y)$			

1. Complete the marginal probabilities in the final row and column.
2. Verify that the joint probabilities sum to 1.
3. Calculate $P(Y = y_1 \mid X = x_1)$.
4. Calculate $P(X = x_2 \mid Y = y_2)$.
5. Are X and Y independent?

Exercise 1.6.11 Suppose that

$$P(A) = 0.4, \quad P(B) = 0.5, \quad P(A \cap B) = 0.2.$$

1. Calculate $P(A \cup B)$.

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2. Calculate $P(B \mid A)$.
3. Calculate $P(A \mid B)$.
4. Are A and B independent?

Exercise 1.6.12 For each statement, decide whether it concerns a joint, marginal or conditional probability.

1. The probability that an observed clause is both interrogative and verb-final
2. The probability that an observed clause is interrogative, irrespective of word order
3. The probability that an observed clause is verb-final, given that it is interrogative
4. The distribution of meanings obtained after summing over all syntactic forms
5. The probability assigned to a meaning after a particular syntactic form has been observed

Justify each answer.

Exercise 1.6.13 Let A represent a syntactic analysis and B a semantic analysis. Explain what must be assumed in order for $P(A, B)$ to be a genuine joint distribution. Why is merely having separate estimates for $P(A)$ and $P(B)$ insufficient?

Exercise 1.6.14 Suppose that a model estimates

$$P(\text{meaning} \mid \text{expression}).$$

1. What is treated as given?
2. What is being estimated?
3. Does this conditional distribution by itself specify a joint space of admissible expression–meaning correspondences?
4. What additional theoretical knowledge would be needed to interpret it as arising from such a joint space?

Exercise 1.6.15 Assess the following claim:

If a model assigns a high conditional probability to a meaning given an expression, it has thereby shown that the expression and the meaning form an admissible linguistic correspondence.

Distinguish probabilistic prediction from a theory of admissibility in your answer.

Exercise 1.6.16 Section 1.6 warns that similar-looking notation may conceal different assumptions. Explain the differences among

$$P(A, B), \quad P(A), \quad P(B \mid A).$$

For each expression, state what knowledge about the outcome space is presupposed and what question the expression answers.

1.7 Logic

Exercise 1.7.1 Using 1 for true and 0 for false, complete the truth table for implication.

A	B	$A \rightarrow B$
1	1	
1	0	
0	1	
0	0	

Which assignment is the only one that makes $A \rightarrow B$ false?

Exercise 1.7.2 Evaluate the truth of each implication under the interpretation given.

1. A is true and B is true.
2. A is true and B is false.
3. A is false and B is true.
4. Both A and B are false.

1 Basics revisited

Why should $A \rightarrow B$ not be understood as asserting that A actually causes B ?

Exercise 1.7.3 Let

$A = \text{"The expression is a sentence"}$

and

$B = \text{"The expression contains a verb."}$

Assume that $A \rightarrow B$ is true. Determine what can be validly concluded in each case.

1. A is true.
2. B is false.
3. B is true.
4. A is false.

Distinguish valid conclusions from conclusions that do not follow.

Exercise 1.7.4 Complete the following instances of modus ponens and modus tollens.

1. $A, A \rightarrow B \vdash \underline{\hspace{2cm}}$
2. $\neg B, A \rightarrow B \vdash \underline{\hspace{2cm}}$

Explain why concluding A from B would not be an instance of either rule.

Exercise 1.7.5 Identify the error in each invalid inference.

1.

$$A \rightarrow B, \quad B \quad \vdash \quad A$$

2.

$$A \rightarrow B, \quad \neg A \quad \vdash \quad \neg B$$

For each inference, give truth values for A and B under which the premises are true but the conclusion is false.

Exercise 1.7.6 Construct truth tables for:

1. $\neg A$,
2. $A \wedge B$,
3. $A \vee B$, and
4. $A \rightarrow B$.

Use the tables to show that

$$A \rightarrow B$$

and

$$\neg A \vee B$$

have the same truth conditions.

Exercise 1.7.7 Determine whether each pair of formulas is logically equivalent. Justify your answer with a truth table.

1. $\neg(A \wedge B)$ and $\neg A \vee \neg B$
2. $\neg(A \vee B)$ and $\neg A \wedge \neg B$
3. $A \rightarrow B$ and $\neg B \rightarrow \neg A$
4. $A \rightarrow B$ and $B \rightarrow A$

Exercise 1.7.8 Frege's judgment sign distinguishes judgeable content from judging that content to be true. Explain this distinction by comparing:

1. the expression of a proposition,
2. the truth value assigned to that proposition, and
3. the assertion or judgment of that proposition.

Why are these not three names for the same thing?

Exercise 1.7.9 Consider the following statements from (56).

1. The morning star is the evening star.
2. The morning star is the morning star.

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Assuming that *the morning star* and *the evening star* refer to the same object, explain how the two statements can have the same truth value but differ in cognitive significance.

Exercise 1.7.10 Classify each expression as a proposition, a predicate or a quantified formula.

1. *rain*
2. *child*(x)
3. *read*(x, y)
4. $\forall x \text{ sleep}(x)$
5. $\exists y \text{ book}(y)$

For every predicate, state its arity.

Exercise 1.7.11 Translate the following sentences into first-order logic. State the domain of quantification and the interpretation of every predicate and constant you use.

1. Every child slept.
2. A child slept.
3. Every professor read *Ulysses*.
4. A professor read every book.
5. No child read *Ulysses*.

Use parentheses to make the scope of every quantifier and connective explicit.

Exercise 1.7.12 Explain why

$$\forall x(\text{child}(x) \rightarrow \text{sleep}(x))$$

does not assert that anything is a child. Construct a model containing no children and determine the truth value of the formula in that model.

Exercise 1.7.13 Let the domain be

$$U = \{\text{Ada}, \text{Bora}, \text{Cem}\}.$$

Suppose that Ada and Bora are professors, Bora and Cem are students, and only Ada slept. Determine the truth values of:

1. $\forall x(\text{professor}(x) \rightarrow \text{sleep}(x))$,
2. $\exists x(\text{professor}(x) \wedge \text{sleep}(x))$,
3. $\forall x(\text{student}(x) \rightarrow \neg \text{sleep}(x))$, and
4. $\exists x(\text{student}(x) \wedge \text{professor}(x))$.

Exercise 1.7.14 The sentence

Every professor read a book

has at least two scope readings. Represent:

1. the reading in which professors may have read different books, and
2. the reading in which one book was read by every professor.

Explain how the order and scope of $\forall$ and $\exists$ distinguish the readings.

Exercise 1.7.15 Compare:

$$\forall x \exists y (\text{professor}(x) \rightarrow (\text{book}(y) \wedge \text{read}(x, y)))$$

and

$$\exists y (\text{book}(y) \wedge \forall x (\text{professor}(x) \rightarrow \text{read}(x, y))).$$

1. Which formula permits a different book for each professor?
2. Which formula requires a single book shared by all professors?
3. Does either formula entail the other? Discuss the role of an empty domain of professors.

Exercise 1.7.16 Why can the meaning of

Every child slept

not be represented by a single unanalyzed propositional symbol without losing information? Explain what first-order predicates and quantification make explicit.

Exercise 1.7.17 Compare the following formulas:

$$\forall x(\text{child}(x) \rightarrow \text{sleep}(x))$$

and

$$\forall P(P(\text{you}) \rightarrow P(I)).$$

1. What kind of object does x range over in the first formula?
2. What kind of object does P range over in the second?
3. Why is the first formula first-order but the second higher-order?

Exercise 1.7.18 Consider the sentence:

I do everything that you do.

Discuss what is required to interpret the variable P in

$$\forall P(P(\text{you}) \rightarrow P(I)).$$

Would P range over actions, properties, predicates or linguistic descriptions? Explain how the choice affects the proposed representation.

Exercise 1.7.19 Section 1.7 distinguishes logical admissibility from the number of readings assigned to an expression. Explain why the two scope readings of

Every professor read a book

do not constitute two different grammaticalities. What varies, and what remains invariant?

Exercise 1.7.20 Assess the following claim:

A more expressive logical notation necessarily provides a better linguistic theory.

In your answer, distinguish:

1. expressive power,
2. logical admissibility,
3. computability, and
4. empirical adequacy for linguistic data.

1.8 Mathematics and computation

Exercise 1.8.1 Section 1.8 distinguishes mathematics from computation in terms of an encoding–decoding relation. Explain:

1. why a mathematical abstraction need not be encoded in a physical medium,
2. why a computation must be physically realized, and
3. why the result of a computation must be decoded.

Exercise 1.8.2 Consider the mathematical operation

$$f(x) = x + 1.$$

Describe three distinct things:

1. the abstract function f ,
2. one possible encoding of f and its argument in a computer, and
3. the physical process by which the encoded operation is executed.

Which of these belongs to the mathematical invariant, and which belong to a computational realization?

Exercise 1.8.3 Suppose that the letter A is represented by a sequence of bits. Describe the mappings involved in:

1. encoding the abstract letter as bits,
2. physically realizing those bits in a digital computer, and
3. decoding the resulting physical state as the letter A.

Why is none of these mappings supplied by the physical medium itself?

Exercise 1.8.4 The same abstract function can be implemented:

1. in a program running on an electronic computer,
2. by moving tokens on a board,
3. by manipulating marks on paper, or

4. by a mechanical device.

What must remain invariant across these implementations? What can vary?

Exercise 1.8.5 Explain Knuth’s requirement that an idea placed in a computer should not require an external interpreter during its execution. Why does this require the relevant operations and distinctions to be made explicit before execution begins?

Exercise 1.8.6 A program produces an output that a human must reinterpret at every intermediate step before the program can continue. Does this satisfy the “no homunculi” requirement discussed in Section 1.8? Explain your answer.

Exercise 1.8.7 Consider the claim:

A computer understands the symbols that it manipulates because those symbols have meanings for its programmer.

Assess the claim by distinguishing:

1. the physical states of the computer,
2. the encoded symbols, and
3. the human interpretation used to decode them.

Exercise 1.8.8 Horsman and colleagues characterize computation in terms of the abstract evolution of a physical system. Explain the roles of:

1. an abstract description,
2. encoding,
3. physical evolution, and
4. decoding.

Why is a merely changing physical system not necessarily a computer?

Exercise 1.8.9 A stone rolls down a hill and eventually stops. Its physical state has changed over time.

1. Is this change by itself a computation?

2. What encoding, abstract evolution and decoding would have to be specified before the process could be treated as a computation?
3. Who supplies these specifications?

Exercise 1.8.10 A calculator displays 4 after being given the input $2+2$. Identify:

1. the abstract mathematical relation,
2. the encoding of the input,
3. the physical execution,
4. the encoded output, and
5. the decoding of that output.

Why should the abstract equality $2 + 2 = 4$ not be identified with this particular computation?

Exercise 1.8.11 Section 1.8 argues that analog computers also require encoding and decoding. Suppose that the voltage in a device represents the value of a linguistic variable.

1. What constitutes the encoding?
2. What constitutes the physical evolution?
3. What constitutes the decoding?
4. How can accumulated physical error affect the result without changing the abstract relation being modeled?

Exercise 1.8.12 Classify each item primarily as an abstract mathematical object, a computational representation, or a physical realization.

1. A function from strings to syntactic structures
2. A sequence of bits representing a string
3. Electrical states in a processor
4. A finite-state automaton defined as a tuple

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5. A program simulating that automaton
6. The program's changing memory states
7. A human interpretation of the program's output as a parse

Explain any case in which more than one classification is possible.

Exercise 1.8.13 Suppose that two programs implement the same finite-state automaton, but one uses a transition table and the other uses a sequence of conditional instructions.

1. What mathematical structure do the programs share?
2. Which properties belong to their distinct encodings?
3. Must the two programs take the same amount of time or memory?
4. Could both nevertheless compute the same function?

Exercise 1.8.14 A linguist proposes a mathematical grammar, and a programmer implements it as a parser.

1. Which questions concern the admissibility conditions imposed by the grammar?
2. Which questions concern the encoding of the grammar?
3. Which questions concern the execution of the parser?
4. Which questions concern decoding its output as a linguistic analysis?

Why does successful execution not by itself establish that the grammar is linguistically adequate?

Exercise 1.8.15 Explain the distinction between:

1. proving that an abstract relation has a property, and
2. running a program that produces instances compatible with that property.

Can repeated successful computations replace the mathematical proof? Can a mathematical proof determine every practical property of an implementation?

Exercise 1.8.16 Section 1.8 relates mathematics and computation to the distinction between invariants and variants. Assess the following claim:

Mathematics studies the invariant and computation merely studies its variants.

Why is the relation more subtle than a complete separation? Discuss the shared problem of connectedness and the special burden imposed on computation by encoding and decoding.

1.9 Linguistic structures

Exercise 1.9.1 For each of the following, state whether the indicated division is primarily phonological, morphological, syntactic, semantic or information-structural.

1. Dividing a word into syllables
2. Dividing a word into stem and affixes
3. Identifying the subject and object of a clause
4. Determining who performed an action
5. Determining which constituent is presented as the topic

Why should these analyses not be assumed to identify the same structure?

Exercise 1.9.2 Consider the Turkish expression:

ev-ler-e house-PL-DAT ‘to the houses’.

Its syllabification is:

ev.le.re.

1. Identify the morpheme boundaries.
2. Identify the syllable boundaries.
3. Which sequence belongs partly to more than one morpheme?
4. Which syllable is not itself a morpheme?
5. What does the example show about the relation between phonological and morphological structure?

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Exercise 1.9.3 Find an example from a language you know in which a morpheme boundary and a syllable boundary do not coincide. Give:

1. the expression,
2. its morphological segmentation,
3. its syllabification, and
4. a brief explanation of the mismatch.

Exercise 1.9.4 Explain why the mapping from morphemes to phonemes is not generally one-to-one.

1. Can one morpheme contain several phonemes?
2. Can the same phoneme occur in several morphemes?
3. Can one morpheme have phonologically different realizations?

Why might a relation be more appropriate than a simple function for connecting morphological and phonological analyses?

Exercise 1.9.5 Compare the Turkish words:

<i>ev-ler</i>	house-PL	'houses',
<i>kiler</i>		'cellar'.

1. Why is *ler* a morpheme in *evler* but not in *kiler*?
2. Why is the plural of *kiler* formed as *kiler-ler*?
3. What knowledge must a learner possess to distinguish a recurring sound sequence from a morpheme?
4. What does the example show about the distinction between morphology and the lexicon?

Exercise 1.9.6 Find two words in a language you know which contain the same sequence of sounds or letters, but in which that sequence is a morpheme in only one word. Explain what evidence distinguishes the morphological analysis from mere segmentation.

Exercise 1.9.7 In the English form *was*, tense and person-number agreement are expressed together.

1. What tense information does *was* express?
2. With which persons and numbers can it occur?
3. Can a separate part of *was* be identified as the tense marker and another part as the agreement marker?
4. Why is this an example of portmanteau realization?

Exercise 1.9.8 Compare the portmanteau realization in *was* with the contractions in:

I wouldn't try and *I wanna try*.

Why does phonological fusion not necessarily imply that the component functions are morphologically inseparable? Identify the contributions of the components wherever possible.

Exercise 1.9.9 Find a portmanteau form in a language you know. State:

1. the form,
2. the grammatical functions expressed by it, and
3. why those functions cannot be assigned to separately identifiable parts of the form.

Exercise 1.9.10 Compare the following sentences:

1. I persuaded her to study.
2. I promised her to study.
3. I expected her to study.
4. I wanted her to study.

For each sentence, determine:

1. who is understood as studying,
2. whether *her* is an argument of the matrix verb, and
3. whether the understood subject of *study* is controlled by the matrix subject or by another expression.

What does the comparison show about the relation between syntactic form and semantic dependency?

Exercise 1.9.11 The expressions in Exercise 1.9.10 have broadly similar surface syntax and can participate in some of the same syntactic constructions. Why does this not entail that they have the same semantic structure? Explain what information must come from the lexical properties of the matrix verb.

Exercise 1.9.12 Construct two sentences with similar syntactic structures but different semantic dependencies. Identify:

1. the shared syntactic pattern,
2. the relevant lexical difference, and
3. the resulting semantic difference.

Exercise 1.9.13 Compare:

I like this book

and

This book I like.

1. Do the sentences differ in their basic truth conditions?
2. Do the expressions refer to different individuals or objects?
3. Which constituent is presented as topic or contrastive material in each sentence?
4. What does the pair show about the distinction between semantics and information structure?

Exercise 1.9.14 Construct two expressions which have approximately the same truth conditions but package their information differently. Describe the topic, focus and background of each expression.

Exercise 1.9.15 Consider the sound /z/ in:

*The boysz like the video,
She toysz with the idea.*

1. What is the syntactic category of the word containing /z/ in each sentence?
2. What morphological function does /z/ express in each case?
3. Is its phonological realization sufficient to determine its morphological function?
4. What does the comparison show about reducing morphology or syntax to phonological form?

Exercise 1.9.16 For each proposed reduction, give one reason from Section 1.9 why it fails.

1. Morphological structure is reducible to syllable structure.
2. Lexical structure is reducible to recurring sound sequences.
3. Morphological structure is reducible to syntactic structure.
4. Semantic structure is reducible to syntactic structure.
5. Information structure is reducible to truth-conditional semantics.

Exercise 1.9.17 Suppose that the same finite-state method is used to model both phonological and morphological patterns.

1. Does the use of the same method make phonology and morphology the same linguistic level?
2. What would differ between the two models?
3. What relations would be required to connect their analyses?

Distinguish the mathematical method from the object to which it is applied.

Exercise 1.9.18 Sign languages and spoken languages differ in their physical realization. Explain why this does not necessarily require different invariant conditions on linguistic admissibility. What belongs to the invariant, and what may belong to the modality-specific variants?

Exercise 1.9.19 A theory maps every phonological representation to exactly one morphological representation. Give two kinds of linguistic evidence that

could show that this mapping is inadequate. Would replacing the function with a relation solve every theoretical problem? Explain.

Exercise 1.9.20 Section 1.9 concludes that linguistic levels require their own admissibility conditions. Choose two linguistic levels and answer:

1. What kinds of objects belong to each level?
2. What counts as a well-formed object at each level?
3. What evidence shows that neither level is reducible to the other?
4. How might the choice of mathematical method be affected by the distinction?

1.10 Linguistics and philosophy

Exercise 1.10.1 Distinguish the following relations between conditions A and B .

1. A is necessary for B .
2. A is sufficient for B .
3. A is both necessary and sufficient for B .
4. A is neither necessary nor sufficient for B .

Express each relation using implication wherever possible.

Exercise 1.10.2 For each statement, identify the condition presented as necessary and the condition presented as sufficient.

1. Being a square is sufficient for being a rectangle.
2. Being a rectangle is necessary for being a square.
3. Having an inflammable material nearby is not sufficient for a fire.
4. Producing a meaningful expression may require more than satisfying the grammar.

Exercise 1.10.3 Consider Mackie's fire example. Let:

- S = a short circuit occurs,
- F = inflammable material is present,
- R = an effective sprinkler operates,
- H = the house catches fire.

Suppose that

$$S \wedge F \wedge \neg R$$

is sufficient for H .

1. Why is S alone insufficient for H ?
2. Why is S necessary within this particular sufficient condition?
3. Why is the whole sufficient condition unnecessary for H ?
4. What alternative sufficient condition might also produce H ?

Exercise 1.10.4 Expand the acronym INUS. For the short circuit in Exercise 1.10.3, identify:

1. the insufficient part,
2. the complex condition within which it is necessary,
3. why that complex condition is unnecessary, and
4. why that complex condition is sufficient.

Exercise 1.10.5 Let an outcome O arise under either of the following conditions:

$$(A \wedge B \wedge C) \vee (D \wedge E).$$

Assume that each conjunction is sufficient for O and that neither is necessary.

1. Under what circumstances is A an INUS condition for O ?
2. Why is A not sufficient for O ?
3. Why is A not necessary for O ?

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4. In what restricted sense is A nevertheless necessary?

Exercise 1.10.6 Construct an everyday example of an INUS condition other than fire. Specify:

1. the outcome to be explained,
2. one sufficient complex of conditions,
3. the factor selected as a cause,
4. why that factor is insufficient by itself,
5. why it is necessary within the complex, and
6. one alternative sufficient complex.

Exercise 1.10.7 Counterfactual reasoning asks what would have happened if some condition had not occurred. Apply this test to the sufficient complex

$$S \wedge F \wedge \neg R$$

in the fire example. Consider separately what would happen if:

1. S were absent,
2. F were absent, and
3. R were present.

What does this reveal about the role of each condition within the complex?

Exercise 1.10.8 Explain why observing that A is regularly followed by B is not by itself sufficient to establish that A causes B . What additional work is done by:

1. comparison with alternative conditions,
2. counterfactual reasoning, and
3. discussion among parties interested in the explanation?

Exercise 1.10.9 Section 1.10 treats causal explanation partly as a process of persuasion. Does this mean that a condition becomes a cause merely because everyone agrees to call it one? Explain the difference between:

1. arbitrary agreement,
2. publicly assessable evidence, and
3. agreement about the role of a factor in a sufficient complex.

Exercise 1.10.10 Consider the proposal in (71):

Grammar is an INUS cause of meaningfulness.

Identify the proposed analogues of:

1. the outcome,
2. the insufficient causal factor,
3. the sufficient complex containing that factor, and
4. alternative sufficient complexes.

Which parts of the analogy are explicitly supplied in Section 1.10, and which require further linguistic theory?

Exercise 1.10.11 Give an example of:

1. a grammatical expression that is difficult to interpret,
2. an ungrammatical expression whose intended meaning can nevertheless be recovered, and
3. an expression whose interpretation depends essentially on context.

Explain how the examples support the claim that grammar contributes to meaningfulness without being sufficient for it.

Exercise 1.10.12 Suppose that meaningfulness depends on the conjunction

$$G \wedge L \wedge C \wedge K,$$

where G is grammatical structure, L lexical knowledge, C context and K background knowledge.

1. Is G sufficient for meaningfulness?

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2. Under what assumptions is *G* necessary within this complex?
3. Give an alternative complex which might permit meaningful communication despite defective grammatical structure.
4. Explain how *G* could consequently be an INUS condition.

Exercise 1.10.13 Native speakers often agree that an expression is grammatically defective while still understanding what its producer intended.

1. Does recoverability of the intended message establish grammaticality?
2. Does grammaticality guarantee a determinate interpretation?
3. What does this show about the relation between grammaticality and meaningfulness?

Exercise 1.10.14 Section 1.10 describes grammar as an abstract object. Explain why “abstract” need not mean:

1. imaginary,
2. merely verbal,
3. a name for a collection of utterances, or
4. a physical object located in the speaker.

In what sense can an abstract object have causal relevance under the proposal in (71)?

Exercise 1.10.15 Compare the following broad positions.

1. Linguistic structures are merely names assigned to collections of observable forms.
2. Linguistic structures are mental objects.
3. Linguistic admissibility belongs to mathematical reality and has mental and physical consequences in its variants.

What kinds of evidence or argument would each position require?

Exercise 1.10.16 Why does realism about grammar not require every grammatical theory to be correct? Distinguish:

1. claiming that grammatical conditions are real,
2. proposing a particular description of those conditions, and
3. empirically assessing the proposed description.

Exercise 1.10.17 Explain why conventionalism need not amount to antirealism. How can a scientific model be:

1. partly constrained by the object being studied,
2. partly dependent on human choices of representation, and
3. publicly assessed as an explanation?

Exercise 1.10.18 Section 1.10 associates different orientations toward knowledge with Socrates, Aristotle, Plato and Bhartṛhari. Choose two of the following activities and explain how they differ:

1. examining and convincing,
2. organizing and asserting,
3. inviting and building, and
4. cooperating and sharing.

How might an INUS explanation combine rather than choose among these activities?

Exercise 1.10.19 Assess the following claim:

Because an INUS explanation permits several sufficient causal complexes, it is simply a form of theoretical pluralism.

Distinguish the existence of alternative sufficient conditions from the claim that all explanations are equally adequate.

Exercise 1.10.20 Develop an INUS analysis of one linguistic phenomenon, such as:

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1. successful reference,
2. interpretation of an ambiguous sentence,
3. acquisition of a grammatical construction, or
4. acceptance of a novel expression.

Identify the relevant conditions, organize them into at least two possible sufficient complexes, and explain why no single condition constitutes the complete explanation.

1.11 Typology and mathematics

Exercise 1.11.1 In your own words, explain the informal characterization of restrictive relative clauses in (73). Identify:

1. the restricting clause,
2. the restricted expression, and
3. the element shared between them

in:

the book that I read yesterday.

Exercise 1.11.2 For each expression, identify the head noun, the relative clause and the grammatical role of the shared element inside the relative clause.

1. the student who solved the problem
2. the problem that the student solved
3. the book that I gave to the student
4. the city in which the student lives

Exercise 1.11.3 Construct restrictive relative clauses in which the shared element corresponds to:

1. the subject of the relative clause,

2. its direct object,
3. its indirect object, and
4. the complement of a preposition.

Underline the head and mark the position associated with it inside the relative clause.

Exercise 1.11.4 Section 1.11 distinguishes several strategies for realizing relative clauses. Match each description with the appropriate strategy.

1. An argument position is left unexpressed in the relative clause.
2. A pronoun occupies the position associated with the head.
3. Verbal morphology indexes the relevant argument.
4. Both extraction and resumption are available.

Give one language mentioned in Section 1.11 for each strategy.

Exercise 1.11.5 Compare the English and Turkish examples in (74).

1. Where does the relative clause occur with respect to the head noun in each language?
2. How is relativization overtly marked in each language?
3. What element is shared between the relative clause and the head?
4. Which properties vary, and which dependency remains invariant?

Exercise 1.11.6 The Navajo examples in (75) use verbal indexing rather than an extracted gap alone.

1. What information about the arguments is expressed on the verbs?
2. Why is verbal indexing by itself not sufficient to identify a relative clause in Navajo?
3. What additional contribution is made by the relative marker?
4. How does this strategy realize the same sharing relation as an extracting relative clause?

Exercise 1.11.7 Consider the Irish expression in (76):

an fear [a labhair mé leis] ‘the man that I spoke with’.

1. Identify the head noun.
2. Identify the relative marker.
3. Identify the resumptive element.
4. What grammatical relation connects the resumptive element to the head?
5. How would a corresponding extracting construction differ?

Exercise 1.11.8 Explain why extraction, resumption and verbal indexing should not be treated as three distinct definitions of a restrictive relative clause. What common dependency must a mathematical invariant admit?

Exercise 1.11.9 Let

$$P = \lambda x. read'(I, x) \quad \text{and} \quad Q = \lambda x. book'(x).$$

Using (77), calculate

$$rel(P, Q) = \lambda x. P(x) \wedge Q(x).$$

Give an informal English description of the resulting property.

Exercise 1.11.10 Give lambda terms for the restricting property P and the nominal property Q in each expression. Then use (77) to state the resulting property.

1. the student who slept
2. the book that Mary bought
3. the city in which Ali lives

Exercise 1.11.11 In

$$rel(P, Q) = \lambda x. P(x) \wedge Q(x),$$

explain the contribution of:

1. P ,

2. Q ,
3. the shared variable x , and
4. conjunction.

Which parts represent descriptive content, and which part represents the sharing dependency?

Exercise 1.11.12 Suppose that two distinct variables are used:

$$\lambda x.\lambda y.P(x) \wedge Q(y).$$

Why does this expression fail to capture the ordinary restrictive relative-clause interpretation? What additional condition would be needed to connect x and y ?

Exercise 1.11.13 A hypothetical construction simultaneously shares two independent arguments between two clauses. Propose a semantic representation using two shared variables. How does this representation differ from (77), and why would such a construction challenge the proposed single-sharing invariant?

Exercise 1.11.14 Compare:

the man I know and *the man that I know.*

1. Which expression contains an overt relative marker?
2. Do the two expressions differ in the property denoted by the relative construction?
3. What morphosyntactic information is overt in one variant but not the other?
4. What remains constant under the analysis in (77)?

Exercise 1.11.15 Consider the headless relative:

What I know can change the outcome.

1. What is missing in comparison with *the thing that I know*?
2. What supplies the referentially dependent element?

1 Basics revisited

3. What descriptive content is contributed by the relative clause?
4. Why can *what* not be treated simply as the relative marker *that*?

Exercise 1.11.16 Construct:

1. one headed relative clause with an overt relative marker,
2. one headed relative clause without an overt relative marker, and
3. one headless relative clause.

For each expression, identify how P , Q and the shared element in (77) are realized.

Exercise 1.11.17 In (80), the syntactic categories are represented as functions:

$$\text{rel}(S^{\text{NP}} : P, \text{NP} : Q) = \text{NP}^{\text{NP}} : \lambda x.P(x) \wedge Q(x).$$

Explain:

1. why S^{NP} represents a clause dependent on an NP,
2. why NP^{NP} represents an NP-dependent nominal function,
3. how the syntactic dependency corresponds to the semantic sharing of x ,
and
4. what the one-to-one correspondence marked by the colon contributes to the analysis.

Exercise 1.11.18 Apply the schema in (80) to:

the book that Mary read.

Supply:

1. an S^{NP} representation for the relative clause,
2. the nominal property Q ,
3. the resulting NP^{NP} function, and
4. its semantic representation using shared x .

Exercise 1.11.19 English and Turkish place the relative clause and the restricted noun in different surface orders. Explain why this difference does not require a different semantic invariant. Which configurational requirements belong to the language-specific variants?

Exercise 1.11.20 Section 1.11 proposes that an overt relative head determines the order and types of its arguments. For a relative marker in a language you know, describe:

1. whether the marker is free or bound,
2. whether it precedes or follows the relative clause,
3. where the restricted nominal occurs,
4. whether extraction, resumption or indexing is used, and
5. how the marker identifies the shared argument.

Exercise 1.11.21 Why may a relative construction without an overt marker require a relational rather than a straightforward functional analysis of its surface realization? Identify the two unlike syntactic categories that must nevertheless be connected.

Exercise 1.11.22 Classify each property as belonging primarily to the proposed invariant or to a morphosyntactic variant.

1. A single element is shared.
2. The relative clause precedes the noun.
3. A resumptive pronoun is present.
4. The restricting and restricted properties are conjoined.
5. The shared argument is indexed on the verb.
6. The construction denotes a property of the shared element.
7. The relative marker is phonologically null.

Justify each classification.

Exercise 1.11.23 Assess the following explanation:

Languages permit only one shared element in a relative clause because sharing two elements would be difficult to process.

How does a processing explanation differ from the admissibility account developed in Section 1.11? What evidence would be required to choose between them?

Exercise 1.11.24 Section 1.11 argues that typology should begin with an invariant and then study its realizations. Choose two relative-clause strategies from different languages and explain:

1. how their forms differ,
2. how argument sharing is realized in each,
3. what mathematical abstraction they share, and
4. why neither realization should be taken as the universal template.

1.12 Theories and models: a brief foray into modeling

Exercise 1.12.1 Explain why defining a scientific model as a “simplified representation of reality” presupposes answers to at least the following questions:

1. What counts as reality?
2. What relation constitutes representation?
3. Which properties may be omitted in the simplification?
4. Which properties must be preserved?

Exercise 1.12.2 Compare the observations attributed to Heisenberg and Poincaré in Section 1.12.

1. What role does the scientist’s method of questioning play for Heisenberg?
2. What role do invariants across facts play for Poincaré?
3. How do both observations make scientific knowledge indirect?
4. Why does this indirectness motivate modeling?

Exercise 1.12.3 Distinguish a theory from a narrative using the following properties:

1. chronological organization,
2. explanatory vocabulary,
3. explicit admissibility conditions,
4. consequences that can be assessed,
5. hypotheses, and
6. constraints on possible models.

Why need a narrative be neither scientifically worthless nor a theory?

Exercise 1.12.4 Consider the claim:

Not every theorist must construct models, but every scientific theory must be modelable.

Explain what *modelable* means here. Why would a theory that cannot in principle give rise to assessable models be difficult for nonbelievers to evaluate?

Exercise 1.12.5 Distinguish a thought experiment from an unconstrained imaginary scenario. What theoretical assumptions and admissibility conditions must a thought experiment preserve if it is to contribute to scientific inquiry?

Exercise 1.12.6 Put the following stages of the modeling process in an appropriate order:

1. admissible models,
2. conceptual vocabulary,
3. theory or idea,
4. modeling language,
5. model verification.

For each transition, state what is added or made more explicit.

Exercise 1.12.7 Explain the difference between conceptual vocabulary and modeling language.

1 Basics revisited

1. Which one introduces the concepts supplied by the theory?
2. Which one makes those concepts precise enough to construct models?
3. Why do the methods and tools chosen by the modeler matter at this stage?

Exercise 1.12.8 Suppose that a theory distinguishes nouns and verbs, and claims that every well-formed clause contains a verbal head.

1. Identify part of the theory's conceptual vocabulary.
2. Propose a simple modeling language for representing the claim.
3. Give one model admissible under the theory.
4. Give one model excluded by it.
5. What would model verification check?

Exercise 1.12.9 Distinguish model verification from model validation.

1. Which asks whether a model conforms to the theory?
2. Which asks whether the theory, through its models, withstands comparison with the world?
3. Can a model be verified but not validated?
4. Can a model that violates its own theory count as validated evidence for that theory?

Give an example to support your answers.

Exercise 1.12.10 An implemented grammar accepts exactly the structures licensed by its formal rules, but performs badly on previously unseen linguistic data.

1. Has the implementation been verified?
2. Has the model been validated?
3. What kind of revision might be required?

4. Why should successful verification not be confused with empirical adequacy?

Exercise 1.12.11 A parser produces excellent results on the same data used to develop it but performs poorly on unseen data. Locate the following in the modeling workflow:

1. training data,
2. provisional models,
3. candidate models,
4. unseen data,
5. assessment, and
6. diagnostics.

What does the difference between the two performances suggest?

Exercise 1.12.12 Why is model validation represented as a feedback process rather than a single final test? Describe at least three kinds of result that might send the modeler back to:

1. model development,
2. the modeling language, or
3. the original theory.

Exercise 1.12.13 Classify each question as principally one of verification or validation.

1. Does the program implement every rule stated by the grammar?
2. Does the grammar predict the judgments of speakers?
3. Does the model satisfy the declared type restrictions?
4. Does the analysis extend to languages not used in its development?
5. Does the output conform to the formal definition of a relative clause?

6. Do the proposed relative-clause invariants survive cross-linguistic comparison?

Exercise 1.12.14 Explain why modelers, supporters, critics and agnostics must be able to use the same modeling language. What would happen to critical assessment if each group interpreted the theoretical vocabulary differently?

Exercise 1.12.15 The term *decoder-only* does not mean that the system contains no encoding. Distinguish:

1. encoding an input into an internal representation,
2. an architecturally distinct encoder stack, and
3. decoding or generating an output.

Why does every computational architecture still require some input–output encoding and decoding?

Exercise 1.12.16 Compare the following transformer architectures.

1. encoder–decoder,
2. encoder-only, and
3. decoder-only.

For each, state whether its developing representation can use information from both directions in an already available input or is constrained to earlier material when predicting a continuation.

Exercise 1.12.17 Why might an encoder-only system be described as a contextual model and a decoder-only system as a language model? Explain the relevance of:

1. bidirectional interpretation,
2. continuation probabilities, and
3. causal access to an expanding prefix.

Exercise 1.12.18 Traditional computational linguistics often treats parsing and generation as related but distinct problems.

1. What does parsing attempt to determine?
2. What does generation attempt to determine?
3. How can a decoder-only model generate a continuation while reanalyzing its existing prefix?
4. Does this show that parsing and generation are identical tasks?
5. What possible shared admissibility problem does it reveal?

Exercise 1.12.19 Section 1.12 warns against equating *analysis* with *parsing*. Explain how analysis can be understood more generally as the structuring of a comparison. How can parsing and generation then be treated as different directional variants of analysis?

Exercise 1.12.20 Consider:

Mary persuaded the student to study.

Identify:

1. the expression understood as the subject of *study*,
2. the head that determines this dependency,
3. the local domain in which the dependency is established, and
4. why the dependency is bounded.

Exercise 1.12.21 Compare:

Mary persuaded the student to study,
Mary persuaded the student to begin to try to study.

In both cases the student is understood as studying. Trace the dependency in each sentence. Why is the second dependency mediated through *begin* and *try* rather than formed directly across all intervening structure?

Exercise 1.12.22 Compare the dependencies in:

The student who you think I believe Mary persuaded to study arrived.

Identify:

1 Basics revisited

1. the bounded control dependency involving *persuaded*,
2. the unbounded relative dependency,
3. the clauses crossed by the unbounded dependency, and
4. why these two dependencies should not be represented in the same way.

Exercise 1.12.23 Explain why increasing a model's attention window cannot by itself provide a general account of unbounded dependencies. What happens when a dependency exceeds any fixed window size?

Exercise 1.12.24 Section 1.12 suggests that a model must retain continuation-relevant information and discard less useful information.

1. What makes an unresolved dependency relevant to a future continuation?
2. Why is remembering every earlier token equally well neither necessary nor sufficient?
3. How might the number of unresolved dependencies affect the internal state required by the model?

Exercise 1.12.25 Compare:

Mary persuaded the student to begin to try to study,
**Mary persuaded the student to study to begin to try.*

The two expressions contain much of the same lexical material.

1. Trace the selectional dependencies in the grammatical sentence.
2. At what point does the dependency structure fail in the ungrammatical sentence?
3. Why can token sequence similarity not determine grammatical admissibility?
4. What would a successful model have to retain about the growing prefix?

Exercise 1.12.26 Suppose that adding more training data improves a language model only slightly.

1.12 Theories and models: a brief foray into modeling

1. What would diminishing returns from additional data suggest?
2. How might architectural assumptions rather than data quantity limit the model?
3. What linguistic knowledge about dependencies could motivate a new modeling language or architecture?

Exercise 1.12.27 Apply the modeling workflow in Figure 1.1 to a linguistic problem of your choice. Specify:

1. the theory or initial idea,
2. its conceptual vocabulary,
3. a modeling language,
4. the admissibility conditions,
5. one provisional model,
6. the data used for model development,
7. the validation procedure,
8. the unseen data used for assessment, and
9. the possible feedback to the theory.

Exercise 1.12.28 Assess the following claim:

A model with the highest performance score is necessarily the best scientific model.

In your answer, distinguish:

1. predictive performance,
2. theoretical admissibility,
3. model verification,
4. model validation,
5. interpretability of diagnostics, and
6. what the model allows us to learn beyond its performance.

2 Mathematics and philosophy of linguistics

2.1 Philosophy of linguistics as philosophy of science

Exercise 2.1.1 Compare the following possible conceptions of linguistics:

1. linguistics as a branch of psychology,
2. linguistics as the study of acoustic or gestural events, and
3. linguistics as a branch of mathematics.

What kind of reality does each conception treat as primary?

Exercise 2.1.2 Explain why using mathematics in linguistics does not by itself establish that linguistics is a branch of mathematics. Give an analogous example from another science.

Exercise 2.1.3 Section 2.1 distinguishes questions about methods from questions about the conditions under which methods become meaningful. Classify the following questions accordingly.

1. Which parsing algorithm should be used?
2. What counts as a possible grammatical dependency?
3. How much memory does an implementation require?
4. Which structures must a theory admit?
5. Which statistical model predicts the data most accurately?

Exercise 2.1.4 Why must a theory specify admissibility conditions before its alternative models can be compared? Explain what would happen if each model imposed a different understanding of the object being modeled.

Exercise 2.1.5 Distinguish:

2 *Mathematics and philosophy of linguistics*

1. physical reality,
2. mental reality, and
3. mathematical reality.

Why does Section 2.1 assign mathematical reality a connecting role rather than treating it as a replacement for the other two?

Exercise 2.1.6 Explain how model selection differs from a winner-takes-all conception of scientific debate. What must competing models share before their differences can be meaningfully assessed?

Exercise 2.1.7 Assess the following inference:

Theoretical physics is formulated mathematically; therefore physics is a branch of mathematics.

Why does the corresponding inference about linguistics require further argument?

Exercise 2.1.8 The language of mathematics itself contains presuppositions concerning such matters as sets, rules and countability. Choose one such presupposition and explain how adopting it could restrict the linguistic objects that a model can represent.

Exercise 2.1.9 Section 2.1 proposes that grammars are necessary for understanding intralinguistic structure but insufficient for explaining cross-linguistic variation.

1. What does a grammar contribute?
2. What additional factors might be needed to explain variation?
3. Why should the grammar not be discarded merely because it is insufficient?

Exercise 2.1.10 Connectedness is proposed as sufficient for studying language acquisition but not necessary for explaining every aspect of linguistic diversity. Explain this asymmetry. What different questions are being asked about acquisition and diversity?

Exercise 2.1.11 Distinguish:

1. what happens to a child in language acquisition, and
2. what a learner deliberately does in language learning.

Give one example of each. Why might the distinction matter for the choice of explanatory methods?

Exercise 2.1.12 Section 2.1 claims that linguistic diversity has common structural ground and is not a domain in which anything goes. Give one cross-linguistic difference and propose one invariant that its variants might share.

2.2 Ontogeny of grammaticality

Exercise 2.2.1 Distinguish *ontogeny* from *phylogeny*. Classify each question.

1. How does grammatical knowledge develop in a child?
2. How did the human capacity for language evolve?
3. When does a child begin producing novel verbs?
4. What changes in the species made linguistic communication possible?

Exercise 2.2.2 Suppose that the admissibility conditions of grammaticality remain stable while children's linguistic knowledge develops.

1. What belongs to the invariant?
2. What can change in the variants?
3. Does increasing complexity of children's productions by itself show that grammatical admissibility has changed?

Exercise 2.2.3 Consider the child productions in (1):

*It colds my bottom,
Mummy trousers me.*

1. Which words are used as verbs?

2. What evidence suggests that the child is not merely repeating adult expressions?
3. What sentence–world correspondences might support each innovation?
4. Why are the innovations relevant to grammatical development?

Exercise 2.2.4 Construct two novel denominal verbs in English or another language. For each, provide a context in which a listener could recover the intended meaning. What grammatical and contextual knowledge makes the innovation interpretable?

Exercise 2.2.5 Explain the difference between learning word–world pairings and learning sentence–world pairings. Why might the latter permit a child to infer the meaning and grammatical behavior of an unfamiliar word?

Exercise 2.2.6 Gestalt approaches begin with an organized whole and investigate the parts that make it possible. Apply this perspective to one child production in (1). What is the meaningful whole, and how might its parts be analyzed in relation to it?

Exercise 2.2.7 Consider the nonstandard expressions in (2), including:

We fucking chins them with bottles.

1. What morphological pattern is extended?
2. What social meaning is associated with the pattern?
3. Why might an analysis based only on standard adult grammar classify the expression inadequately?
4. What additional semiotic knowledge is relevant?

Exercise 2.2.8 Distinguish information from knowledge as these notions are used in Section 2.2. Why does arriving at a reasoned explanation through dialogue involve more than collecting observations?

Exercise 2.2.9 Explain the social-semiotic claim that meaning is not first created in an isolated mind and only subsequently projected onto society. How can

speakers jointly maintain a meaning network while still differing in their individual judgments?

Exercise 2.2.10 Ryle asks whether clarifying a concept amounts merely to replacing a philosophically poor expression with a better one. Explain his alternative: investigating the nature of the thing rather than treating the concept as an internally confused object.

Exercise 2.2.11 Apply Ryle's approach to a disputed linguistic category such as *subject*, *word*, *tense* or *voice*.

1. How is the category put to use?
2. What evidence would show that an analyst has misunderstood it?
3. Why is stipulating a clearer definition not necessarily enough?

Exercise 2.2.12 Explain the proposal that grammar is a necessary but insufficient condition for meaningfulness. Identify at least three other conditions that may participate in a sufficient complex for interpreting an utterance.

Exercise 2.2.13 Ryle's statement

Knowing that presupposes knowing how

is applied in Section 2.2 to linguistic categories. Give a linguistic example in which a speaker's ability to use a category is evident even if the speaker cannot state an explicit definition of it.

Exercise 2.2.14 Compare:

Colorless green ideas sleep furiously,
**Green furiously ideas sleep colorless.*

1. Why can an interpretation be constructed for the first expression?
2. What structural conditions fail in the second?
3. Why does using the same words not give the expressions the same grammatical status?

4. What does the comparison show about probability based only on word occurrence?

Exercise 2.2.15 Consider the expressions in (6):

**She won the competition for an hour, She won the competition for a year,
She played the piano in a year, *She played the piano in an hour.*

For each interpretation discussed in the text:

1. identify the relevant event category,
2. explain the effect of the temporal phrase, and
3. describe any categorial shift required to make the expression meaningful.

Exercise 2.2.16 Construct a context in which one of the starred expressions in (6) becomes acceptable. Does supplying such a context show that the original judgment was mistaken, or does it show that reference and event category can shift? Explain.

Exercise 2.2.17 The contrast in (7) shows that flexible word order does not mean that every order is admissible.

1. Which dependencies remain recoverable in (7a)?
2. Why are they not recoverable in (7b)?
3. What does the example show about identifying grammaticality with surface-order frequency?

Exercise 2.2.18 Distinguish grammaticality from acceptability.

1. Give a grammatical but difficult-to-process expression.
2. Give a structurally defective expression whose intended message is recoverable.
3. Identify the extragrammatical factors affecting each judgment.

Exercise 2.2.19 Why is an acceptability judgment not automatically a direct report of grammaticality? Discuss the possible effects of context, processing, frequency, prescriptive education and the analyst's instructions.

Exercise 2.2.20 Husserl distinguishes sensuous intuition from categorial intuition. Apply the distinction to an expression whose literal or most immediate interpretation is anomalous but which becomes meaningful after a change of event category or reference.

Exercise 2.2.21 Section 2.2 contrasts closed-loop and open-loop control.

1. How does a closed-loop system depend on environmental feedback?
2. How does an open-loop system select an output or niche?
3. In what sense can an open-loop system be said to select desirable input?
4. How might category choice be modeled as a distinct control?

Exercise 2.2.22 Reconsider (5)–(7) in terms of categorial recoverability. Classify each relevant expression as:

1. structurally and categorially recoverable,
2. recoverable only after a reference or category shift, or
3. hopelessly unrecoverable under the intended grammar.

Justify every classification.

Exercise 2.2.23 Assess the following claim:

Because grammaticality judgments are psychological responses, grammaticality is nothing more than acceptability.

Distinguish the phenomenon to be explained, the means used to gather evidence and the speaker's report about the phenomenon.

Exercise 2.2.24 Why does identifying biological or computational mechanisms for language not make grammar identical with those mechanisms? Explain the distinction between an abstract grammatical condition and a candidate realization of that condition.

2.3 Some questions that make linguistics a special science

Exercise 2.3.1 Why is language acquisition presented as a problem for reducing linguistics entirely to biology or psychology? Compare children's development of language with their development of music, painting or complex problem-solving abilities.

Exercise 2.3.2 Section 2.3 does not deny the biological or psychological dimensions of language. What stronger claim would be required to reduce linguistic explanation to either biology or psychology?

Exercise 2.3.3 Consider the relative clause:

the cat that the dog chased ____.

1. Which argument has been extracted?
2. Construct a corresponding subject relative.
3. State the subject–object extraction asymmetry described in Section 2.3.
4. What hypothetical language pattern would reverse that asymmetry?

Exercise 2.3.4 Suppose that a language permits direct-object relativization by extraction. What prediction follows from the accessibility asymmetry discussed in Section 2.3? Would the prediction concern one construction or a range of argument types?

Exercise 2.3.5 Why is a gap in the logically possible space of languages scientifically interesting? Explain why the nonattestation of object-only extraction cannot simply be inferred from the logical definition of a relation or function.

Exercise 2.3.6 Section 2.3 observes that nonverbal constituents can undergo extraction in many languages, whereas genuine verb extraction is unattested.

1. What would genuine verb extraction have to involve?
2. Why is moving a phrase containing a verb not automatically verb extraction?
3. What kind of admissibility condition might the asymmetry motivate?

2.3 Some questions that make linguistics a special science

Exercise 2.3.7 State the agreement asymmetry described in Section 2.3. Construct the paradigm of a hypothetical language that would violate it by exhibiting object agreement without subject agreement.

Exercise 2.3.8 Why does the occasional blocking of agreement with a particular noun type not necessarily refute an agreement hierarchy? Distinguish a systematic reversal of an asymmetry from a restricted lacuna within a paradigm.

Exercise 2.3.9 Compare:

John cut himself and **Heself cut John*.

1. Which argument binds the reflexive in the grammatical expression?
2. What would a genuine subject reflexive require?
3. Why does the logical representation $cut(x, x)$ fail to exclude the unattested pattern?
4. What does this show about logical possibility and linguistic admissibility?

Exercise 2.3.10 Consider the Turkish expression:

uyu-du-m sleep-PAST-1SG ‘I slept’.

1. Which information makes the form propositional?
2. Why is the verbal root alone not a full proposition?
3. How does this differ from saying that an overt subject noun phrase is always required?

Exercise 2.3.11 Find another example from a pro-drop language in which an inflected verb forms a complete clause. Identify the contribution of the root and each piece of inflectional morphology.

Exercise 2.3.12 Determine the arguments of *bet* in:

I bet him five euros that the final goes overtime.

Which argument is clausal? Why does the number of total arguments not contradict the proposed restriction on the number of full clausal arguments?

Exercise 2.3.13 Compare:

*I convinced him that the World Cup is this year if not next year,
She told me that if the World Cup is this year, we should watch it,
She said that we may win that we are strong.

Identify the matrix arguments and the internal dependencies of the subordinate material. Why does the final example resist analysis as one clause with two independent full clausal arguments?

Exercise 2.3.14 Construct a hypothetical lexical entry for a verb taking two independent full clausal arguments. Explain what evidence would be needed to show that both clauses are selected directly by the same verb rather than one being embedded within or dependent on the other.

Exercise 2.3.15 The asymmetries in Section 2.3 include extraction, agreement, reflexivization, verbal morphology and clausal argument structure. Choose three and explain:

1. the logically possible but unattested pattern,
2. the attested asymmetry, and
3. why a language-specific admissibility principle is required to explain the difference.

Exercise 2.3.16 Assess the following claim:

If a pattern can be represented in logic, it should be possible in a human language.

Use at least two phenomena from Section 2.3 to distinguish logical expressibility from linguistic admissibility.

2.4 Mathematics in linguistics versus mathematicizing linguistics

Exercise 2.4.1 Explain the distinction stated in (12):

Mathematicizing linguistics begins with methods,
mathematical thinking begins with admissibility.

2.4 *Mathematics in linguistics versus mathematicizing linguistics*

Give one example of each approach to the same linguistic problem.

Exercise 2.4.2 A linguist begins a project by deciding to use neural networks. Another begins by asking what dependencies a theory of relative clauses must admit.

1. Which project begins with a method?
2. Which begins with admissibility?
3. What questions must each researcher eventually ask to produce a scientifically assessable model?

Exercise 2.4.3 Why is mathematics privileged in Section 2.4 for distinguishing necessary structure from contingent realization rather than simply for performing calculations? Give a linguistic example of this distinction.

Exercise 2.4.4 Katz treats grammars as abstract objects belonging to mathematical reality. Explain why this position does not by itself account for the unexpected typological gaps discussed in Section 2.3.

Exercise 2.4.5 Compare the roles of mathematical methods and mathematical thinking in physics or another science.

1. How can a mathematical method be used to study a variant?
2. How can mathematical abstraction reveal an invariant?
3. Why does neither use turn the science into a branch of mathematics?

Exercise 2.4.6 Distinguish weak generative capacity from strong generative capacity.

1. Which concerns the strings generated by a grammar?
2. Which concerns the structural descriptions assigned to them?
3. Can two grammars have the same weak generative capacity but differ in strong generative capacity?
4. Why would that difference matter empirically?

Exercise 2.4.7 Explain Chomsky's claim that an adequate linguistic theory must restrict the class of possible grammars. What would happen to explanatory adequacy if every grammar compatible with the observed strings remained equally possible?

Exercise 2.4.8 What role does an evaluation measure play in choosing among descriptively adequate grammars? Explain why the measure must operate within a restricted class of admissible grammars.

Exercise 2.4.9 Chomsky warns that questions may be selected because they are mathematically feasible rather than because they are empirically central. Give an example of:

1. a mathematically tractable linguistic question,
2. an empirically important question that may be harder to formalize, and
3. a research program that mistakenly equates tractability with significance.

Exercise 2.4.10 Why would choosing a more powerful formal mechanism than a phenomenon requires be theoretically undesirable? Consider both:

1. the explanatory commitments of the theory, and
2. the feasibility of comparing computational models.

Exercise 2.4.11 Section 2.4 notes that formal-language theory and automata theory largely disappeared from many linguistics curricula while remaining central in computer science. What intellectual consequences might follow if linguists encounter only the implementations of formal ideas and not their mathematical foundations?

Exercise 2.4.12 Parameters, feature geometries, categorial function spaces and monostructural or multistructural theories employ different descriptive vocabularies. Explain how comparison of their underlying abstractions could reveal shared admissibility questions without erasing theoretical differences.

Exercise 2.4.13 What is reification in the context of linguistic theory? Give an example in which a useful abstraction is mistakenly treated as a concrete mechanism. What theoretical commitments follow from the mistake?

2.4 Mathematics in linguistics versus mathematicizing linguistics

Exercise 2.4.14 Explain why modelability is necessary even for a theorist who does not personally construct computational models. What must be made available to support criticism by practitioners, agnostics and rival theorists?

Exercise 2.4.15 Assess the following claim:

The purpose of mathematics in linguistics is to provide methods for solving already identified linguistic problems.

How can mathematics instead reveal that the original problem was incorrectly formulated or that an unnoticed admissibility problem exists?

Exercise 2.4.16 Choose one phenomenon from Section 2.3 and design two research programs:

1. one that begins by selecting a familiar method, and
2. one that begins by specifying the admissibility problem.

Compare the questions, possible models and explanatory risks of the two programs.

3 Degrees of structure from string admissibility

3.1 Cutting the Chomsky hierarchy at different seams

Exercise 3.1.1 Explain why the Chomsky hierarchy classifies families of formal languages rather than individual linguistic expressions. Distinguish:

1. a string,
2. a formal language, and
3. a family of formal languages.

Exercise 3.1.2 A formal language is a set of strings. Explain what it means to say that the family of context-free languages is a set of sets of strings. Give a small finite example of:

1. an alphabet,
2. two languages over that alphabet, and
3. a family containing those languages.

Exercise 3.1.3 Reproduce the principal inclusion relations in the Chomsky hierarchy using the subset symbol. Include:

1. regular languages,
2. context-free languages,
3. context-sensitive languages, and
4. recursively enumerable languages.

3 Degrees of structure from string admissibility

What does each inclusion claim, and what does it not claim?

Exercise 3.1.4 Explain the statement:

More structure does not mean more strings.

Why can a smaller family of languages impose less structural control even though every language in that family also belongs to the larger families containing it?

Exercise 3.1.5 Suppose that a regular language L is also classified as context-free. Does this mean that:

1. L has acquired additional strings,
2. its strings must receive context-free analyses, or
3. the context-free family contains every regular language?

Explain your answer.

Exercise 3.1.6 Why should a model normally use the least powerful mechanism sufficient for the structural dependencies under investigation? Discuss:

1. computational resources,
2. model selection, and
3. unnecessary theoretical commitments.

Exercise 3.1.7 For each research decision, state whether it concerns principally an invariant or a computational variant.

1. Determining the degree of dependency that strings must support
2. Selecting an automaton capable of recognizing those strings
3. Deciding whether dependencies may cross
4. Measuring the running time of a parser
5. Determining whether a language requires more than context-freeness

3.1 Cutting the Chomsky hierarchy at different seams

Exercise 3.1.8 The Swiss German pattern in (2) gives a dependency sequence of the form

$$s_1 s_2 s_3 v_1 v_2 v_3.$$

Draw lines connecting every s_i with its corresponding v_i .

1. Do the dependency lines nest or cross?
2. How would the pattern continue for four subject–verb pairs?
3. Why is the possibility of extending the pattern important?

Exercise 3.1.9 Compare the dependency patterns:

$$a_1 a_2 \cdots a_k b_k \cdots b_2 b_1$$

and

$$a_1 a_2 \cdots a_k b_1 b_2 \cdots b_k.$$

1. Which pattern contains nested dependencies?
2. Which contains crossing dependencies?
3. Why can identical unindexed strings conceal this structural difference?
4. Which pattern resembles the Swiss German example?

Exercise 3.1.10 Explain the reasoning behind the claim that evidence from one natural language can constrain the admissibility assumptions for human language generally. Which additional premise is required to move from Swiss German to a claim about the human language capacity?

Exercise 3.1.11 Classify the following morphological processes as concatenative or nonconcatenative.

1. suffixation,
2. prefixation,
3. infixation,
4. circumfixation,

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5. templatic morphology, and
6. reduplication.

Why might some nonconcatenative processes motivate a class above regular languages?

Exercise 3.1.12 Consider Turkish:

<i>mavi</i>	‘blue’,
<i>masmavi</i>	‘deep blue’,
<i>açık</i>	‘clear’,
<i>apaçık</i>	‘obvious, completely clear’.

1. What formal material is repeated or introduced?
2. What semantic contribution accompanies the process?
3. Why does the unavailability of indefinite repetition matter for classifying the dependency?
4. Why is this not obviously the same kind of embedding found in syntax?

Exercise 3.1.13 Let

$$L = \{a^n b^n \mid n \in \mathbb{N}\}.$$

List its first five strings, assuming $0 \in \mathbb{N}$. What dependency between the two blocks must be preserved? Why can the set not be characterized merely as a^*b^* ?

Exercise 3.1.14 String admissibility is described in Section 3.1 as an INUS condition on grammars for meaningfulness. Explain:

1. why string admissibility is insufficient for meaningfulness,
2. why it may be necessary within a larger explanatory complex, and
3. what other conditions that complex might contain.

3.2 Structure: admissibility of strings

Exercise 3.2.1 Explain why Section 3.2 classifies admissible structural possibilities over strings rather than classifying particular grammar formalisms. What is the difference between:

1. a condition that every sufficiently long string must satisfy, and
2. a method for describing or recognizing such strings?

Exercise 3.2.2 What is the role of a pumping lemma in the study of a family of formal languages? Explain why a pumping lemma:

1. concerns sufficiently long strings,
2. identifies repeatable structure, and
3. does not prescribe a particular grammar or automaton.

Exercise 3.2.3 A pumping lemma gives a necessary property of every language in a particular family. Explain why finding one decomposition that pumps successfully does not by itself prove that a language belongs to that family.

3.2.1 Context-free admissibility

Exercise 3.2.1.1 State the pumping lemma for context-free languages in (3), including all conditions on:

1. the length of z ,
2. the decomposition $z = uvwxy$,
3. the pumped portions,
4. the bounded central region, and
5. the strings obtained for every $i \geq 0$.

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Exercise 3.2.1.2 In the decomposition

$$z = uvwxy,$$

state the role of each component under context-free pumping.

1. Which parts remain fixed?
2. Which parts are repeated together?
3. Which part serves as the center or head?
4. Which condition prevents pumping from being vacuous?

Exercise 3.2.1.3 Why does the condition

$$|vx| \geq 1$$

permit $v = \epsilon$ or $x = \epsilon$ but prohibit both from being empty? What would happen to

$$uv^iwx^iy$$

if both were empty?

Exercise 3.2.1.4 Explain the purpose of:

$$|vwx| \leq n.$$

Why must the two pumped portions and their center occur within a bounded region of the sufficiently long string?

Exercise 3.2.1.5 Let

$$L = \{a^n b^n \mid n \geq 0\}$$

and choose:

$$z = a^4 b^4, \quad u = a, \quad v = a, \quad w = a^2 b^2, \quad x = b, \quad y = b.$$

Construct the strings obtained for:

1. $i = 0$,
2. $i = 1$,

3. $i = 2$, and

4. $i = 3$.

Verify that each contains the same number of as and bs .

Exercise 3.2.1.6 In Exercise 3.2.1.5, what dependency connects v and x ? Why must they be pumped the same number of times? What would happen if they could be pumped independently?

Exercise 3.2.1.7 For the general context-free pumping form:

$$uv^iwx^iy,$$

draw the dependency between v and x around the center w . Why is this an embedding relation rather than a single unconstrained iteration?

Exercise 3.2.1.8 Explain why the center w need not be the geometrical midpoint of the string. In what functional sense can it nevertheless be treated as a head?

Exercise 3.2.1.9 The term *context-free* does not mean that a string has no context. Explain instead what is locally free in a context-free dependency and what remains structurally controlled.

Exercise 3.2.1.10 Suppose that a language contains no strings of length at least its proposed pumping length n .

1. What follows about the size of the language?
2. Why is no pumping cycle forced?
3. Why is every finite language regular?

Exercise 3.2.1.11 A student says:

A context-free language permits arbitrary structure because its contexts are free.

Correct the statement. What local control remains, and what kind of nonlocal control is unavailable?

Exercise 3.2.1.12 Explain why the context-free pumping lemma is an admissibility condition rather than:

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1. a phrase-structure grammar,
2. a parsing algorithm, or
3. a pushdown automaton.

How are those later methods related to the invariant?

3.2.2 Regular admissibility

Exercise 3.2.2.1 State the pumping lemma for regular languages in (4), including the conditions on:

1. the length of z ,
2. the decomposition $z = uvw$,
3. the nonempty pumped part,
4. the initial bounded region, and
5. pumping for every $i \geq 0$.

Exercise 3.2.2.2 Compare:

$$uv^i w$$

with the context-free form:

$$uv^i wx^i y.$$

1. How many portions are pumped in each?
2. Which form establishes a dependency across a center?
3. Which form represents iteration without paired control?

Exercise 3.2.2.3 Let

$$L = \{a^n b \mid n \geq 0\}.$$

Choose $z = a^4 b$ and the decomposition:

$$u = a, \quad v = a^2, \quad w = ab.$$

Construct the pumped strings for $i = 0, 1, 2, 3$. Verify that they remain in L .

Exercise 3.2.2.4 Why does the pumped portion v in:

$$uv^i w$$

not depend on another portion of the string? Explain why Section 3.2.2 therefore treats regular structure as iteration rather than embedding.

Exercise 3.2.2.5 The regular pumping lemma is often written informally using v^* . Explain what the Kleene star captures and what structural information it does not capture.

Exercise 3.2.2.6 Does the notation uvw imply that regular iteration must occur before the final material w ? Explain why equivalent descriptions can place the iterating portion elsewhere without changing the degree of structural control.

Exercise 3.2.2.7 Let:

$$L_1 = \{a^n \mid n \geq 0\} \quad \text{and} \quad L_2 = \{a^n b^n \mid n \geq 0\}.$$

Explain why an independent iteration is sufficient for L_1 but insufficient for L_2 .

Exercise 3.2.2.8 Give a linguistic example of a process that appears iterative rather than embedding. State:

1. what may repeat,
2. what remains fixed, and
3. why no paired dependency around a center is required.

Exercise 3.2.2.9 A language admits the strings:

$$b, ab, aab, aaab, \dots$$

1. Express the language using set-builder notation.
2. Express it using a regular expression.
3. Identify suitable u , v and w in a sufficiently long string.

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4. What happens when $i = 0$?

Exercise 3.2.2.10 Explain why regular languages admit less structure rather than fewer strings. Could a regular language contain arbitrarily many strings? Could a finite language be context-free? Justify both answers.

3.2.3 Linear admissibility

Exercise 3.2.3.1 State the pumping lemma for linear languages in (5). Which condition differs from the context-free pumping lemma?

Exercise 3.2.3.2 Compare:

$$|vwx| \leq n$$

in context-free pumping with:

$$|uvxy| \leq n$$

in linear pumping. Which component is omitted from the second length bound, and why is that difference structurally significant?

Exercise 3.2.3.3 Which properties do the context-free and linear pumping forms share? Discuss:

1. the decomposition of z ,
2. the requirement that pumping be nonempty, and
3. the paired pumping of v and x .

Exercise 3.2.3.4 Why is linear admissibility more structurally powerful than regular admissibility? Why is it nevertheless more restricted than general context-free admissibility?

Exercise 3.2.3.5 Let:

$$L = \{a^n b^n \mid n \geq 0\}.$$

Explain why its matching dependency can be represented by linear admissibility. Identify:

1. the left pumped context,

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2. the right pumped context, and
3. the material between them.

Exercise 3.2.3.6 A student argues:

Linear languages must contain only strings whose symbols occur in a straight line.

Explain why this interpretation of *linear* is mistaken. What does linearity restrict in the organization of dependencies?

Exercise 3.2.3.7 Compare the three pumping patterns:

regular:	$uv^i w$,
linear:	$uv^i wx^i y$,
context-free:	$uv^i wx^i y$.

The latter two have the same pumped shape. Explain why their different bounds nevertheless give them different degrees of structural admissibility.

Exercise 3.2.3.8 Place the following families in order from least to most structural control:

regular, linear, context-free.

Explain why increasing control produces a larger family of admissible structural possibilities.

3.3 String admissibility: nonreified mathematics with implications for order and structure

Exercise 3.3.1 Distinguish iteration from embedding using the pumping forms developed in Section 3.2. What feature of the mathematical representation turns two repeated portions into a dependency?

Exercise 3.3.2 Explain why the common exponent i in:

$$uv^i wx^i y$$

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represents structural control. How would the interpretation change if the two portions had independent exponents:

$$uv^iwx^jy?$$

Exercise 3.3.3 Let:

$$A = \{a^n b^n \mid n \in \mathbb{N}\}.$$

1. List several strings in A .
2. State the ordering condition.
3. State the numerical dependency.
4. Explain why A is linear.
5. Explain why A is not regular.

Exercise 3.3.4 Let:

$$B = \{a^n b^n c^m d^m \mid n, m \geq 1\}.$$

1. List the strings obtained for $(n, m) = (1, 1), (1, 2), (2, 1)$ and $(2, 2)$.
2. Identify the two independent dependencies.
3. Draw the dependency arcs for $aabbccdd$.
4. Explain why B is context-free but not linear.

Exercise 3.3.5 Let:

$$C = \{a^n b^m c^m d^n \mid n, m \geq 1\}.$$

1. List the strings obtained for $(n, m) = (1, 1), (1, 2), (2, 1)$ and $(2, 2)$.
2. Identify the dependencies between symbol blocks.
3. Draw the dependency arcs for $aabbccdd$.
4. Explain why the dependencies are nested.
5. Why can C be linear although it contains two matching dependencies?

3.3 *String admissibility: nonreified mathematics with implications for order and structure*

Exercise 3.3.6 Compare the languages:

$$B = \{a^n b^n c^m d^m \mid n, m \geq 1\}$$

and

$$C = \{a^n b^m c^m d^n \mid n, m \geq 1\}.$$

Why does the number of dependencies alone not determine linearity? What difference in their organization is decisive?

Exercise 3.3.7 For each pattern, classify the dependencies as independent, nested or crossing.

1. $a_1 a_2 b_2 b_1$
2. $a_1 b_1 c_1 d_1$
3. $a_1 a_2 b_1 b_2$
4. $a_1 b_1 a_2 b_2$

Draw dependency arcs to justify your answers.

Exercise 3.3.8 Explain the claim that structural admissibility concerns order and structure jointly. Why is it inadequate to say that:

1. order is derived entirely from structure,
2. structure is derived entirely from order, or
3. order and structure have unrelated admissibility conditions?

Exercise 3.3.9 Two strings contain exactly the same number of *as* and *bs*:

$$aabb \quad \text{and} \quad abab.$$

Give two possible dependency interpretations for them. Why is symbol count alone insufficient to determine degree of structure?

Exercise 3.3.10 Explain why the formal properties in Chapter 3 concern structured meaningfulness rather than the lexical content of strings. Could the same admissibility conditions apply to:

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1. parentheses,
2. musical symbols,
3. actions, and
4. linguistic expressions?

What would have to be preserved?

Exercise 3.3.11 Section 3.3 suggests that bound morphemes do not ordinarily contain bound morphemes through iterative embedding. Give an example of:

1. ordinary affixation,
2. repeated affixation, and
3. syntactic embedding.

Explain why repetition of affixes is not automatically embedding within an affix.

Exercise 3.3.12 Phonemes and syllables can be decomposed into smaller units or features. Why does decomposition not by itself constitute embedding? What additional dependency would be required?

Exercise 3.3.13 Compare a phonological phrase with an individual phoneme. Why might phonological or prosodic phrases support embedding even if individual segments do not?

Exercise 3.3.14 The cross-linguistic definition of *word* remains controversial. Explain why asking whether words embed like syntactic phrases requires an independent understanding of wordhood. What evidence might bear on the question?

Exercise 3.3.15 Distinguish transparent reference transmission from correlational reference. Use one of the following contrasts:

1. syntax and morphology,
2. morphology and the lexicon,
3. plural marking and pluralizable objects,
4. possession and possessible objects, or

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5. viewpoint marking and viewable events.

Exercise 3.3.16 Suppose that a morphological process marks possession only on objects culturally classified as ownable.

1. What formal dependency does the marking express?
2. What referential or semantic condition restricts it?
3. Why would a description based solely on string shape be insufficient?

Exercise 3.3.17 Phrase-structure grammars, categorial grammars and dependency grammars use different descriptive vocabularies. Why may they nevertheless be assessed against the same string-admissibility conditions?

Exercise 3.3.18 For each item, classify it as primarily an invariant condition or a variant method.

1. regular pumping,
2. context-free pumping,
3. a phrase-structure grammar,
4. a dependency grammar,
5. a finite automaton,
6. a parsing algorithm,
7. the distinction between iteration and embedding.

Explain any classification that might appear ambiguous.

Exercise 3.3.19 Reconstruct the progression summarized in Figure 3.2:

strings $\longrightarrow$ admissibility conditions $\longrightarrow$ degree of structure $\longrightarrow$ choice of grammar $\longrightarrow$ choice of

For every arrow, explain why the item on the left constrains the choice on the right.

Exercise 3.3.20 A researcher selects a powerful parsing algorithm before determining the degree of structure required by the linguistic phenomenon.

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1. At which point has the progression in Figure 3.2 been reversed?
2. What unnecessary theoretical commitments might result?
3. What computational costs might result?
4. How should the investigation be reorganized?

Exercise 3.3.21 Assess the following claim:

Because a context-free mechanism can describe every regular language, regular admissibility is linguistically unnecessary.

Distinguish descriptive possibility from explanatory adequacy and minimal sufficient power.

Exercise 3.3.22 Choose a linguistic pattern involving iteration, embedding or cross-serial dependency. Describe:

1. the relevant strings,
2. the dependencies assigned to them,
3. the least family in the hierarchy that appears sufficient,
4. an appropriate descriptive method, and
5. an appropriate computational method.

State clearly which parts of your proposal concern the invariant and which concern its variants.

4 Formal grammars: describing admissible strings

4.1 Context-free grammars

4.1.1 Definitions

Exercise 4.1.1.1 Let

$$G = (V, \Sigma, R, S)$$

be a context-free grammar. State the role of each member of the quadruple. Which sets must be finite, and which member of the quadruple must belong to V ?

Exercise 4.1.1.2 For each rule, determine whether it has the form permitted in a context-free grammar.

1. $S \rightarrow aSb$
2. $A \rightarrow BCd$
3. $AB \rightarrow a$
4. $aA \rightarrow Bb$
5. $B \rightarrow \epsilon$
6. $C \rightarrow C$

For every inadmissible rule, identify the violated condition.

Exercise 4.1.1.3 Explain why the term *context-free* concerns the lefthand side of a rewrite rule. Why may the righthand side contain a complex string even though the grammar is context-free?

Exercise 4.1.1.4 Explain the difference between:

G is a context-free grammar

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and

$\mathcal{L}(G)$ is a context-free language.

Which statement concerns a description, and which concerns the stringset described?

Exercise 4.1.1.5 Interpret:

$$\mathcal{L}(G) = \{w \in \Sigma^* \mid S \Rightarrow^* w\}.$$

Explain:

1. why w must contain terminals only,
2. why a derivation begins with S , and
3. what the Kleene star over $\Rightarrow$ means.

4.1.2 Examples

Exercise 4.1.2.1 For the grammar in (6):

$$\begin{aligned} S &\rightarrow AB, \\ A &\rightarrow aAb \mid ab, \\ B &\rightarrow cBd \mid cd, \end{aligned}$$

give the sets V , Σ and R , and identify the start symbol.

Exercise 4.1.2.2 Using the grammar in Exercise 4.1.2.1, give complete derivations of:

1. $abcd$,
2. $aabbcd$,
3. $abccdd$, and
4. $aaabbbcccd$.

For the last string, state the values of n and m in:

$$\mathcal{L}(G) = \{a^n b^n c^m d^m \mid n, m \geq 1\}.$$

Exercise 4.1.2.3 Determine which strings belong to the language of the grammar in (6). Justify every answer by giving a derivation or explaining why no derivation is possible.

1. ab
2. $abcd$
3. $aabbbcd$
4. $aabbccdd$
5. $cdab$
6. ϵ

Exercise 4.1.2.4 Why are the a/b and c/d dependencies in (6) independent? Construct:

1. a string for which $n = m$,
2. a string for which $n < m$, and
3. a string for which $n > m$.

Exercise 4.1.2.5 Draw a derivation tree for:

$$S \Rightarrow^* aabbccdd$$

using the grammar in (6). Verify that:

1. every local branch corresponds to a rule,
2. the root is S , and
3. every leaf is a terminal.

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Exercise 4.1.2.6 The grammar in (9) describes all strings over $\{a, b\}$ containing the same number of a s and b s:

$$\begin{aligned} S &\rightarrow aB \mid bA \mid \epsilon, \\ A &\rightarrow aS \mid bAA, \\ B &\rightarrow bS \mid aBB. \end{aligned}$$

Derive:

1. ab ,
2. ba ,
3. $aabb$, and
4. $abba$.

Exercise 4.1.2.7 Explain the bookkeeping roles of A and B in the grammar in (9). Why does encountering another b while A is active introduce two occurrences of A ?

Exercise 4.1.2.8 Consider:

$$\begin{aligned} S &\rightarrow AB, \\ A &\rightarrow aA \mid \epsilon, \\ B &\rightarrow bB \mid \epsilon. \end{aligned}$$

1. Describe its language using set notation.
2. Derive $aaabbb$.
3. Is its language context-free?
4. Is its language also regular?
5. What does this show about the difference between a language class and the descriptive form chosen for it?

Exercise 4.1.2.9 Using:

$$S \rightarrow (S)S \mid \epsilon,$$

derive:

1. $()$,
2. $(())$,
3. $()()()$, and
4. $((()))$.

Identify which occurrence of S provides nesting and which provides iteration.

4.2 An excursion to linear-indexed grammars

4.2.1 Definitions

Exercise 4.2.1.1 Explain the difference among:

$$A[], \quad A[i \dots], \quad A[\dots], \quad A.$$

What does each notation say about the index associated with A ?

Exercise 4.2.1.2 For each rule type, explain how the index changes:

$$\begin{aligned} A[i \dots] &\rightarrow \beta[\dots], \\ A[\dots] &\rightarrow \beta[i \dots], \\ A[] &\rightarrow \beta, \\ A &\rightarrow \beta. \end{aligned}$$

Classify each as pushing an index, removing an index, requiring an empty index or making an ordinary context-free rewrite.

Exercise 4.2.1.3 Why is an index passed to only one nonterminal on the right-hand side of a linear-indexed rule? What additional power would be introduced if the entire index were copied to several nonterminals?

Exercise 4.2.1.4 Explain the significance of:

$$\mathcal{L}(G) = \{w \mid S[] \Rightarrow^* w\}.$$

Why must the index associated with the start-symbol dependency be exhausted when a terminal string is admitted?

4.2.2 Examples

Exercise 4.2.2.1 Consider the grammar in (14):

$$\begin{aligned} S[\dots] &\rightarrow s_i S[i \dots] \mid A[\dots], \\ A[i \dots] &\rightarrow A[\dots] v_i, \\ A[] &\rightarrow \epsilon. \end{aligned}$$

Give a derivation of:

$$s_1 s_2 s_3 v_1 v_2 v_3.$$

Show the index at every stage.

Exercise 4.2.2.2 In the grammar in (14), what index is stored after reading:

$$s_1 s_2 s_3?$$

In what order are the indices removed, and why does this yield the dependency sequence:

$$s_1 s_2 s_3 v_1 v_2 v_3$$

rather than:

$$s_1 s_2 s_3 v_3 v_2 v_1?$$

Exercise 4.2.2.3 Draw a derivation tree for:

$$s_1 s_2 v_1 v_2$$

using the grammar in (14). Connect every s_i to its corresponding v_i . Which dependencies cross?

Exercise 4.2.2.4 Why can the Swiss German dependency pattern not be described by an ordinary context-free grammar? Explain what information the index preserves that the terminal string alone does not.

Exercise 4.2.2.5 Consider:

$$\begin{aligned} S[\dots] &\rightarrow aS[1 \dots]d \mid A[\dots], \\ A[1 \dots] &\rightarrow bA[\dots]c, \\ A[] &\rightarrow \epsilon. \end{aligned}$$

Derive:

1. ϵ ,
2. $abcd$,
3. $aabbccdd$, and
4. $aaabbbccdd$.

Exercise 4.2.2.6 Using the grammar in Exercise 4.2.2.5, explain why the following strings cannot be derived:

1. ad ,
2. $aabccdd$,
3. $aabbccdd$, and
4. $aabdd$.

For each failure, identify the unmatched index or terminal dependency.

Exercise 4.2.2.7 The existence of a linear-indexed grammar for a stringset establishes formal admissibility at that level. Why does it not establish that the stringset is a possible natural language? Distinguish model verification from empirical model validation.

Exercise 4.2.2.8 Assess Montague's claim that there is no important theoretical difference between natural and artificial languages. How does the distinction between formal admissibility and empirical attestation qualify that claim?

4.3 Regular expressions (grammars)

4.3.1 Definitions

Exercise 4.3.1.1 Let $\Sigma = \{a, b\}$. State which of the following are regular expressions according to (19).

1. ϵ
2. $\emptyset$
3. a

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4. ab
5. $a \cup b$
6. a^*
7. $a \cap b$
8. $a - b$

For every excluded expression, identify the unavailable operation.

Exercise 4.3.1.2 Explain the difference among:

$$\epsilon, \quad \{\epsilon\}, \quad \emptyset.$$

Which is a string, which is a language containing one string, and which is a language containing no strings?

Exercise 4.3.1.3 Describe the languages denoted by:

1. a ,
2. ab ,
3. $a \cup b$,
4. a^* ,
5. aa^* , and
6. $(a \cup b)^*$.

Exercise 4.3.1.4 List the strings of length at most three in:

1. $\mathcal{L}(a^*)$,
2. $\mathcal{L}((ab)^*)$,
3. $\mathcal{L}((a \cup b)^*)$, and
4. $\mathcal{L}(a^*b)$.

Exercise 4.3.1.5 State the difference between:

$$u^* \quad \text{and} \quad u^+ = uu^*.$$

Which admits ϵ ? Give examples with $u = a$.

4.3.2 Examples

Exercise 4.3.2.1 Let:

$$\alpha = ((a \cup b \cup c)^2 b)^* (a \cup b \cup c \cup \epsilon)^2.$$

Determine whether each string belongs to $\mathcal{L}(\alpha)$.

1. ϵ
2. a
3. ac
4. aab
5. $acba$
6. $acbac$
7. $aacab$

Check the symbol in every third position.

Exercise 4.3.2.2 Why does the regexp in (21) admit strings of length zero, one and two? Explain why this follows from the informal condition that every third symbol is b .

Exercise 4.3.2.3 Write regular expressions for the following languages over $\Sigma = \{a, b\}$.

1. Strings beginning with a
2. Strings ending with b
3. Strings containing exactly one a
4. Strings containing at least one a
5. Strings of length exactly three
6. Strings in which every a is immediately followed by b

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Exercise 4.3.2.4 The expression:

$$(un \cup \epsilon)(happy \cup fair)$$

describes four English word forms. List them. Why does adding *do* to the second parenthesis raise a problem that is not merely orthographic?

Exercise 4.3.2.5 Explain why:

undoes

is morphologically ordinary whereas:

unhappies

requires an unusual interpretation. What lexical and categorial information is absent from the regexp alone?

Exercise 4.3.2.6 Let:

$$\alpha = (1 \cup 2 \cup 0^+2)^*(0 \cup \epsilon)^*.$$

Generate five strings in $\mathcal{L}(\alpha)$ and verify that none contains the substring 01.

Exercise 4.3.2.7 Over the alphabet $\{0, 1\}$, explain why:

$$1^*0^*$$

describes all and only the strings containing no substring 01. Why does the corresponding expression over $\{0, 1, 2\}$ require a different structural analysis?

Exercise 4.3.2.8 Write a regular expression for each language.

1. Strings containing no 1
2. Strings containing no consecutive 1s
3. Strings containing at least two 0s
4. Strings of even length
5. Strings ending in either 00 or 11

Exercise 4.3.2.9 Why is a regular expression for strings containing an even number of 0s and an odd number of 1s considerably less transparent than the informal description? What does this show about the relation between descriptive simplicity and structural simplicity?

Exercise 4.3.2.10 Explain why the absence of intersection and set difference from the recursive definition in (19) does not imply that no regular language can be described using those set-theoretic notions. Distinguish the language from the permitted regexp vocabulary.

4.4 An excursion to linear grammars

4.4.1 Definitions

Exercise 4.4.1.1 State the permitted form of a rule in a linear grammar:

$$A \rightarrow \beta.$$

How many nonterminals may occur in β , and where may that nonterminal occur?

Exercise 4.4.1.2 Classify each rule as linear, regular or neither.

1. $A \rightarrow aBb$
2. $A \rightarrow abB$
3. $A \rightarrow Bab$
4. $A \rightarrow aBC$
5. $AB \rightarrow ab$
6. $A \rightarrow abc$
7. $A \rightarrow \epsilon$

Some rules may belong to both the linear and regular descriptions.

Exercise 4.4.1.3 Why is every regular grammar linear, while not every linear grammar is regular? Give a rule that is linear but neither right-regular nor left-regular.

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Exercise 4.4.1.4 Explain why restrictions on the shape of linear rules are consequences of linear admissibility rather than arbitrary stylistic restrictions on grammar writing.

4.4.2 Examples

Exercise 4.4.2.1 Using:

$$S \rightarrow aSb \mid \epsilon,$$

derive:

$$\epsilon, \quad ab, \quad aabb, \quad aaabbb.$$

Explain how every recursive use of the rule creates a dependency between one a and one b .

Exercise 4.4.2.2 Using:

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon,$$

derive:

1. $abba$,
2. $baab$,
3. aba , and
4. $aabbba$.

Why can this grammar derive only palindromes?

Exercise 4.4.2.3 Give two palindromes and two nonpalindromes over $\{a, b\}$. For each nonpalindrome, indicate where a derivation using the grammar in (29) would fail.

Exercise 4.4.2.4 For:

$$\begin{aligned} S &\rightarrow 0S1 \mid A, \\ A &\rightarrow aAb \mid \epsilon, \end{aligned}$$

derive:

1. ϵ ,

2. 01,
3. *ab*,
4. 0*ab*1, and
5. 00*aabb*11.

Identify the 0/1 and *a/b* dependencies.

Exercise 4.4.2.5 Explain why:

$$\{0^n a^m b^m 1^n \mid n, m \geq 0\}$$

has nested dependencies and is describable by the linear grammar in (30), whereas:

$$\{0^n 1^n a^m b^m \mid n, m \geq 0\}$$

contains two sequentially independent dependencies.

Exercise 4.4.2.6 The language of palindromes is linearly admissible but is not attested as the complete word or sentence pattern of a natural language. Why does formal admissibility not entail natural-language attestation?

Exercise 4.4.2.7 Compare the English compound in (23) with an ordinary syntactic phrase. Why does substitution by *one* fail to make all the internal referents independently accessible?

Exercise 4.4.2.8 Trace the indexed dependencies in the Turkish compound in (25):

Kadıköy belediyesi vergi dairesi çöp vergisi yatırma gişeleri binası.

Why do the questions in (25b–d) suggest that morphological embedding does not transmit reference as freely as syntactic embedding?

Exercise 4.4.2.9 Compare the compound in (25) with the syntactic possessive construction in (26). Which deeply embedded referent is accessible to questioning in the syntactic construction? What does the contrast suggest about morphological and syntactic admissibility?

4.5 Methods for context-free grammars

4.5.1 Structure-preserving and stringset-preserving CFGs

Exercise 4.5.1.1 Distinguish weak equivalence from strong equivalence between two grammars.

1. What must weakly equivalent grammars preserve?
2. What additional property must strongly equivalent grammars preserve?
3. Can two grammars generate the same strings but assign different structures?

Exercise 4.5.1.2 Using the grammar in (31), give two derivation trees for:

$$9 \div 3 \div 8.$$

Identify the bracketings represented by the trees. Which one corresponds to ordinary left associativity?

Exercise 4.5.1.3 Using the grammar in (31), give two derivation trees for:

$$9 + 3 \times 8.$$

Identify the arithmetic value associated with each tree. Which tree respects ordinary operator precedence?

Exercise 4.5.1.4 Why is the grammar in (31) adequate in weak generative capacity but inadequate as a structural description of ordinary arithmetic? Identify the two kinds of overinterpretation it permits.

Exercise 4.5.1.5 Consider the revised grammar in (34):

$$\begin{aligned} S &\rightarrow S + T \mid S - T \mid T, \\ T &\rightarrow T \times F \mid T \div F \mid F, \\ F &\rightarrow (S) \mid -S \mid 0 \mid 1 \mid \cdots \mid 9. \end{aligned}$$

Draw the unique intended tree for:

$$9 + 3 \times 8.$$

Explain how the categories S , T and F encode precedence.

Exercise 4.5.1.6 Using the grammar in (34), draw the intended tree for:

$$9 \div 3 \div 8.$$

How does left recursion enforce left associativity?

Exercise 4.5.1.7 Give derivations under (34) for:

1. $2 + 3$,
2. $2 \times 3 + 4$,
3. $2 + 3 \times 4$, and
4. $(2 + 3) \times 4$.

Explain how parentheses override ordinary precedence.

Exercise 4.5.1.8 Why can a descriptively simpler but structurally ambiguous grammar make later model selection and error diagnosis harder? Give an example in which a failure could be blamed either on the grammar or on the computational method.

4.5.2 Normal forms and binary CKY derivation

Exercise 4.5.2.1 What is a normal form for context-free grammars? Explain why converting a grammar to a normal form must preserve its language but need not preserve its original derivation trees.

Exercise 4.5.2.2 Determine whether each rule conforms to Greibach normal form.

1. $A \rightarrow aBC$
2. $A \rightarrow a$
3. $A \rightarrow Ba$
4. $A \rightarrow abB$
5. $A \rightarrow \epsilon$

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6. $S \rightarrow \epsilon$, when $\epsilon \in \mathcal{L}(G)$

Exercise 4.5.2.3 Determine whether each rule conforms to Chomsky normal form.

1. $A \rightarrow BC$
2. $A \rightarrow a$
3. $A \rightarrow B$
4. $A \rightarrow aB$
5. $A \rightarrow BCD$
6. $S \rightarrow \epsilon$, when $\epsilon \in \mathcal{L}(G)$

Exercise 4.5.2.4 Why does Greibach normal form reveal the constant-growth property of nonempty context-free strings? How many terminals does each derivational step introduce?

Exercise 4.5.2.5 Explain why Chomsky normal form shows that every context-free structure can be represented by binary branching. Does this establish that every empirical linguistic structure is intrinsically binary branching?

Exercise 4.5.2.6 Convert the following grammar to Chomsky normal form:

$$\begin{aligned} S &\rightarrow aAB, \\ A &\rightarrow a, \\ B &\rightarrow b. \end{aligned}$$

Introduce any additional nonterminals required.

Exercise 4.5.2.7 For a string of length 5, state:

1. the number of rows in its CKY chart,
2. the number of cells in each row,
3. the total number of cells, and
4. the span covered by the single cell in the final row.

Exercise 4.5.2.8 A CKY cell covers the substring from position i to position j . Explain how every possible division point k , where $i < k < j$, supplies two earlier cells that can be combined using a rule:

$$A \rightarrow BC.$$

Exercise 4.5.2.9 For a four-symbol string, list the possible span-length combinations used to construct:

1. a cell of length 2,
2. a cell of length 3, and
3. a cell of length 4.

Exercise 4.5.2.10 Why can CKY decide whether a string belongs to the language of a grammar in Chomsky normal form? What must appear in the cell covering the entire input string for the answer to be positive?

Exercise 4.5.2.11 Explain why a recursive start symbol together with:

$$S \rightarrow \epsilon$$

causes difficulty for the ordinary CKY chart. How does replacing the empty-string endpoint with a special terminal such as \$ relax and then restore the original problem?

Exercise 4.5.2.12 Using G'' in (37c), fill the lexical row of a CKY chart for:

baab\$.

Then identify the binary combinations required to derive the start symbol over the complete string.

Exercise 4.5.2.13 Reconstruct the CKY derivation in Figure 4.3 for:

baab\$.

For each nonlexical entry, record:

1. the grammar rule used,

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2. the two shorter chart entries combined, and
3. the span covered by the result.

Exercise 4.5.2.14 Figure 4.4 contains two complete derivations of:

$$n + n * n.$$

Identify the two bracketings. Which one corresponds to standard arithmetic precedence?

Exercise 4.5.2.15 A CKY chart contains several nonterminals in one cell. Does this show that the CKY algorithm is nondeterministic, that the grammar is ambiguous, or both? Explain the distinction between the completeness of the algorithm and the ambiguity of the grammar.

Exercise 4.5.2.16 Explain why CKY is a computational method rather than a grammatical description. What information comes from the grammar, and what operations are contributed by the algorithm?

4.6 Methods for regular expressions

4.6.1 Some descriptions are stylistically simpler than others

Exercise 4.6.1.1 Factor the following regular expressions while preserving their languages.

1. $uv \cup uw$
2. $vu \cup wu$
3. $auv \cup auw$
4. $uvx \cup uvy$

Exercise 4.6.1.2 Using the identities in (39), simplify:

$$u(u \cup v)^* \cup v(u \cup v)^*.$$

Explain each step and give the final expression using $+$.

Exercise 4.6.1.3 Verify the identity:

$$(uv)^*u = u(vu)^*$$

by listing strings obtained from each side for zero, one, two and three iterations.

Exercise 4.6.1.4 Use the identities in (39) to give alternative regular expressions for:

$$(u \cup v)^*.$$

Explain informally why each expression describes every string over the two component languages.

Exercise 4.6.1.5 Give two weakly equivalent regular expressions that differ substantially in length or readability. Why does a simpler expression not necessarily imply a simpler computation?

Exercise 4.6.1.6 Distinguish:

1. simplicity of admissibility,
2. simplicity of description, and
3. simplicity of computation.

Why can none of these be inferred automatically from either of the other two?

4.6.2 Regular relations

Exercise 4.6.2.1 Interpret the phonological rule schema:

$$A \rightarrow B/C \underline{\hspace{1cm}} D.$$

Identify:

1. the input pattern,
2. the output pattern,
3. the left context, and
4. the right context.

3. $(a : x)(b : y)$

4. $(a : x \cup a : y)$

Which relation is not functional?

Exercise 4.6.2.7 Construct regular relations that:

1. replace every a with b ,
2. preserve every a ,
3. relate every a to ϵ ,
4. insert b after every a , and
5. optionally relate a to either a or b .

Exercise 4.6.2.8 The terms *insertion* and *deletion* describe the relation from an input-oriented perspective. Explain why:

$$a : \epsilon$$

does not literally destroy a mathematical object.

Exercise 4.6.2.9 Consider the relation in Section 4.6.2 which retains every c and maps all occurrences of a and b to ϵ while requiring every third input symbol to be b . Determine whether the following pairs belong to the relation:

1. $aabc : c$
2. $aabc : ac$
3. $aabcc : cc$
4. $aacc : cc$
5. $aab : \epsilon$

For every rejected pair, state whether the input condition or the output correspondence fails.

Exercise 4.6.2.10 Explain why a regular relation has both an input language and an output language. How does recognizing a pair differ from recognizing a string in one regular language?

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Exercise 4.6.2.11 Compare:

$$L = \{a^n b^n \mid n \in \mathbb{N}\}$$

with:

$$R = (a : b)^*.$$

1. Why is L not a regular language?
2. Why is R a regular relation?
3. How are the a/b correspondences organized differently?
4. What category mistake results from treating the two objects as equivalent?

Exercise 4.6.2.12 A relation may associate one input with several outputs. Explain why this does not make it a defective function. Give a linguistic example in which one phonological or morphological input has more than one possible output.

Exercise 4.6.2.13 Use the informal description of Sanskrit nasal retroflexion following (43) to identify:

1. the segment that changes,
2. its output,
3. the preceding trigger,
4. the prohibited intervening material,
5. the following context, and
6. the domain of application.

Exercise 4.6.2.14 Why must irrelevant phonological features be preserved between the input and output descriptions of a regular relation? What incorrect predictions could result if they were simply omitted?

Exercise 4.6.2.15 Explain why a human reader may successfully interpret an underspecified phonological rule whereas a computer cannot be expected to supply the missing conditions without an explicit representation. Relate your answer to the “no homunculi” principle.

Exercise 4.6.2.16 Distinguish:

1. a regular expression,
2. a regular relation,
3. a finite automaton, and
4. a finite-state transducer.

Which objects describe, which compute, and which involve one or two languages?

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5.1 Pushdown automata

5.1.1 Definitions

Exercise 5.1.1.1 Let:

$$M = (Q, \Sigma, \Gamma, \Delta, s, F)$$

be a pushdown automaton. State the role of each component. Which components distinguish a PDA from a finite automaton?

Exercise 5.1.1.2 Interpret the transition:

$$(p, a, A), (q, B) \in \Delta.$$

State:

1. the current state,
2. the input symbol passed over,
3. the stack symbol removed,
4. the new state, and
5. the stack material added.

Exercise 5.1.1.3 Explain the effects of the following PDA transitions.

1. $(p, a, A), (q, B)$
2. $(p, \epsilon, A), (q, B)$
3. $(p, a, \epsilon), (q, B)$

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4. $(p, a, A), (q, \epsilon)$

5. $(p, \epsilon, \epsilon), (q, A)$

Which transitions consume input, pop the stack or push onto it?

Exercise 5.1.1.4 Interpret a PDA configuration:

$$[q, \alpha, \gamma].$$

Why do the current state, remaining input and complete stack contents jointly constitute a complete computational snapshot?

Exercise 5.1.1.5 For input w , compare:

$$[s, w, \epsilon] \quad \text{and} \quad [f, \epsilon, \epsilon].$$

Why does the acceptance condition in (2) require:

1. an accepting state,
2. exhausted input, and
3. an empty stack?

Exercise 5.1.1.6 Suppose:

$$(q, a, A), (p, BC) \in \Delta.$$

Complete the one-step computation:

$$[q, axy, Ay] \vdash \underline{\hspace{2cm}}.$$

Identify which parts of the input and stack remain unchanged.

Exercise 5.1.1.7 Explain the difference between:

$$\Rightarrow \quad \text{and} \quad \vdash.$$

Which symbol describes a grammatical derivation, and which describes a computational step involving a machine configuration?

Exercise 5.1.1.8 Interpret:

$$\mathcal{L}(M) = \{w \in \Sigma^* \mid [s, w, \epsilon] \vdash^* [f, \epsilon, \epsilon], f \in F\}.$$

Why is this a procedural definition of a language?

5.1.2 Examples

Exercise 5.1.2.1 Trace the computation in (3c) for:

aabbcd.

At every step, record:

1. the state,
2. the remaining input,
3. the stack contents, and
4. the transition used.

Exercise 5.1.2.2 In the PDA in (3), explain the respective functions of the stack symbols *A* and *C*. Which terminal dependencies do they allow the machine to verify?

Exercise 5.1.2.3 Use the PDA in (3) to determine whether the following strings are accepted.

1. *abcd*
2. *aabbcd*
3. *abccdd*
4. *aabbccdd*
5. *aabbbcd*
6. *aabb*

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For every rejected string, identify the first failed acceptance condition.

Exercise 5.1.2.4 Explain why:

$$[q_1, \epsilon, C]$$

is not an accepting configuration in the PDA in (3), even though the input has been exhausted.

Exercise 5.1.2.5 Explain why:

$$[p_2, \epsilon, \epsilon]$$

is not an accepting configuration in the PDA in (3), even though both the input and stack have been exhausted.

Exercise 5.1.2.6 The PDA in (4) recognizes:

$$\{wcw^r \mid w \in \{0, 1\}^*\}.$$

List the strings obtained when:

$$w = \epsilon, \quad w = 0, \quad w = 01, \quad w = 110.$$

Exercise 5.1.2.7 Trace the PDA computation in (4) for:

$$10c01.$$

Explain:

1. what is stored before c ,
2. what changes when c is encountered, and
3. how the material after c is checked.

Exercise 5.1.2.8 Why can the PDA in (4) reject:

$$01c01$$

even though the same number of 0s and 1s occurs on both sides of c ? What structural condition is stronger than symbol counting?

Exercise 5.1.2.9 Using the PDA in (5), trace the computation of:

1. ab ,
2. $aabb$,
3. $abab$, and
4. $aaabbb$.

Interpret a as an opening parenthesis and b as a closing parenthesis.

Exercise 5.1.2.10 Determine why the PDA in (5) rejects:

1. aab ,
2. abb ,
3. ba , and
4. $aababb$.

Distinguish premature closing from unmatched opening.

Exercise 5.1.2.11 Explain the statement:

A pushdown automaton does not understand a dependency; it possesses enough memory to verify one.

What is represented by the stack, and what semantic interpretation is not supplied by the machine?

Exercise 5.1.2.12 The palindrome PDA in (4) has fewer states than the PDA in (3), although its dependency may appear more complex. Why does this show that the number of states alone is not a general measure of structural complexity?

5.2 Finite automata

5.2.1 Definitions

Exercise 5.2.1.1 Let:

$$M = (Q, \Sigma, \Delta, s, F)$$

be a finite automaton. State the role of each component and compare the definition with that of a PDA.

Exercise 5.2.1.2 Why can an FA configuration be represented as:

$$[q, \alpha]$$

rather than:

$$[q, \alpha, \gamma]?$$

What computational resource has disappeared?

Exercise 5.2.1.3 Interpret the transition:

$$(p, a), (q) \in \Delta$$

and the corresponding computational step:

$$[p, ax] \vdash [q, x].$$

Exercise 5.2.1.4 Distinguish a DFA from an NFA in terms of the mathematical status of Δ .

1. When is Δ a function?
2. When is it merely a relation?
3. What must be unique for every state–input pair in a DFA?

Exercise 5.2.1.5 For each situation, state whether the automaton is deterministic.

1. From q on a , exactly one transition is available.
2. From q on a , two transitions are available.

3. From q , an ϵ -transition is available.
4. From q on a , no transition is available.

Distinguish determinism from completeness of a transition function.

Exercise 5.2.1.6 Interpret:

$$\mathcal{L}(M) = \{w \in \Sigma^* \mid [s, w] \vdash^* [f, \epsilon], f \in F\}.$$

Why is merely reaching an accepting state before the input is exhausted insufficient?

5.2.2 Examples

Exercise 5.2.2.1 In Figure 5.1, each state records whether the numbers of a s and b s read so far are even or odd. Give the parity represented by:

$$s, \quad q_1, \quad q_2, \quad q_3.$$

Why is q_2 the only accepting state?

Exercise 5.2.2.2 Trace the DFA in Figure 5.1 on:

1. ab ,
2. aab ,
3. abb ,
4. $abab$, and
5. $aaabbb$.

State the final state and whether each string is accepted.

Exercise 5.2.2.3 Construct a transition table equivalent to Figure 5.1. Verify from the table that the transition relation is a function.

Exercise 5.2.2.4 Figure 5.2 recognizes strings in which every third symbol is b . Determine whether it accepts:

1. ϵ ,

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2. *a*,
3. *ac*,
4. *aab*,
5. *acbca*,
6. *aacab*, and
7. *acbacb*.

Exercise 5.2.2.5 Explain what information the states in Figure 5.2 retain about the position of the next input symbol. Why is a stack unnecessary?

Exercise 5.2.2.6 Identify two distinct sources of nondeterminism in Figure 5.3:

1. more than one transition on the same input, and
2. empty transitions.

Give an example of each from the automaton.

Exercise 5.2.2.7 Trace all relevant paths of the NFA in Figure 5.3 for:

1. *baa*,
2. *bca*,
3. *bcabc*, and
4. *cab*.

A string is accepted if at least one complete path ends in an accepting configuration.

Exercise 5.2.2.8 All states in Figure 5.3 are accepting. Explain why this does not imply that every input string is accepted.

Exercise 5.2.2.9 Compare the descriptions of the language in Figure 5.2 as:

1. an informal property of strings,
2. a regular expression, and

3. a finite automaton.

Which is an informal characterization, which is a formal description, and which computes membership?

Exercise 5.2.2.10 Explain why every FA may be regarded as a PDA that never uses its stack. Why does the converse fail for strictly context-free languages?

Exercise 5.2.2.11 Compare the computational resources of:

1. a finite automaton,
2. a pushdown automaton, and
3. a Turing machine.

Relate each difference to the kinds of dependencies it can preserve.

Exercise 5.2.2.12 Why does adding more tapes or more than two stacks not enlarge the class of Turing-computable functions? Why does this not mean that all such machine designs are equally efficient?

5.3 Methods for pushdown automata

5.3.1 Obtaining a PDA from CFG: top-down parsing

Exercise 5.3.1.1 For a CFG:

$$G = (V, \Sigma, R, S),$$

state the components of the two-state PDA constructed by the top-down translation in (10). What is its stack alphabet?

Exercise 5.3.1.2 Explain the purpose of the initial transition:

$$(s, \epsilon, \epsilon), (q, S).$$

Why does a top-down computation begin by placing the grammar's start symbol on the stack?

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Exercise 5.3.1.3 For every terminal $a \in \Sigma$, the construction introduces:

$$(q, a, a), (q, \epsilon).$$

What does this transition check, and why is it required even though terminal matching may remain implicit in an ordinary grammatical derivation?

Exercise 5.3.1.4 A grammar rule:

$$A \rightarrow \beta$$

is translated into:

$$(q, \epsilon, A), (q, \beta).$$

Why does this transition consume no input? How does it turn a description into a possible machine action?

Exercise 5.3.1.5 Apply the top-down construction to:

$$S \rightarrow aSb \mid \epsilon.$$

Write:

1. the states,
2. the stack alphabet,
3. the initial transition,
4. the terminal-matching transitions, and
5. the rule-expansion transitions.

Exercise 5.3.1.6 Using the PDA constructed in Exercise 5.3.1.5, give accepting computations for:

$$\epsilon, \quad ab, \quad aabb.$$

Give a failed computation for aab .

Exercise 5.3.1.7 In the top-down computation in (12), identify every step that:

1. expands a nonterminal,

2. matches a terminal, and
3. initializes or completes the computation.

Exercise 5.3.1.8 Why is the direct top-down CFG-to-PDA translation generally nondeterministic? What choices arise when several grammar rules have the same lefthand side?

Exercise 5.3.1.9 Suppose that the grammar contains:

$$A \rightarrow A\beta.$$

Explain why its direct top-down transition:

$$(q, \epsilon, A), (q, A\beta)$$

can be repeated indefinitely without consuming input.

Exercise 5.3.1.10 Eliminate immediate left recursion from:

$$A \rightarrow Aa \mid Ab \mid c \mid d.$$

Use the general transformation in (13).

Exercise 5.3.1.11 Show that the original and transformed grammars in Exercise 5.3.1.10 generate the same strings. Give derivations of:

$$c, \quad ca, \quad cba, \quad daab$$

under both descriptions.

Exercise 5.3.1.12 Why does elimination of left recursion preserve weak equivalence but not necessarily strong equivalence? Draw corresponding left- and right-branching trees for one string.

Exercise 5.3.1.13 Why might a linguist object to eliminating left recursion merely to accommodate a top-down parsing method? Discuss associativity, scope and structure preservation.

5.3.2 Obtaining a PDA from CFG: bottom-up parsing

Exercise 5.3.2.1 Explain why bottom-up parsing begins with the terminal string and attempts to reach the start symbol. How is this direction related to:

$$S \Rightarrow^* w?$$

Exercise 5.3.2.2 In the construction in (15), a grammar rule:

$$A \rightarrow \beta$$

produces the transition:

$$(s, \epsilon, \beta), (s, A).$$

Why is this operation traditionally called a reduction?

Exercise 5.3.2.3 Explain the purpose of:

$$(s, \epsilon, S), (q, \epsilon).$$

Why can the machine enter its accepting state only after the complete stack has been reduced to S ?

Exercise 5.3.2.4 Apply the bottom-up construction to:

$$\begin{aligned} S &\rightarrow AB, \\ A &\rightarrow a, \\ B &\rightarrow b. \end{aligned}$$

Write all PDA transitions and give an accepting computation for ab .

Exercise 5.3.2.5 Using the PDA from Exercise 5.3.2.4, show why the following strings are rejected:

$$a, \quad b, \quad ba, \quad aab.$$

Exercise 5.3.2.6 Why can a bottom-up parser handle a left-recursive grammar without the nontermination problem encountered by direct top-down parsing?

Exercise 5.3.2.7 The righthand side β to be reduced may not occur at the visible top of the stack. Why does this create a search problem? What does an LR method attempt to avoid?

Exercise 5.3.2.8 Trace the two bottom-up computations in (17) for:

$$n + n * n.$$

Identify the reduction at which their structural interpretations first diverge.

Exercise 5.3.2.9 Give the two bracketings represented by the computations in (17). Which one respects ordinary multiplication precedence?

Exercise 5.3.2.10 Compare top-down and bottom-up parsing with respect to:

1. their starting points,
2. the direction in which grammar rules are used,
3. the contents of the stack,
4. left recursion, and
5. nondeterministic choices.

5.4 Methods for finite automata

5.4.1 Eliminating nondeterminism

Exercise 5.4.1.1 Define the ϵ -closure of a state q . Why must it always contain q itself?

Exercise 5.4.1.2 Define the input-transition function $t(q, a)$ used in the subset construction. Why must it include possible ϵ -transitions both before and after consuming a ?

Exercise 5.4.1.3 For an NFA with:

$$q_0 \xrightarrow{\epsilon} q_1, \quad q_1 \xrightarrow{\epsilon} q_2, \quad q_2 \xrightarrow{a} q_3,$$

find:

1. ϵ -closure(q_0),
2. ϵ -closure(q_1), and

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3. $t(q_0, a)$.

Assume that no other transitions exist.

Exercise 5.4.1.4 Why are the states of the constructed DFA subsets of the NFA's state set? Explain what it means for the DFA to be in the state:

$$\{q_1, q_2, q_3\}.$$

Exercise 5.4.1.5 In the subset construction, why is a DFA subset-state accepting whenever it contains at least one accepting state of the original NFA?

Exercise 5.4.1.6 Why must the subset construction terminate when the NFA has finitely many states? If the NFA has n states, what is the largest possible number of DFA subset-states?

Exercise 5.4.1.7 Using the table in (19), calculate:

1. $t(\{s, q_1\}, a)$,
2. $t(\{s, q_1\}, b)$,
3. $t(\{s, q_1\}, c)$,
4. $t(\{q_1, q_3\}, a)$, and
5. $t(\{q_1, q_3\}, b)$.

Exercise 5.4.1.8 Explain why the empty subset $\emptyset$ acts as a dead state in the constructed DFA. Where must every transition from $\emptyset$ lead?

Exercise 5.4.1.9 Reconstruct the DFA in Figure 5.4 from the NFA in Figure 5.3. List:

1. every reachable subset-state,
2. all transitions from each subset-state, and
3. all accepting subset-states.

Exercise 5.4.1.10 The DFA in Figure 5.4 has more states than the NFA in Figure 5.3. Why does this not contradict the claim that determinization simplifies computation?

Exercise 5.4.1.11 Assess the following claim:

Nondeterministic finite automata recognize more languages than deterministic finite automata.

What does the subset construction prove?

5.4.2 From regular expressions to automata and back

Exercise 5.4.2.1 For a right-regular grammar, explain how each rule type is translated into an NFA transition.

$$\begin{aligned} A &\rightarrow aB, \\ A &\rightarrow a, \\ A &\rightarrow \epsilon. \end{aligned}$$

Exercise 5.4.2.2 Why is a new accepting state Z introduced in the regular-grammar-to-NFA construction? Why is one such state sufficient even when several grammar rules terminate a derivation?

Exercise 5.4.2.3 Convert the following regular grammar into an NFA:

$$\begin{aligned} S &\rightarrow aA \mid bS \mid \epsilon, \\ A &\rightarrow aS \mid bA. \end{aligned}$$

Identify all states, transitions, the start state and the accepting state.

Exercise 5.4.2.4 What property of strings is described by the grammar in Exercise 5.4.2.3? Use the corresponding NFA to test:

$$\epsilon, \quad a, \quad aa, \quad aba, \quad aabb.$$

Exercise 5.4.2.5 Convert the grammar in (22) into an NFA by applying the construction in (20). Explain the role of states S , A and Z .

Exercise 5.4.2.6 Why can the grammar in (22) describe prohibited continuations only indirectly, whereas the corresponding automaton displays failure as an unavailable transition or a dead state?

Exercise 5.4.2.7 Apply subset construction to the NFA in Figure 5.5 and obtain the DFA in Figure 5.6. Why are:

$$\{S, Z\} \quad \text{and} \quad \{A, Z\}$$

accepting states?

Exercise 5.4.2.8 Describe the procedure in (23) for converting an FA into a right-regular grammar. Why is a new unique state Z useful before the transitions are translated into grammar rules?

Exercise 5.4.2.9 Convert the following FA into a right-regular grammar:

$$q_0 \xrightarrow{a} q_1, \quad q_1 \xrightarrow{b} q_1,$$

where q_0 is the start state and q_1 is accepting. Describe the language of the resulting grammar.

Exercise 5.4.2.10 Explain how the regular-grammar patterns below correspond to regexp operations.

$$\begin{aligned} A &\rightarrow aB, & B &\rightarrow b, \\ A &\rightarrow a \mid b, \\ A &\rightarrow aA \mid \epsilon. \end{aligned}$$

Identify concatenation, union and Kleene iteration.

Exercise 5.4.2.11 For the grammar:

$$\begin{aligned} S &\rightarrow 1S \mid 2S \mid 0A \mid \epsilon, \\ A &\rightarrow 0A \mid 2S \mid \epsilon, \end{aligned}$$

write recurrence-style regular expressions for S and A . Substitute the expression for A into the expression for S .

Exercise 5.4.2.12 Explain informally why:

$$(1 \cup 2 \cup 0^+ 2)^* (0 \cup \epsilon)^*$$

describes all strings over $\{0, 1, 2\}$ containing no substring 01. State the role of each parenthesized component.

Exercise 5.4.2.13 Compare the following routes:

$$\text{regex} \longrightarrow \text{regular grammar} \longrightarrow \text{NFA}$$

and:

$$\text{FA} \longrightarrow \text{regular grammar} \longrightarrow \text{regex}.$$

What is preserved at every stage, and what may become more complicated?

Exercise 5.4.2.14 Why does the existence of an algorithmic conversion not guarantee that the resulting regex, grammar or automaton is stylistically simple? Distinguish equivalence from elegance.

5.4.3 Transducers and regular relations

Exercise 5.4.3.1 Distinguish an ordinary finite automaton from a finite-state transducer. What additional alphabet and output relation must a transducer possess?

Exercise 5.4.3.2 Compare Moore and Mealy machines.

1. Which associates output with states?
2. Which associates output with transitions?
3. Does the distinction change the class of regular relations that can be computed?

Exercise 5.4.3.3 Let:

$$M = (Q, \Sigma_i, \Sigma_o, \Delta, s, F)$$

be a Mealy machine. State the role of Σ_i and Σ_o and interpret:

$$(p, a : b), (q) \in \Delta.$$

Exercise 5.4.3.4 Interpret the starting and accepting configurations:

$$[s, w_i : \epsilon] \quad \text{and} \quad [f, \epsilon : w_o].$$

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Which string is consumed, and which string is accumulated?

Exercise 5.4.3.5 Explain the difference between:

$$\mathcal{L}_i(M)$$

and:

$$\mathcal{L}_o(M, w_i).$$

Why may one accepted input have more than one possible output if the transition structure is relational?

Exercise 5.4.3.6 Construct a one-state Mealy machine computing each relation.

1. Every a is preserved.
2. Every a is related to b .
3. Every a is related to ϵ .
4. Every a may be related to either a or b .

State which machines are functional and which are relational.

Exercise 5.4.3.7 Suppose that H represents an underspecified high vowel and may be realized as:

$$i, \quad i, \quad u, \quad \ddot{u}.$$

Write the four atomic regular relations. What phonological contexts would be required to select among them deterministically?

Exercise 5.4.3.8 Trace the Mealy machine in Figure 5.8 on:

1. aab ,
2. acb ,
3. $aabacb$, and
4. $acbach$.

Give the output string for every accepted input.

Exercise 5.4.3.9 In Figure 5.8, why is c immediately before every required third-position b related to ϵ , while other occurrences of c are preserved?

Exercise 5.4.3.10 Explain why the apparent deletion in:

$$c : \epsilon$$

is mathematically a correspondence between two strings rather than the physical destruction of a symbol.

Exercise 5.4.3.11 Construct a Mealy machine that recognizes strings over $\{a, b\}$ and changes every a to b while preserving every b . Give:

1. its formal components,
2. its transition diagram, and
3. computations for aab and $baba$.

Exercise 5.4.3.12 Explain the three-way distinction summarized in Figure 5.9:

admissibility	$\longrightarrow$	invariant,
grammar	$\longrightarrow$	description,
automaton	$\longrightarrow$	recognition,
transducer	$\longrightarrow$	transformation.

Why should none of the three variant methods be identified with admissibility itself?

Exercise 5.4.3.13 Choose a simple morphological or phonological alternation from a language you know. Specify:

1. the input alphabet,
2. the output alphabet,
3. the relevant regular relations,
4. the conditioning context, and
5. the intended input–output pairs.

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Sketch a transducer capable of computing the relation.

Exercise 5.4.3.14 Assess the following claim:

Because a finite-state transducer produces an output, it explains the linguistic relation between the input and output.

Distinguish:

1. computing a correspondence,
2. describing its formal pattern,
3. stating its admissibility conditions, and
4. explaining its linguistic significance.

6 Mathematics of linguistic embedding and delineation

6.1 Why are syntax and semantics necessarily embedding?

Exercise 6.1.1 For each of the following, identify the embedded constituent at every level. Use brackets to display the resulting structure.

1. The man the dog bit laughed.
2. The man the dog the cat chased bit laughed.
3. I know that she left.
4. You know that I know that she left.
5. The man's book's dust jacket's color faded.

Which examples involve clausal embedding, and which involve nonclausal nesting?

Exercise 6.1.2 Define the *embedding depth* of a clause as follows: a matrix clause has depth zero, a clause immediately embedded in it has depth one, and so forth. Determine the maximum embedding depth of each sentence.

1. She believes that the earth is round.
2. I know that she believes that the earth is round.
3. You think that I know that she believes that the earth is round.
4. I know you know she knows that John believes that the earth is flat.

Is there a grammatical upper bound on the embedding depth generated by the relevant syntactic rule? Distinguish this question from the practical difficulty of processing very deep embeddings.

Exercise 6.1.3 Compare the following two structural patterns:

$$A_1[A_2[A_3B_3]B_2]B_1$$

and

$$A_1B_1[A_2B_2[A_3B_3]].$$

1. Which pattern represents center embedding?
2. Which pattern represents right-branching embedding?
3. Assign the words of the following sentences to the corresponding positions:
 - a) The man the dog the cat chased bit laughed.
 - b) I know you know she knows that John left.
4. Explain why the two patterns may differ sharply in processing difficulty even if both are recursively generable.

Exercise 6.1.4 Consider the following judgments.

1. The man the dog the cat chased bit laughed.
2. I know you know she knows that John believes that the earth is flat.
3. *I know you know she likes that John believes that the earth is flat.

For each sentence, decide whether any difficulty arises principally from:

1. grammatical incompatibility,
2. embedding depth,
3. the position of the embedded material, or
4. a combination of these.

Explain why processing difficulty alone does not establish ungrammaticality.

Exercise 6.1.5 The textbook contrasts English center embedding with a corresponding Turkish construction.

1. State the contrast.

6.1 Why are syntax and semantics necessarily embedding?

2. Does the contrast show that one language possesses recursion while the other does not?
3. What does it instead suggest about the relation among grammatical structure, linear order, and processing?
4. Formulate one empirical prediction that could be tested using speakers of the two languages.

Exercise 6.1.6 Compare the interpretations of the following:

1. She persuaded him to study.
2. She promised him to study.
3. She expected him to study.
4. She wanted him to study.

For each sentence:

1. identify the understood subject of *study*;
2. determine whether *him* is a semantic argument of the matrix verb;
3. describe the relation that the matrix verb establishes between its participants and the embedded proposition.

What remains syntactically similar across the four examples, and what changes semantically?

Exercise 6.1.7 For each pair, determine whether the first sentence entails the second.

1. a) She managed to leave.
b) She left.
2. a) She tried to leave.
b) She left.
3. a) He knows that she left.
b) She left.

4. a) He thinks that she left.
b) She left.

Explain why the syntactic fact that a proposition is embedded does not by itself determine its semantic status.

Exercise 6.1.8 Compare:

1. He does not think that she left.
2. He does not know that she left.
1. Does either sentence entail that she did not leave?
2. Does either sentence presuppose or entail that she left?
3. Describe the contribution of the matrix predicate in each case.
4. Explain why identical-looking syntactic scope need not produce identical semantic consequences.

Exercise 6.1.9 Compare:

1. She remembered to lock the door.
2. She remembered that she locked the door.

For each sentence, state:

1. the temporal relation between the remembering and the locking;
2. whether the sentence presents the locking as an obligation or intended action;
3. whether the embedded material describes a remembered fact.

Explain how the syntactic form of the complement constrains, but does not exhaust, the semantic distinction.

Exercise 6.1.10 The chapter characterizes syntax as the transparent transmission of structured content and semantics as the structured content transmitted.

1. Explain this distinction in your own words.

2. Why must syntax preserve enough structure for embedded meanings to be interpreted?
3. Why does preservation of structure not amount to determination of meaning?
4. Illustrate your answer using one pair of embedding verbs from the preceding exercises.

6.2 Contingent homomorphisms: putting embeddings together

Exercise 6.2.1 Insert parentheses into each expression so that its constituent structure is unambiguous.

1. she played the piano
2. she played the piano for an hour
3. she played the piano in a year
4. the child knows the teacher left

For each expression, explain what licenses the matching of the parentheses. Why is a formally balanced arrangement not necessarily a grammatical linguistic structure?

Exercise 6.2.2 State the rule-to-rule thesis in your own words. Then distinguish the following claims:

1. every syntactic rule has a corresponding semantic operation;
2. syntactic and semantic structures are identical;
3. the syntax–semantics correspondence is an empirically supported regularity;
4. the correspondence holds by definition.

Which claims are compatible with the chapter's conception of a contingent homomorphism?

Exercise 6.2.3 Let S be a set of syntactic structures with a syntactic combination operation $\circ_S$, and let M be a set of meanings with a semantic combination operation $\circ_M$. A mapping $h : S \rightarrow M$ is homomorphic when

$$h(a \circ_S b) = h(a) \circ_M h(b).$$

1. Explain what each side of this equation represents.
2. Translate the equation into an informal claim about phrase construction.
3. What evidence would count against a proposed mapping h ?
4. Why can a mapping satisfy the equation for some constructions but fail for others?

Exercise 6.2.4 The chapter proposes that a syntactic description answers *what* is being combined, whereas a semantic method specifies *how* it is interpreted. Apply this distinction to:

$$VP \longrightarrow V_{tr} NP.$$

1. What information is supplied by the syntactic rule?
2. What additional information must the semantic rule supply?
3. Could two lexical verbs satisfying the same syntactic rule make different semantic contributions? Give an example.

Exercise 6.2.5 Using the rules in Section 6.2, write the sequence of syntactic and semantic actions required to construct *she played the piano*. At every step, display:

1. the syntactic constituent recognized;
2. the semantic expression associated with it;
3. the rule licensing the next combination.

Use the argument order adopted in the textbook.

Exercise 6.2.6 Perform the following reductions symbolically:

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1.

$$(\lambda v. \lambda np. v(np))(V)(N);$$

2.

$$(\lambda v. \lambda np. \lambda pp. v(np)(pp))(V)(N)(P);$$

3.

$$(\lambda np. \lambda vp. vp(np))(N)(V);$$

4.

$$(\lambda p. \lambda np. p(np))(P)(N).$$

At each step, state which variable is replaced and by which expression.

Exercise 6.2.7 Let

$$\llbracket \text{she} \rrbracket = s, \quad \llbracket \text{the piano} \rrbracket = p,$$

and let the transitive use of *played* have denotation

$$\lambda x. \lambda y. \text{play}'(x, y).$$

Following the argument order specified by the rules in the textbook:

1. combine the verb with the object;
2. combine the resulting VP meaning with the subject;
3. give the fully reduced expression;
4. identify where the syntactic derivation and semantic derivation proceed in parallel.

Exercise 6.2.8 Classify the most natural interpretation of *play the piano* in each sentence as an activity or an accomplishment.

1. She played the piano for an hour.
2. She played the piano in a year.
3. The child played the piano every afternoon.
4. The prodigy played the piano in six months.

State what contextual enrichment is needed where the accomplishment interpretation is selected.

Exercise 6.2.9 Explain the contrasts:

1. She won the competition in an hour.
2. *She won the competition for an hour.
3. She was champion for a year.
4. *She played the piano in an hour.

For the last example:

1. construct a context in which it becomes acceptable;
2. state how the lexical category or aspectual interpretation changes;
3. explain why this is not merely a change in elapsed time.

Exercise 6.2.10 Suppose a parser recognizes a verb as V_{tr} but assigns it the semantic operation stipulated for V_{acmp} .

1. Why is this not a harmless notational substitution?
2. What kind of incorrect interpretation might result?
3. At which point in a rule-to-rule derivation should the mismatch be detected?
4. Relate the error to the notion of a category mistake.

Exercise 6.2.11 Design a bottom-up derivation for a simple transitive sentence. Represent a syntactic step by $\Rightarrow$ and its corresponding semantic action by $\vdash$.

1. Begin with the three lexical items.
2. construct the object NP and the verb;
3. construct the VP;
4. combine the VP with the subject NP;

5. show the final syntactic and semantic outputs.

Why is bottom-up construction especially suitable for associating meaning with constituents as their grammaticality is established?

Exercise 6.2.12 Give one hypothetical language pattern for each of the following:

1. a transparent syntax–semantics correspondence;
2. one syntactic construction with two systematically different interpretations;
3. two syntactic constructions with approximately the same interpretation;
4. a syntactically regular expression whose interpretation depends strongly on context.

Explain why the last three do not make formal modelling impossible, but do make the proposed homomorphisms contingent.

6.3 Contingent homomorphisms and fundamental questions

Exercise 6.3.1 For each statement, decide whether it treats the syntax–semantics relation as axiomatic or contingent.

1. Every syntactic derivation has exactly one interpretation by definition.
2. A proposed semantic mapping is tested against speaker judgments.
3. The correspondence between two domains is assumed before data are examined.
4. Some mappings are stable across speakers, while others vary by context.
5. A model is revised when it predicts an unattested interpretation.

Justify each answer.

Exercise 6.3.2 A child encounters utterances in situations containing sounds, objects, actions, intentions, and other participants.

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1. What regularities must the child discover?
2. Which of these concern form?
3. Which concern structure?
4. Which concern meaning and action?
5. Why does successful acquisition not require the child to begin by explicitly assuming a syntax–semantics homomorphism?

Exercise 6.3.3 Construct a small hypothetical dataset containing:

1. three utterances;
2. the situations in which they occur;
3. a possible syntactic analysis;
4. a possible interpretation.

Propose two competing mappings between structure and meaning. State what further observation would distinguish them.

Exercise 6.3.4 Consider possible invariants across:

1. individual speakers;
2. spoken and signed modalities;
3. unrelated languages;
4. child and adult grammars;
5. different communicative situations.

For each comparison, propose one candidate invariant and one property that might vary. Explain how the claim could be investigated empirically.

Exercise 6.3.5 Match each question with the relevant area of investigation:

1. How is a grammatical system acquired?
2. In what sense are child and adult grammars related?

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3. Why do languages differ despite constraints on possible dependencies?

The areas are:

1. continuity;
2. diversity;
3. ontogeny.

Then explain why a theory of contingent mappings must address all three.

Exercise 6.3.6 Distinguish three versions of the continuity hypothesis:

1. complete continuity;
2. partial continuity;
3. domain-specific continuity.

For each version:

1. formulate a precise claim;
2. give a possible supporting observation;
3. give a possible counterexample.

Exercise 6.3.7 The admissibility of grammaticality may remain stable while the content entering grammatical distinctions develops.

1. Explain this claim in your own words.
2. Illustrate it with the acquisition of achievements or accomplishments.
3. Illustrate it with the development of joint attention.
4. Why would it be a mistake to infer from developing content that the entire grammatical system changes in kind?

Exercise 6.3.8 Two models fit the same initial acquisition data:

1. Model A assumes a fixed one-to-one mapping between syntactic rules and meanings.

2. Model B permits several mappings whose applicability depends on lexical and contextual evidence.

Design two additional observations that could favor one model over the other. Explain why model selection becomes substantive once homomorphism is not guaranteed in advance.

6.4 Why are morphology and phonology sufficiently decomposing and necessarily nonembedding?

Exercise 6.4.1 Distinguish:

1. repetition;
2. iteration;
3. hierarchical decomposition;
4. embedding.

For each pattern below, state which notions are involved:

1. ha ha ha;
2. repeated plural marking;
3. a word divided into stem and affixes;
4. a clause contained in another clause;
5. a phonological form divided into syllables and segments.

Exercise 6.4.2 Evaluate the following claims.

1. A word may contain several morphemes.
2. A morpheme normally contains another morpheme as one of its grammatical arguments.
3. A syllable may contain several phonemes.

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4. A phoneme normally contains another phoneme as a referentially accessible dependent.
5. Any hierarchical analysis is therefore an instance of embedding.

Correct the false claims and explain the distinction on which the section relies.

Exercise 6.4.3 Consider the Turkish nominal construction discussed in Section 6.4:

Kadıköy belediye-si_i vergi daire-si_j çöp vergi-si_k yatırma_k gişe-ler-i_j bina-sı_i.

1. Mark the three dependency pairs.
2. Describe the nesting pattern using brackets.
3. Which element supplies the outermost referent?
4. Why does the formal presence of several nested dependencies not establish unrestricted semantic embedding?

Exercise 6.4.4 The construction in Exercise 6.4.3 can answer a question corresponding to *What building is this?*, but not, without restructuring, questions corresponding to *What office is this?* or *What tax is this?*

1. What does this show about the accessibility of the inner nominal material?
2. How does it differ from the accessibility of an embedded clause in syntax?
3. Explain the notion of referential encapsulation.

Exercise 6.4.5 Suppose the outer dependency in the Turkish construction is changed from third-person possession to first-person plural possession.

1. Which two positions must display the corresponding change?
2. Why may those positions be linearly far apart?
3. Does long-distance agreement by itself prove syntactic embedding?
4. Repeat the analysis for a change to the inner dependency indexed *j*.

Exercise 6.4.6 Compare the nominal construction with:

Where is the child who I know you know?

1. Identify the syntactic dependencies in the sentence.
2. From which depth is *the child* interpreted as an argument?
3. Why is the dependency accessible through the nested clauses?
4. Contrast this with the inaccessible inner referents of the Turkish nominal construction.

Exercise 6.4.7 For each phenomenon, choose the more appropriate initial model: a finite automaton, a finite-state transducer, or an embedding grammar.

1. recognizing a fixed inventory of permissible syllable shapes;
2. mapping a singular noun to a regularly suffixed plural;
3. tracking a clause inside a clause inside a clause;
4. recognizing a sequence of morphotactically ordered affixes;
5. assigning tense information while reading an inflected verb.

Defend each choice.

Exercise 6.4.8 Design a small transducer that maps:

$$\begin{aligned} cat &\mapsto cat' \wedge sg', \\ cats &\mapsto cat' \wedge pl', \\ dog &\mapsto dog' \wedge sg', \\ dogs &\mapsto dog' \wedge pl'. \end{aligned}$$

1. Specify the states and transitions.
2. State what the machine recognizes.
3. State what semantic information it produces.
4. Explain why production of correlated meaning does not require the machine to construct an embedded semantic object.

Exercise 6.4.9 Explain and defend the following paired characterization:

Morphology and phonology are sufficiently decomposing and necessarily nonembedding.

Your answer should:

1. explain *sufficiently*;
2. explain *necessarily*;
3. distinguish decomposition from absence of internal structure;
4. identify one empirical observation that could force refinement of the characterization.

6.5 Mathematics of delineation

Exercise 6.5.1 Explain the complementary roles of embedding and delineation.

1. What question about connectedness does embedding answer?
2. What question about identity does delineation answer?
3. Why must a linguistic system both connect to other systems and remain distinguishable from them?
4. Give one linguistic example of each process.

Exercise 6.5.2 Classify each boundary as primarily physical, functional, representational, social, or conventional. More than one classification may be appropriate.

1. the membrane of a cell;
2. the boundary between syntax and music in one mind;
3. the boundary of a speech community;
4. the boundary between a word and a neighboring word;
5. the boundary of a person's private vocabulary;

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6. the boundary between a language and a closely related dialect.

Explain why an abstract boundary need not be unreal.

Exercise 6.5.3 Define homeostasis and describe a minimal homeostatic system.

1. What quantity is maintained?
2. What detects deviation?
3. What acts to correct it?
4. What distinguishes the system from its environment?

Suggest a cautious linguistic analogy and state one respect in which the analogy may fail.

6.5.1 The idea of system as an abstraction

Exercise 6.5.1.1 Define:

1. a system;
2. an element;
3. an interconnection;
4. a control system;
5. a desired response.

Why is a mere collection of objects not necessarily a system?

Exercise 6.5.1.2 For a thermostat-controlled heating system, identify:

1. the desired response;
2. the sensor;
3. the controller;
4. the actuator;
5. the process;

6. the input;
7. the output;
8. a possible disturbance from the environment.

Then repeat the analysis for a linguistic interaction of your choice.

Exercise 6.5.1.3 Analyze a sweeping robot as a control system.

1. What does the sensor measure?
2. What does the motor actuator control?
3. What environmental conditions can affect the process?
4. Give one internal failure and one environmental obstruction.
5. Why is the distinction between system and environment useful for diagnosis?

Exercise 6.5.1.4 The chapter lists five properties relevant to complexity:

1. many components;
2. many interactions;
3. nonlinearity;
4. asymmetry;
5. nonholonomy.

For each property:

1. define it;
2. provide a nonlinguistic example;
3. suggest a possible linguistic example.

Exercise 6.5.1.5 Explain why having many components and many interactions may make a system large without making it complex in the stronger sense used in the chapter. Compare:

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1. a warehouse containing a million identical, noninteracting boxes;
2. a small ecosystem containing ten mutually dependent species.

Which complexity criteria does each satisfy?

Exercise 6.5.1.6 Consider the two systems:

$$y = 3x \quad \text{and} \quad y = x^2.$$

1. Which is linear?
2. Test whether doubling the input doubles the output.
3. Explain why nonlinear interaction can amplify small differences in initial conditions.
4. Why does nonlinearity complicate prediction even when the governing rule is known?

Exercise 6.5.1.7 A growing organism does not merely become a larger collection of identical cells: its cells differentiate.

1. Explain why this is asymmetric development.
2. Contrast differentiation with simple addition.
3. Suggest an analogous case in which parts of a linguistic system become functionally differentiated.
4. State the limits of your analogy.

Exercise 6.5.1.8 Explain the chapter's use of *nonholonomic*.

1. How can a component temporarily fall outside the control of the whole?
2. Why can the system still remain a control system?
3. What follows if a component is permanently outside all control?
4. Give one biological and one linguistic or communicative illustration.

Exercise 6.5.1.9 Distinguish:

1. procedural modularity;
2. representational modularity.

Classify each example and justify your answer:

1. an organ performing a specialized bodily process;
2. a subsystem with access to only a restricted type of representation;
3. a parser passing a structural description to an interpreter through a limited interface;
4. several processes using the same unrestricted database.

Exercise 6.5.1.10 Starting with

$$e^{ix} = \cos x + i \sin x,$$

substitute $x = \pi$ and derive

$$e^{i\pi} + 1 = 0.$$

At each step, state:

1. the relevant trigonometric value;
2. the algebraic substitution;
3. why the imaginary term disappears.

Explain briefly how growth, rotation, and geometric structure are brought together in the identity.

Exercise 6.5.1.11 Consider

$$P(f, x) = \sum_{n \in \mathbb{N}} f(n)x^n.$$

Let $f(n) = 1$.

1. Write the first five terms.
2. Identify the resulting series.
3. State the condition under which it converges.

4. Give its closed form.
5. Evaluate it at $x = \frac{1}{2}$.

Exercise 6.5.1.12 The exponential function has the series representation

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

1. Write the terms through $n = 4$.
2. Approximate e by setting $x = 1$ and using these terms.
3. Differentiate the series term by term.
4. Show that the resulting series is again the series for e^x .
5. Explain why this makes the exponential function especially useful for modelling continuous change.

Exercise 6.5.1.13 Show that

$$x^t = e^{(\ln x)t}$$

for $x > 0$. If $-s = \ln x$, derive

$$x^t = e^{-st}.$$

Explain why this reformulation is useful when continuous time is introduced into a model.

Exercise 6.5.1.14 Using

$$F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt,$$

verify the following transforms:

1. for the unit-step function $u(t)$,

$$\mathcal{L}\{u(t)\} = \frac{1}{s};$$

2. for $f(t) = e^{-at}$,

$$\mathcal{L}\{e^{-at}\} = \frac{1}{s+a}.$$

State the conditions on s required for the integrals to converge.

Exercise 6.5.1.15 Without repeating the integrations, use the transform correspondences in the textbook to complete:

$f(t)$	$F(s)$
1	
e^{-3t}	
t	
t^2	
t^n	

Then explain what information is changed, and what information is preserved, when a function is represented in the s -domain.

Exercise 6.5.1.16 A system has input transform

$$X(s) = \frac{1}{s}$$

and output transform

$$Y(s) = \frac{1}{s(s+2)}.$$

1. Compute the transfer function

$$H(s) = \frac{Y(s)}{X(s)}.$$

2. What does the transfer function describe?
3. Why does it not by itself reveal the internal physical construction of the system?
4. Explain the distinction between modelling a system's dynamics and modelling its internal structure.

6.5.2 Open systems, semantics, and choice

Exercise 6.5.2.1 Using the open-system diagram in Section 6.5.2, define the function of:

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1. the desired response;
2. the comparator and error signal;
3. the controller;
4. the actuator;
5. the process;
6. the sensor;
7. the environmental input.

Describe the complete path from desired response to measured output.

Exercise 6.5.2.2 Suppose the desired output is r , while the sensor reports y .

1. Express a simple error signal e in terms of r and y .
2. What is the sign of e when the measured output is too small?
3. What is its sign when the measured output is too large?
4. Explain how negative feedback can reduce the error.
5. Why must the sensor provide a measurement rather than merely return the unprocessed output?

Exercise 6.5.2.3 Give three ways in which an environment can affect the controller of an open linguistic system. Consider:

1. the participants;
2. their goals;
3. the physical or social situation;
4. their previous discourse.

Explain why these influences introduce genuine degrees of freedom.

Exercise 6.5.2.4 A speaker wishes to get a window closed. Compare:

1. Close the window.

2. Could you close the window?
3. It is cold in here.
1. What desired response can the three utterances share?
2. How do their grammatical and semantic strategies differ?
3. What environmental or social information may guide the speaker's choice?
4. Why can the choice not be determined from truth conditions alone?

Exercise 6.5.2.5 Treat a controller as implementing a policy that maps a perceived situation to an action.

1. Write a simple policy for choosing between a declarative, interrogative, and imperative.
2. Identify the variables on which the policy depends.
3. Which variables are internal to the linguistic system?
4. Which arrive from the environment?
5. Explain how categorial choices can be discrete even when the relevant contextual measurements are continuous.

Exercise 6.5.2.6 Consider the expressions:

1. that cup;
2. the cup beside the book;
3. the cup we discussed yesterday.

Explain how an open referential system may use:

1. perceptual information;
2. linguistic structure;
3. discourse history;
4. speaker choice.

Why is reference adjustment naturally modelled as interaction with an environment?

Exercise 6.5.2.7 Choose one naturally occurring request, correction, or clarification and analyze it as an open system. Identify:

1. the desired response;
2. the initial linguistic action;
3. the observed response;
4. the error, if any;
5. the environmental information;
6. the speaker's adjusted action.

6.5.3 Closed systems, semantics, and decision-making

Exercise 6.5.3.1 Define a closed system as it is used in this section.

1. What happens to external inputs?
2. What happens to the system's degrees of freedom?
3. How is the desired response determined?
4. Why can feedback still occur in a closed system?

Exercise 6.5.3.2 Compare the open and closed systems represented in Figures 6.2 and 6.3. Complete the following comparison:

	Open system	Closed system
Environmental input		
Controller		
Degrees of freedom		
Adjustment of reference		
Status of the mapping		

Explain which contrast is most important for the chapter's account of semantics.

Exercise 6.5.3.3 Explain how an axiomatic syntax–semantics homomorphism can be treated as a closed system.

1. What counts as the input?
2. What fixes the semantic operation?
3. What counts as the output?
4. What role is assigned to possible worlds or models?
5. Why is online adjustment of reference not part of the mapping itself?

Exercise 6.5.3.4 Distinguish:

1. calculating the truth conditions of an expression in a model;
2. deciding what a speaker refers to in a changing environment.

Give an example in which the first task can be completed only after the second has supplied an interpretation for a context-sensitive expression.

Exercise 6.5.3.5 The chapter argues that open and closed systems both require study.

1. Give one question best addressed by a closed formal model.
2. Give one question best addressed by an open interactive model.
3. Give one linguistic phenomenon requiring contributions from both.
4. Explain why success at one level does not eliminate the other.

Exercise 6.5.3.6 Explain the importance of modelability.

1. What makes a linguistic proposal precise enough to be modelled?
2. Must every linguist personally construct mathematical models?
3. What must a nonmodelling researcher provide so that a proposal remains modelable?
4. How can an attempted model reveal an unnoticed problem in a verbal theory?

Exercise 6.5.3.7 A hearer encounters the utterance *She is ready*.

6 Mathematics of linguistic embedding and delineation

1. List at least three possible referents of *she* in a suitable situation.
2. List at least two possible interpretations of what she is ready for.
3. Which aspects can be represented in a closed compositional analysis?
4. Which require information from an open system?
5. Describe the hearer's eventual interpretation as a decision among admissible alternatives.

Exercise 6.5.3.8 Reconstruct the argument represented in Figure 6.4:

linguistic reality $\longrightarrow$ mathematical thinking $\longrightarrow$ admissibility $\longrightarrow \begin{cases} \text{embedding} \\ \text{delineation} \end{cases} \longrightarrow \text{explan}$

For every arrow:

1. explain the relation it represents;
2. cite one example from the book;
3. state what explanatory information would be lost if that step were omitted.

Exercise 6.5.3.9 Choose one linguistic phenomenon examined in the book and give a synthetic analysis using the chapter's principal distinctions:

1. admissibility and grammaticality;
2. embedding and decomposition;
3. syntax and semantics;
4. contingent and axiomatic homomorphism;
5. open and closed systems;
6. connectedness and delineation.

Conclude by stating what the mathematical treatment explains that an informal description leaves unresolved.

Exercise 6.5.3.10 Formulate a research problem arising from Chapter 6.

1. State the linguistic observation.
2. Identify the relevant invariant or dependency.
3. Specify whether the phenomenon is principally one of embedding, decomposition, or delineation.
4. Propose an open-system or closed-system model.
5. State what evidence could falsify the model.
6. Explain what discovering that the model fails would teach us.

Exercises for Mathematics of Linguistic Reality

This companion workbook provides exercises cross-referenced by section to *Mathematics of Linguistic Reality: An Introduction*. It is intended for linguistics students learning to connect mathematical structures, formal grammars, automata, and the empirical study of language.